PREMIUM EXAM EDITION

THE IKKISH 21 QUANT MASTERY SERIES

500 ARITHMETIC WORD PROBLEMS

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VOLUME 1 • TOP 250 FOUNDATIONAL QUESTIONS

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THE IKKISH 21 ARITHMETIC MASTERY (VOLUME 1: Q1 - Q250)

Foundational Chapters: Percentage, Profit & Loss, SI-CI, Ratio & Ages

250 Practice Questions
With Step-by-Step Smart Solutions

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CHAPTER 01: PERCENTAGE & CALCULATION TECHNIQUES

45 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Fraction to %: 1/2 = 50%, 1/3 = 33.33%, 1/4 = 25%, 1/5 = 20%, 1/6 = 16.66%, 1/7 = 14.28%, 1/8 = 12.5%, 1/9 = 11.11%, 1/12 = 8.33%
- % Change = (Final - Initial) / Initial × 100
- Successive % Change: a + b + (ab / 100)
- If A is r% more than B, then B is [r / (100 + r)] × 100% less than A.

Q.001 Percentage

The monthly income of Vikram is Rs. 30000 more than that of Aman. Vikram spends 35% of his income, while Aman spends 45% of his income. If Vikram saves Rs. 24000 more than Aman, find Aman's monthly income.

(A) 60000 (B) 40000
(C) 50000 (D) 36000
(E) 45000

Q.002 Percentage

Aniket spends 20%, 15% and 15% of his monthly salary on rent, food and travel respectively. His remaining money increases by 25% when his friend gives him Rs. 4000. Find Aniket's monthly salary.

(A) 32000 (B) 28000
(C) 36000 (D) 40000
(E) 24000

Q.003 Percentage

The larger number is 20 less than the square of 24. Find the sum of twice 40% of the smaller number and half of the larger number.

(A) 344 (B) 358
(C) 352 (D) 368
(E) 372

Q.004 Percentage

In an election contested by only two candidates, 10% of the total voters did not cast their votes. Out of the votes cast, 25% were declared invalid. The winning candidate received votes equal to 45% of the total voters and won by 4500 votes. Find the total number of voters.

(A) 20000 (B) 24000
(C) 18000 (D) 16000
(E) 22000

Q.005 Percentage

Gifts were planned to be distributed equally among 900 students at an annual function. However, 150 students were absent, so each present student received 4 extra gifts. If all 900 students had been present, how many gifts would each student have received?

(A) 15 (B) 18
(C) 20 (D) 24
(E) 25

Q.006 Percentage

If 60% of 50% of X is 18 and 40% of 75% of Y is 24, then find 25% of 20% of $(X + Y)^2$.

(A) 880 (B) 940
(C) 980 (D) 1040
(E) 1080

Q.007 Percentage

Rs. X is distributed among Megha, Pihu, Garima and Avani. Megha receives 20% more than Garima, and Pihu receives 25% less than Megha. Avani receives Rs. 600 less than Garima. If the difference between the amounts received by Megha and Avani is Rs. 1200, find X.

(A) 10500 (B) 12600
(C) 9600 (D) 13200
(E) 11700

Q.008 Percentage

In a class, there are 360 boys and 180 girls. The numbers of boys and girls who failed are equal. If the number of boys who passed is 150% more than the number of girls who passed, find the total number of students who passed.

(A) 390 (B) 460
(C) 360 (D) 420
(E) 480

Q.009 Percentage

Nandini spends 30% of her monthly salary on groceries. Of the salary remaining after groceries, she spends 20% on rent and 30% on transportation, both calculated on that same balance. If she deposits the remaining Rs. 21000 in a savings account, find her monthly salary.

(A) Rs. 70000 (B) Rs. 60000
(C) Rs. 84000 (D) Rs. 50000
(E) Rs. 75000

Q.010 Percentage

Sameer and Mayank together can complete a project in 16 days. Mayank alone takes 24 days more than Sameer alone to complete the same project. In how many days can Sameer alone complete the project?

(A) 20 days (B) 28 days
(C) 24 days (D) 32 days
(E) 36 days

Q.011 Percentage

Rs. X is distributed among Avani, Meenal, Pallavi and Nandini. Avani receives 30% more than Pallavi, and Meenal receives 20% less than Avani. Nandini receives Rs. 700 less than Pallavi. If the difference between the amounts received by Avani and Nandini is Rs. 1600, find X.

(A) Rs. 11200 (B) Rs. 12320
(C) Rs. 13500 (D) Rs. 10800
(E) Rs. 12640

Q.012 Percentage

A candidate scores 32% marks and fails by 18 marks. Another candidate scores 47% marks and gets 72 marks more than the minimum passing marks. Find the total marks of the examination.

(A) 500 (B) 700
(C) 550 (D) 600
(E) 800

Q.013 Percentage

In an election, two candidates contest against each other. 20% of the votes polled are invalid. The winner gets 70% of the valid votes and wins by 160 votes. Find the number of invalid votes.

(A) 80 (B) 120
(C) 140 (D) 100
(E) 160

Q.014 Percentage

In 2021, the price of diesel was Rs. X per liter, and it increased by 25% every year. If the price of diesel in 2023 is Rs. 250 per liter, find the difference between the prices of diesel in 2022 and 2023.

(A) Rs. 40 (B) Rs. 45
(C) Rs. 60 (D) Rs. 55
(E) Rs. 50

Q.015 Percentage

In a colony, 35% of the residents are male and the rest are female. If 30% of the females are graduates and the number of non-graduate females is 910, find the number of males in the colony.

(A) 650 (B) 900
(C) 700 (D) 800
(E) 1000

Q.016 Percentage

The population of a city was 40,000. It increased by 20% in the first year and decreased by 15% in the second year. Find the population at the end of two years.

(A) 41600 (B) 38000
(C) 43200 (D) 40000
(E) 40800

Q.017 Percentage

The number of students in School N is 25% more than that in School O, and the number of students in School M is 30% less than that in School O. If the total number of students in Schools M and O together is 204, find the total number of students in Schools N and O together.

(A) 240 (B) 260
(C) 270 (D) 280
(E) 300

Q.018 Percentage

In an election between two candidates, Lakshya gets 25% fewer votes than Sandeep. If Sandeep wins by 900 votes, find the total number of votes polled. All votes polled were valid.

(A) 6300 (B) 7200
(C) 5400 (D) 6000
(E) 8100

Q.019 Percentage

Ananya spends 30% of her monthly salary on rent and 20% of the remaining salary on groceries. She saves the rest of the salary. If her monthly saving is Rs. 11200, then find her monthly salary.

(A) Rs. 18000
(B) Rs. 20000
(C) Rs. 22000
(D) Rs. 24000
(E) Rs. 25000

Q.020 Percentage

A student scored 108 marks in Science. He scored 72% marks in Hindi and X marks in English. The maximum marks for each subject are 150. If his overall percentage in all three subjects together is 76%, find the marks scored by him in English.

(A) 120
(B) 132
(C) 118
(D) 126
(E) 140

Q.021 Percentage

The sum of three consecutive odd numbers and three consecutive even numbers is 177. The smallest even number is 7 less than the smallest odd number. Find the sum of the largest odd number and the largest even number.

(A) 59
(B) 61
(C) 65
(D) 63
(E) 67

Q.022 Percentage

In a sports camp, there are 32 boys and 25 girls. The weight of 60% of the girls is less than or equal to 45 kg, and the remaining girls have weight more than 45 kg. The weight of 75% of the boys is greater than or equal to 45 kg, and the remaining boys have weight less than 45 kg. The number of boys having weight less than 45 kg is what percent of the number of girls having weight more than 45 kg?

(A) 70%
(B) 80%
(C) 90%
(D) 60%
(E) 100%

Q.023 Percentage

The difference between the squares of two positive numbers is 192. The square of the smaller number is 4 times the larger number. Find the sum of the two numbers.

(A) 20
(B) 22
(C) 24
(D) 28
(E) 30

Q.024 Percentage

The length and breadth of a rectangular garden are (x + 7) m and (x − 5) m, respectively. If the cost of levelling the garden at Rs. 12 per m² is Rs. 6480, find the cost of fencing the garden at Rs. 18 per meter.

(A) Rs. 1512
(B) Rs. 1620
(C) Rs. 1728
(D) Rs. 1800
(E) Rs. 1944

Q.025 Percentage

Two candidates appeared in a test. One candidate scored 24 marks more than the other. If the marks of the higher-scoring candidate are 56% of the total marks scored by both candidates together, find the marks scored by the higher-scoring candidate.

(A) 104
(B) 112
(C) 118
(D) 124
(E) 128

Q.026 Percentage

The LCM of two natural numbers is 24 times their HCF. If the product of the two numbers is 21600, then find the maximum possible difference between the two numbers.

(A) 630
(B) 720
(C) 570
(D) 690
(E) 600

Q.027 Percentage

There are four consecutive even numbers. If the sum of the first three numbers is 150, find the product of the smallest number and the second largest number.

(A) 2304
(B) 2600
(C) 2392
(D) 2704
(E) 2496

Q.028 Percentage

The monthly income of Ankit is Rs. 70000. He spends x% of his monthly income on rent, 25% of the remaining income on his children's education and 40% of the next remaining income on travelling. If Ankit saves Rs. 27000, then find by what percentage the amount spent on travelling exceeds the amount spent on rent.

(A) 60%
(B) 100%
(C) 120%
(D) 80%
(E) 140%

Q.029 Percentage

Rashmi spends 40% of her monthly income on rent, then 25% of the remaining amount on groceries, and then 20% of the remaining amount on travelling. She saves the remaining amount, which is Rs 14,400. If Vinay's monthly income is 20% less than Rashmi's monthly income, find 50% of Vinay's monthly income.

(A) Rs 14,000
(B) Rs 18,000
(C) Rs 15,000
(D) Rs 16,000
(E) Rs 20,000

Q.030 Percentage

Priya spends 25% of her monthly income on groceries, 20% of the remaining amount on rent, and then 10% of the amount left after rent on travel. She saves the rest. If her monthly savings are Rs. 43200, then find the amount she spends on groceries.

(A) Rs. 20000
(B) Rs. 16000
(C) Rs. 24000
(D) Rs. 18000
(E) Rs. 22000

Q.031 Percentage

Aman spends 30% of his monthly income on rent and 20% of the remaining amount on groceries. He then spends 25% of the remaining amount on travel and saves Rs. 12600. Find his monthly income.

(A) Rs. 28000
(B) Rs. 36000
(C) Rs. 32000
(D) Rs. 25000
(E) Rs. 30000

Q.032 Percentage

A hypothetical dataset gives the populations of three cities: Udaipur, Indore and Surat. The population of Indore is 20% more than the population of Udaipur, while the population of Surat is 75% of the population of Indore. If the sum of the populations of Udaipur and Surat is 95,000, then find the population of Indore.

(A) 58,000
(B) 72,000
(C) 55,000
(D) 60,000
(E) 64,000

Q.033 Percentage

1728 notebooks were distributed equally among the students of a class. Each student received three times as many notebooks as the number of students in the class. Find the number of students in the class.

(A) 24
(B) 18
(C) 28
(D) 30
(E) 32

Q.034 Percentage

On a farm, there are only goats and ducks. The total number of heads is 140 and the total number of legs is 380. Find the number of goats on the farm.

(A) 40
(B) 45
(C) 50
(D) 60
(E) 55

Q.035 Percentage

Sameer spends 20% of his monthly income on rent and 25% of the remaining amount on groceries. He then spends 10% of the remaining amount on travel and saves Rs. 16200. Find his monthly income.

(A) Rs. 24000
(B) Rs. 27000
(C) Rs. 32000
(D) Rs. 36000
(E) Rs. 30000

Q.036 Percentage

250% of the first integer is 5 more than the third integer, and the second integer is 4 less than the third integer. If the sum of the three integers is 34, then find the second- largest integer among them.

(A) 15
(B) 8
(C) 11
(D) 9
(E) 13

Q.037 Percentage

The sum and difference of the LCM and HCF of two positive integers are 348 and 324, respectively. If one of the numbers is 48, then find the other number.

(A) 66
(B) 78
(C) 96
(D) 84
(E) 108

Q.038 Percentage

In an election between two candidates, 20% of the voters did not cast their votes. The winning candidate got 65% of the votes polled and won by 4800 votes. Find the total number of voters. All votes polled were valid.

(A) 18000
(B) 24000
(C) 20000
(D) 16000
(E) 22000

Q.039 Percentage

Aditi got 432 marks in an annual exam. Meena got 52% marks in the same exam, which is 36 marks more than Aditi's marks. The minimum passing score is 36% of the maximum marks. Find how many marks Aditi got more than the passing marks.

(A) 96
(B) 120
(C) 84
(D) 108
(E) 112

Q.040 Percentage

The population of city P decreases by 25% every year, and the population of city Q decreases by 10% every year. Initially, city Q had 8000 more people than city P. After two years, the population of city Q was 12420 more than that of city P. Find the initial population of city Q.

(A) 28000
(B) 30000
(C) 32000
(D) 36000
(E) 40000

Q.041 Percentage

There are five consecutive even numbers. The sum of the smallest and the largest numbers is 56. The smallest number of three consecutive odd numbers is 17 more than the smallest even number. Find the largest odd number.

(A) 43
(B) 47
(C) 45
(D) 49
(E) 41

Q.042 Percentage

When the digits of a two-digit number are reversed, the new number is 2 more than twice the original number. If the reversed number is 208% of the original number, find the sum of the digits of the original number.

(A) 9
(B) 8
(C) 11
(D) 7
(E) 10

Q.043 Percentage

Vinay's monthly income is 20% more than Rahul's monthly income. Vinay spends 30% of his income on rent, 20% of the remaining amount on food, and saves the rest. If Rahul's monthly income is Rs. 6,560 more than Vinay's monthly savings, then find Vinay's monthly income.

(A) Rs. 20,000 (B) Rs. 28,000
(C) Rs. 24,000 (D) Rs. 30,000
(E) Rs. 22,000

Q.044 Percentage

If m and 5 are the two roots of the equation $p^2 - ap + b = 0$, and m and 2 are the two roots of the equation $p^2 - (a - 3)p + (b - 12) = 0$, then find the value of (a + b).

(A) 29 (B) 24
(C) 32 (D) 36
(E) 27

Q.045 Percentage

Sahil's monthly salary is Rs. 60,000. He spends 25% of his salary on food and 20% of the remaining salary on house rent. He also spends Rs. X on insurance and 15% of his salary on travel. These are his only expenses. If his monthly savings are Rs. 18,000, then find the value of X.

(A) Rs. 6,000 (B) Rs. 10,000
(C) Rs. 9,000 (D) Rs. 8,000
(E) Rs. 12,000

📖 Chapter 01 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.001 Correct Option: (E)

- Let Aman's monthly income be Rs. x.
- Then Vikram's monthly income = Rs. (x + 30000).
- Vikram spends 35% of his income, so he saves 65% of his income.
- Vikram's savings = 65% of (x + 30000)
- Aman spends 45% of his income, so he saves 55% of his income.
- Aman's savings = 55% of x
- Given that Vikram saves Rs. 24000 more than Aman.
- So,
- 65% of (x + 30000) - 55% of x = 24000
- 65% of x + 19500 - 55% of x = 24000
- 10% of x + 19500 = 24000
- 10% of x = 4500
- x = 4500 × 100 / 10
- x = 45000

Solution Q.002 Correct Option: (A)

- Aniket spends 20%, 15% and 15% of his salary on rent, food and travel respectively.
- Total expenditure = 20% + 15% + 15% = 50%
- So, remaining money = 100% - 50% = 50% of salary
- When his friend gives him Rs. 4000, his remaining money increases by 25%.
- This means Rs. 4000 is equal to 25% of his remaining money.
- So,
- 25% of remaining money = 4000
- Remaining money = 4000 × 100 / 25
- Remaining money = Rs. 16000
- Now, this remaining money is 50% of his salary.
- So,
- 50% of salary = 16000
- Salary = 16000 × 100 / 50
- Salary = Rs. 32000

Solution Q.003 Correct Option: (E)

- Average age of Divya, Leena and Meena = 36 years.
- So, total age of Divya, Leena and Meena = 36 × 3 = 108 years.
- Average age of Divya and Leena = 30 years.
- So, total age of Divya and Leena = 30 × 2 = 60 years.
- Therefore, Meena's age = 108 - 60 = 48 years.
- Reema's age is 50% of Meena's age.
- So, Reema's age = 50% of 48 = 24 years.
- Reema's age is also 75% of Leena's age.
- So, 75% of Leena's age = 24.
- Leena's age = 24 × 100 / 75 = 32 years.
- Now, Divya + Leena = 60.
- Divya's age = 60 - 32 = 28 years.

Solution Q.004 Correct Option: (A)

- Let the total number of voters = x
- 10% voters did not cast their votes.
- So, votes cast = 90% of x.
- Out of the votes cast, 25% were invalid.
- Invalid votes = 25% of 90% of x = 22.5% of x.
- Valid votes = 90% of x - 22.5% of x = 67.5% of x.
- The winning candidate received votes equal to 45% of total voters.
- So, winner's votes = 45% of x.
- Since there are only two candidates, loser's votes = valid votes - winner's votes.
- Loser's votes = 67.5% of x - 45% of x = 22.5% of x.
- Winning margin = 45% of x - 22.5% of x.
- Winning margin = 22.5% of x.
- Given winning margin = 4500.
- So, 22.5% of x = 4500.
- x = 4500 × 100 / 22.5.
- x = 20000.

Solution Q.005 **Correct Option: (C)**

• Let each student receive x gifts if all 900 students are present.
• Total gifts = 900x.
• Since 150 students were absent, present students = 900 - 150 = 750.
• Each present student received 4 extra gifts, so each received x + 4 gifts.
• Now total gifts remain the same.
• So, 900x = 750(x + 4).
• 900x = 750x + 3000.
• 150x = 3000.
• x = 20.
• Hence, each student would have received 20 gifts if all 900 students had been present.

Solution Q.006 **Correct Option: (C)**

• 60% of 50% of X = 18
• Now, 60% of 50% means:
• (60/100) × (50/100) = 30/100 = 30%
• So, 30% of X = 18
• X = 18 × 100/30
• = 60
• Now, given:
• 40% of 75% of Y = 24
• 40% of 75% = (40/100) × (75/100)
• = 30%
• So, 30% of Y = 24
• Y = 24 × 100/30
• = 80
• Now, X + Y = 60 + 80 = 140
• So, $(X + Y)^2$ = 140 × 140 = 19600
• We need to find 25% of 20% of $(X + Y)^2$.
• 25% of 20% means:
• (25/100) × (20/100) = 5/100 = 5%
• Required value = 5% of 19600
• Required value = 19600 × 5/100
• = 980

Solution Q.007 **Correct Option: (E)**

• Let amount received by Garima = y units
• Megha receives 20% more than Garima.
• Megha's amount = 120y/100 = 6y/5
• Pihu receives 25% less than Megha.
• Pihu's amount = 75/100 × 6y/5
• Pihu's amount = 9y/10
• Avani receives Rs. 600 less than Garima.
• Avani's amount = y - 600
• Difference between Megha's amount and Avani's amount = Rs. 1200
• So:
• 6y/5 - (y - 600) = 1200
• 6y/5 - y + 600 = 1200
• 6y/5 - y = 600
• y/5 = 600
• y = 600 × 5
• y = 3000
• Garima's amount = Rs. 3000
• Megha's amount = 6 × 3000/5 = Rs. 3600
• Pihu's amount = 9 × 3000/10 = Rs. 2700
• Avani's amount = 3000 - 600 = Rs. 2400
• Total amount = 3600 + 2700 + 3000 + 2400
• Total amount = Rs. 11700

Solution Q.008 **Correct Option: (D)**

• Let the number of boys who failed and the number of girls who failed = x each.
• Total boys = 360
• Total girls = 180
• Number of boys who passed = 360 - x
• Number of girls who passed = 180 - x
• The number of boys who passed is 150% more than the number of girls who passed.
• So, number of boys who passed = 250% of the number of girls who passed = 5/2 ×
• number of girls who passed
• Therefore, 360 - x = 5/2 × (180 - x)
• Multiply both sides by 2.
• 2 × (360 - x) = 5 × (180 - x)
• 720 - 2x = 900 - 5x
• 5x - 2x = 900 - 720
• 3x = 180
• x = 180/3 = 60
• Number of boys who passed = 360 - 60 = 300
• Number of girls who passed = 180 - 60 = 120
• Total number of students who passed = 300 + 120 = 420

Solution Q.009 **Correct Option: (B)**

• Rs. 60000
• Solution:
• Let Nandini's monthly salary be Rs. 100.
• She spends 30% on groceries.
• Remaining salary = 100 - 30 = Rs. 70
• From the remaining Rs. 70, she spends 20% on rent and 30% on transportation.
• Total spending from remaining salary = 20% + 30% = 50%
• So, remaining amount after these expenses = 50% of Rs. 70
• Remaining amount = 70 × 50/100 = Rs. 35
• This means Rs. 35 is saved out of Rs. 100 salary.
• So, saved amount = 35% of salary
• Given saved amount = Rs. 21000
• Therefore, 35% of salary = 21000
• Salary = 21000 × 100/35
• Salary = Rs. 60000

Solution Q.010 **Correct Option: (C)**

• days
• Solution:
• Let Sameer alone complete the project in x days.
• Mayank takes 24 days more than Sameer.
• So, Mayank alone can complete the project in x + 24 days.
• Sameer's 1-day work = 1/x
• Mayank's 1-day work = 1/(x + 24)
• Together they complete the project in 16 days.
• So, their combined 1-day work = 1/16
• Now:
• 1/x + 1/(x + 24) = 1/16
• (x + 24 + x)/(x(x + 24)) = 1/16
• (2x + 24)/(x^2 + 24x) = 1/16
• 16(2x + 24) = x^2 + 24x
• 32x + 384 = x^2 + 24x
• x^2 + 24x - 32x - 384 = 0
• x^2 - 8x - 384 = 0
• x^2 - 24x + 16x - 384 = 0
• x(x - 24) + 16(x - 24) = 0
• (x - 24)(x + 16) = 0
• x = 24. The other root is -16 and is rejected because time must be positive.

Solution Q.011 **Correct Option: (B)**

• Rs. 12320
• Solution:
• Let Pallavi's amount be Rs. y.
• Avani receives 30% more than Pallavi.
• Avani's amount = 130y/100 = 13y/10
• Meenal receives 20% less than Avani.
• Meenal's amount = 80/100 × 13y/10
• Meenal's amount = 26y/25
• Nandini receives Rs. 700 less than Pallavi.
• Nandini's amount = y - 700
• Difference between Avani's amount and Nandini's amount = Rs. 1600
• So:
• 13y/10 - (y - 700) = 1600
• 13y/10 - y + 700 = 1600
• 13y/10 - 10y/10 = 900
• 3y/10 = 900
• y = 900 × 10/3
• y = 3000
• Pallavi's amount = Rs. 3000
• Avani's amount = 13 × 3000/10 = Rs. 3900
• Meenal's amount = 26 × 3000/25 = Rs. 3120
• Nandini's amount = 3000 - 700 = Rs. 2300
• Total amount = 3900 + 3120 + 3000 + 2300
• Total amount = Rs. 12320
• Therefore, X = Rs. 12320.

Solution Q.012 **Correct Option: (D)**

• Let the total marks of the examination be T.
• First candidate scores 32% marks and fails by 18 marks.
• So, passing marks = 32% of T + 18
• Second candidate scores 47% marks and gets 72 marks more than passing marks.
• So, passing marks = 47% of T - 72
• Both expressions represent the same passing marks.
• Therefore:
• 32% of T + 18 = 47% of T - 72
• 72 + 18 = 47% of T - 32% of T
• 90 = 15% of T
• 15/100 × T = 90
• T = 90 × 100/15
• T = 600

Solution Q.013 **Correct Option: (D)**

• Let the total number of votes = x
• Invalid votes = 20% of x
• Valid votes = 80% of x
• Winner gets 70% of valid votes.
• Loser gets 30% of valid votes.
• Difference between their votes = 70% - 30% = 40% of valid votes
• Given winning margin = 160 votes
• So, 40% of valid votes = 160
• 40/100 × 80x/100 = 160
• 32x/100 = 160
• 32x = 16000
• x = 16000/32
• x = 500
• Total votes = 500
• Invalid votes = 20% of 500
• Invalid votes = 20 × 500/100
• Invalid votes = 100

Solution Q.014 **Correct Option: (E)**

• Rs. 50
• Solution:
• The price increases by 25% every year.
• So, price becomes 125% of the previous year.
• Let the price in 2021 be Rs. X.
• Price in 2022 = 125X/100 = 5X/4
• Price in 2023 = 5X/4 × 5/4 = 25X/16
• Given, price in 2023 = Rs. 250
• So, 25X/16 = 250
• X = 250 × 16/25
• X = 160
• Price in 2021 = Rs. 160
• Price in 2022 = 160 × 5/4 = Rs. 200
• Price in 2023 = Rs. 250
• Required difference = 250 - 200 = Rs. 50

Solution Q.015 **Correct Option: (C)**

• In the colony, males = 35%.
• So, females = 100% - 35% = 65%.
• 30% of females are graduates.
• So, non-graduate females = 100% - 30% = 70% of females.
• Given, non-graduate females = 910.
• So, 70% of females = 910
• Females = 910 × 100/70
• Females = 1300
• Now, 65% of total residents = 1300.
• Total residents = 1300 × 100/65
• Total residents = 2000
• Males = 35% of 2000
• Males = 35/100 × 2000
• Males = 700

Solution Q.016 **Correct Option: (E)**

• Initial population = 40,000.
• In the first year, population increased by 20%.
• Population after first year = 40000 × 120/100
• Population after first year = 48000
• In the second year, population decreased by 15%.
• Population after second year = 48000 × 85/100
• Population after second year = 40800

Solution Q.017 **Correct Option: (C)**

• Let the number of students in School O = x
• Students in School N = 25% more than School O.
• Students in School N = 125% of x = 125x/100 = 5x/4.
• Students in School M = 30% less than School O.
• Students in School M = 70% of x = 70x/100 = 7x/10.
• Given:
• School M + School O = 204
• 7x/10 + x = 204
• 17x/10 = 204
• x = 204 × 10/17
• x = 120
• Students in School O = 120.
• Students in School N = 5x/4 = 5 × 120/4 = 150.
• Students in Schools N and O together = 150 + 120 = 270.

Solution Q.018 **Correct Option: (A)**

• Let Sandeep's votes be x.
• Lakshya gets 25% fewer votes than Sandeep.
• So, Lakshya's votes = 75% of x.
• Lakshya's votes = 75x/100 = 3x/4.
• Winning margin = Sandeep's votes - Lakshya's votes.
• Winning margin = x - 3x/4.
• Winning margin = x/4.
• Given winning margin = 900.
• So,
• x/4 = 900
• x = 900 × 4
• x = 3600.
• Sandeep's votes = 3600.
• Lakshya's votes = 3x/4 = 3 × 3600/4 = 2700.
• Total votes polled = 3600 + 2700 = 6300.

Solution Q.019 **Correct Option: (B)**

• Rs. 20000
• Solution:
• Let Ananya's monthly salary be Rs. 100.
• Amount spent on rent = 30% of 100 = Rs. 30
• Remaining salary = 100 - 30 = Rs. 70
• Amount spent on groceries = 20% of 70
• Amount spent on groceries = 20 × 70/100 = Rs. 14
• Saving = 70 - 14 = Rs. 56
• So, when salary is Rs. 100, saving is Rs. 56.
• Given saving = Rs. 11200
• Monthly salary = 11200 × 100/56
• Monthly salary = Rs. 20000

Solution Q.020 **Correct Option: (D)**

• Maximum marks in each subject = 150
• Total maximum marks for three subjects = 150 × 3 = 450
• Overall percentage = 76%
• Total marks obtained = 450 × 76/100 = 342
• Marks in Science = 108
• Marks in Hindi = 72% of 150
• Marks in Hindi = 150 × 72/100 = 108
• Let marks in English = X
• So, 108 + 108 + X = 342
• 216 + X = 342
• X = 342 - 216 = 126

Solution Q.021 **Correct Option: (D)**

• Let the smallest odd number = x
• Three consecutive odd numbers = x, x + 2, x + 4.
• Smallest even number is 7 less than the smallest odd number.
• So, smallest even number = x - 7.
• Three consecutive even numbers = x - 7, x - 5, x - 3.
• Sum of all six numbers = 177.
• So, x + x + 2 + x + 4 + x - 7 + x - 5 + x - 3 = 177.
• 6x - 9 = 177.
• 6x = 177 + 9
• 6x = 186
• x = 186/6
• x = 31.
• Largest odd number = x + 4 = 31 + 4 = 35.
• Largest even number = x - 3 = 31 - 3 = 28.
• Required sum = 35 + 28 = 63.

Solution Q.022 **Correct Option: (B)**

• %
• Solution:
• Total boys = 32.
• Total girls = 25.
• 60% of girls have weight less than or equal to 45 kg.
• So, girls having weight more than 45 kg = 40% of 25.
• Girls having weight more than 45 kg = 40/100 × 25 = 10.
• 75% of boys have weight greater than or equal to 45 kg.
• So, boys having weight less than 45 kg = 25% of 32.
• Boys having weight less than 45 kg = 25/100 × 32 = 8.
• Required percentage = (Number of boys weighing less than 45 kg / Number of girls
• weighing more than 45 kg) × 100.
• Required percentage = 8/10 × 100 = 80%.

Solution Q.023 **Correct Option: (C)**

• Let the smaller number be x and the larger number = y
• Both numbers are positive.
• Given, $x^2 = 4y$ and $y^2 - x^2 = 192$.
• Substitute $x^2 = 4y$:
• $y^2 - 4y = 192$
• $y^2 - 4y - 192 = 0$
• $(y - 16)(y + 12) = 0$
• Since y is positive, y = 16.
• Then $x^2 = 4 \times 16 = 64$.
• Since x is positive, x = 8.
• Required sum = 8 + 16 = 24.

Solution Q.024 **Correct Option: (C)**

• Rs. 1728
• Solution:
• Cost of levelling = Rs. 6480 and rate = Rs. 12 per m^2.
• Area of garden = 6480 / 12 = 540 m^2.
• So, (x + 7)(x - 5) = 540.
• By checking factor pair with difference 12, 30 × 18 = 540.
• So, x + 7 = 30 and x - 5 = 18.
• Length = 30 m and breadth = 18 m.
• Perimeter = 2 × (30 + 18) = 96 m.
• Cost of fencing = 96 × 18 = Rs. 1728.

Solution Q.025 **Correct Option: (B)**

• Represent the combined marks of both candidates as 100%.
• Higher-scoring candidate = 56% of total.
• Lower-scoring candidate = 44% of total.
• Difference = 56% - 44% = 12% of total.
• Given difference = 24 marks.
• So, 12% of the combined marks = 24 marks.
• Combined marks = 24 × 100 / 12 = 200 marks.
• Marks of higher-scoring candidate = 56% of 200 = 112.

Solution Q.026 **Correct Option: (D)**

• Let the HCF of the two numbers = h.
• Let the two numbers be ha and hb, where a and b are co-prime.
• LCM = hab
• Given, LCM = 24 × HCF
• hab = 24h
• ab = 24
• Product of two numbers = HCF × LCM
• 21600 = h × 24h
• $21600 = 24h^2$
• $h^2 = 21600/24 = 900$
• h = 30
• Now, ab = 24.
• Co-prime factor pairs of 24 are 1 and 24, and 3 and 8.
• For maximum difference, take the pair 1 and 24.
• Numbers = 30 × 1 and 30 × 24
• Numbers = 30 and 720
• Maximum difference = 720 - 30 = 690

Solution Q.027 **Correct Option: (E)**

• Let the four consecutive even numbers = x, x + 2, x + 4 and x + 6.
• Sum of first three numbers = 150
• x + x + 2 + x + 4 = 150
• 3x + 6 = 150
• 3x = 144
• x = 48
• So, the numbers are 48, 50, 52 and 54.
• Smallest number = 48
• Second largest number = 52
• Required product = 48 × 52 = 2496

Solution Q.028 **Correct Option: (D)**

• %
• Solution:
• The monthly income of Ankit is Rs. 70000.
• He spends some amount on rent, and after that some amount remains.
• Let the remaining amount after rent be R.
• He spends 25% of R on his children's education.
• So, remaining amount after education = 75% of R.
• Remaining amount after education = 75R/100
• Remaining amount after education = 3R/4
• Now, he spends 40% of this remaining amount on travelling.
• Travelling amount = 40% of 3R/4
• Travelling amount = 40/100 × 3R/4
• Travelling amount = 3R/10
• After spending 40% on travelling, he saves the remaining 60%.
• Saving amount = 60% of 3R/4
• Saving amount = 60/100 × 3R/4
• Saving amount = 9R/20
• According to the question, saving amount = Rs. 27000.
• So,
• 9R/20 = 27000
• R = 27000 × 20/9
• R = 60000
• So, the amount remaining after rent is Rs. 60000.
• Total income = Rs. 70000.
• Rent amount = 70000 - 60000 = Rs. 10000.
• Travelling amount = 3R/10
• Travelling amount = 3 × 60000/10
• Travelling amount = Rs. 18000.
• Now, we have to find how much percent more travelling amount is than rent amount.
• Required percentage = (Travelling amount - Rent amount)/Rent amount × 100
• Required percentage = (18000 - 10000)/10000 × 100
• Required percentage = 8000/10000 × 100
• Required percentage = 80%

Solution Q.029 **Correct Option: (D)**

• Rs 16,000
• Solution:
• Let Rashmi's monthly income be Rs x.
• She spends 40% of her income on rent.
• So, the remaining income after rent = 100% - 40% = 60%.
• Now, she spends 25% of the remaining amount on groceries.
• So, the amount left after groceries = 100% - 25% = 75% of the previous remaining
• amount.
• After that, she spends 20% of the remaining amount on travelling.
• So, the amount left after travelling = 100% - 20% = 80% of the previous remaining
• amount.
• This final remaining amount is her saving.
• Using the banking exam shortcut:
• Saving = x × 60/100 × 75/100 × 80/100
• Now simplify the percentage fractions:
• 60/100 = 3/5
• 75/100 = 3/4
• 80/100 = 4/5
• So, saving = x × 3/5 × 3/4 × 4/5
• Saving = x × 9/25
• Given saving = Rs 14,400.
• So, x × 9/25 = 14,400
• x = 14,400 × 25/9
• x = 1,600 × 25
• x = 40,000
• So, Rashmi's monthly income = Rs 40,000.
• Vinay's monthly income is 20% less than Rashmi's monthly income.
• So, Vinay's monthly income = 80% of 40,000
• Vinay's monthly income = 40,000 × 80/100 = Rs 32,000
• Now, required value = 50% of Vinay's monthly income.
• 50% of 32,000 = 32,000 × 50/100 = Rs 16,000

Solution Q.030 **Correct Option: (A)**

• Rs. 20000
• Solution:
• Let Priya's monthly income be Rs. x.
• She spends 25% on groceries, so remaining income = 75% of x = 3x/4
• She spends 20% of the remaining amount on rent, so remaining = 80% of 3x/4
• Remaining after rent = 3x/4 × 4/5 = 3x/5
• She spends 10% of the remaining amount on travel, so final saving = 90% of 3x/5
• Saving = 3x/5 × 9/10 = 27x/50
• Given saving = Rs. 43200
• 27x/50 = 43200
• x = 43200 × 50 / 27
• x = Rs. 80000
• Amount spent on groceries = 25% of 80000
• Amount spent on groceries = 80000 × 25/100 = Rs. 20000

Solution Q.031 **Correct Option: (E)**

• Rs. 30000
• Solution:
• Let Aman's monthly income be Rs. 100.
• He spends 30% on rent, so remaining amount = Rs. 70.
• He spends 20% of Rs. 70 on groceries.
• Amount left after groceries = 70 × 80 / 100 = Rs. 56.
• He spends 25% of Rs. 56 on travel.
• Amount saved = 56 × 75 / 100 = Rs. 42.
• So, Rs. 42 corresponds to actual savings of Rs. 12600.
• Monthly income = 12600 × 100 / 42 = Rs. 30000.

Solution Q.032 **Correct Option: (D)**

• Let the population of Udaipur be 100x.
• Population of Indore is 20% more than Udaipur.
• So, population of Indore = 120x.
• Population of Surat = 75% of Indore = 75% of 120x = 90x.
• Given, Udaipur + Surat = 95,000.
• So, 100x + 90x = 95,000.
• 190x = 95,000.
• x = 95,000 / 190 = 500.
• Population of Indore = 120x = 120 × 500 = 60,000.

Solution Q.033 **Correct Option: (A)**

• Let the number of students in the class be n.
• Each student received three times the number of students.
• So, notebooks received by each student = 3n
• Total notebooks distributed = Number of students × notebooks received by each
• student
• 1728 = n × 3n
• 1728 = 3n²
• n² = 1728 / 3 = 576
• n = 24
• So, the number of students in the class is 24.

Solution Q.034 **Correct Option: (C)**

• Let the number of goats be g and the number of ducks be d.
• Total heads = 140.
• So, g + d = 140.
• A goat has 4 legs and a duck has 2 legs.
• Total legs = 380.
• Shortcut method:
• If all 140 animals were ducks, total legs would be 140 × 2 = 280.
• Actual legs = 380.
• Extra legs = 380 - 280 = 100.
• Each goat has 2 extra legs as compared to a duck.
• Number of goats = 100 / 2 = 50.

Solution Q.035 **Correct Option: (E)**

• Rs. 30000
• Solution:
• Let Sameer's monthly income = Rs. 100x
• Amount spent on rent = 20% of 100x = 20x
• Remaining amount = 100x - 20x = 80x
• Amount spent on groceries = 25% of 80x = 20x
• Remaining amount = 80x - 20x = 60x
• Amount spent on travel = 10% of 60x = 6x
• Savings = 60x - 6x = 54x
• Given, savings = Rs. 16200
• 54x = 16200
• x = 300
• Monthly income = 100x = 100 × 300 = Rs. 30000

Solution Q.036 **Correct Option: (C)**

• Let the first integer be x and the third integer = y
• 250% of the first integer is 5 more than the third integer.
• So, 250% of x = y + 5
• 250% means 250/100 = 5/2
• So, 5x/2 = y + 5
• This gives y = 5x/2 - 5
• The second integer is 4 less than the third integer.
• So, second integer = y - 4
• Sum of three integers = 34
• x + y + y - 4 = 34
• x + 2y = 38
• Now put y = 5x/2 - 5
• x + 2 × (5x/2 - 5) = 38
• x + 5x - 10 = 38
• 6x = 48
• x = 8
• Third integer = 5 × 8/2 - 5 = 20 - 5 = 15
• Second integer = 15 - 4 = 11
• The three integers are 8, 11, and 15.
• The second-largest integer = 11

Solution Q.037 **Correct Option: (D)**

• Sum of LCM and HCF = 348.
• Difference of LCM and HCF = 324.
• LCM = (348 + 324) / 2 = 336.
• HCF = (348 - 324) / 2 = 12.
• Product of two numbers = LCM × HCF.
• So, 48 × other number = 336 × 12.
• Other number = 336 × 12 / 48 = 84.

Solution Q.038 **Correct Option: (C)**

• Let the total number of voters = x
• 20% voters did not cast their votes.
• So, votes polled = 80% of x.
• Winning candidate got 65% of the polled votes.
• Losing candidate got 35% of the polled votes.
• Difference = 65% - 35% = 30% of polled votes.
• 30% of 80% of x = 4800.
• 24% of x = 4800.
• x = 4800 × 100 / 24 = 20000.

Solution Q.039 **Correct Option: (D)**

• Aditi's marks = 432.
• Meena's marks are 36 more than Aditi's marks.
• Meena's marks = 432 + 36 = 468.
• Meena got 52% marks in the exam.
• So, 52% of total marks = 468.
• Total marks = 468 × 100 / 52 = 900.
• Passing marks = 36% of 900.
• Passing marks = 900 × 36 / 100 = 324.
• Aditi got more than passing marks = 432 - 324 = 108.
• Therefore, Aditi got 108 marks more than the passing marks.

Solution Q.040 **Correct Option: (C)**

• Let the initial population of city P = x.
• Initial population of city Q = x + 8000.
• City P decreases by 25% every year.
• So, population of P after 2 years = x × 75/100 × 75/100 = 9x/16.
• City Q decreases by 10% every year.
• So, population of Q after 2 years = (x + 8000) × 90/100 × 90/100 = 81/100 × (x +
• 8000).
• Given difference after 2 years = 12420.
• 81/100 × (x + 8000) - 9x/16 = 12420.
• 81x/100 + 6480 - 9x/16 = 12420.
• 81x/100 - 9x/16 = 5940.
• LCM of 100 and 16 = 400.
• 324x/400 - 225x/400 = 5940.
• 99x/400 = 5940.
• x = 5940 × 400 / 99 = 24000.
• Initial population of city Q = 24000 + 8000 = 32000.

Solution Q.041 **Correct Option: (C)**

• Let the five consecutive even numbers be:
• x, x + 2, x + 4, x + 6, x + 8
• Smallest even number = x
• Largest even number = x + 8
• According to the question:
• x + x + 8 = 56
• 2x + 8 = 56
• 2x = 48
• x = 24
• Smallest even number = 24
• Smallest odd number = 24 + 17 = 41
• Three consecutive odd numbers:
• 41, 43, 45
• Largest odd number = 45

Solution Q.042 **Correct Option: (D)**

• Let the original number be N.
• Reversed number = 208% of N
• Reversed number = 208N/100
• Also, reversed number = 2N + 2
• So:
• 208N/100 = 2N + 2
• 208N = 200N + 200
• 208N - 200N = 200
• 8N = 200
• N = 200/8
• N = 25
• Original number = 25
• Reversed number = 52
• Sum of digits = 2 + 5 = 7

Solution Q.043 **Correct Option: (C)**

• Rs. 24,000
• Solution:
• Vinay's monthly income is 20% more than Rahul's monthly income.
• So, Vinay's income : Rahul's income = 6 : 5.
• Let Vinay's income = 6x and Rahul's income = 5x.
• Vinay spends 30% of his income on rent.
• Remaining income = 70% of 6x.
• He spends 20% of the remaining on food, so he saves 80% of the remaining.
• Vinay's savings = 80% of 70% of 6x.
• Vinay's savings = 80/100 × 70/100 × 6x = 84x/25.
• Rahul's income is Rs. 6,560 more than Vinay's savings.
• 5x - 84x/25 = 6,560.
• (125x - 84x)/25 = 6,560.
• 41x/25 = 6,560.
• x = 6,560 × 25/41 = 4,000.
• Vinay's monthly income = 6x = 6 × 4,000 = Rs. 24,000.

Solution Q.044 **Correct Option: (A)**

• For the equation $p^2 - ap + b = 0$, roots are m and 5.
• Sum of roots = a
• m + 5 = a
• Product of roots = b
• 5m = b
• For the equation $p^2 - (a - 3)p + (b - 12) = 0$, roots are m and 2.
• Sum of roots = a - 3
• m + 2 = a - 3
• Product of roots = b - 12
• 2m = b - 12
• From m + 5 = a, we get a - 3 = m + 2.
• So, the sum condition is satisfied.
• Now use product condition:
• 2m = b - 12
• But b = 5m.
• 2m = 5m - 12
• 3m = 12
• m = 4
• a = m + 5 = 4 + 5 = 9
• b = 5m = 5 × 4 = 20
• a + b = 9 + 20 = 29.

Solution Q.045 **Correct Option: (C)**

• Rs. 9,000
• Solution:
• Monthly salary of Sahil = Rs. 60,000
• Amount spent on food = 25% of 60,000 = Rs. 15,000
• Remaining salary after food = 60,000 - 15,000 = Rs. 45,000
• Amount spent on house rent = 20% of 45,000 = Rs. 9,000
• Amount spent on travel = 15% of 60,000 = Rs. 9,000
• Monthly savings = Rs. 18,000
• Total salary = Food + Rent + Travel + Insurance + Savings
• 60,000 = 15,000 + 9,000 + 9,000 + X + 18,000
• 60,000 = 51,000 + X
• X = 9,000
• Therefore, the value of X is Rs. 9,000.

CHAPTER 02: PROFIT, LOSS, MARKED PRICE & DISCOUNT

84 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Profit% = (SP - CP) / CP × 100 | Loss% = (CP - SP) / CP × 100
- Marked Price (MP) = CP + Markup | Discount% = (MP - SP) / MP × 100
- Direct CP to MP Ratio: CP / MP = (100 - Discount%) / (100 + Profit%)
- Successive Discounts d1 and d2: Equivalent = d1 + d2 - (d1 × d2 / 100)

Q.046 Profit, Loss, Marked Price

Priya and Pallavi started a business by investing Rs. 15000 and Rs. 10000 respectively. They decided that 60% of the total profit would be shared equally, while the remaining profit would be divided in the ratio of their investments. If Priya received Rs. 320 more than Pallavi, find the total profit.

(A) 3500 (B) 4500
(C) 5000 (D) 4000
(E) 3200

Q.047 Profit, Loss, Marked Price

A shopkeeper marked a watch 60% above its cost price. Priya purchased it after getting a discount of 25% and then sold it to Shreya at a profit of 20%. If Shreya purchased the watch for Rs. 4320, find the shopkeeper's cost price of the watch.

(A) 2800 (B) 3000
(C) 3600 (D) 3200
(E) 2400

Q.048 Profit, Loss, Marked Price

Raj and Naman started a partnership business by investing Rs. 9000 and Rs. 11000 respectively. After 6 months, Raj withdrew 20% of his investment, while Naman added Rs. 3000 more. If Raj's share of profit at the end of the year was Rs. 12150, find the total profit.

(A) 27600 (B) 32400
(C) 30900 (D) 36000
(E) 29400

Q.049 Profit, Loss, Marked Price

A retailer bought a mixer grinder for Rs. 750. He marked it 40% above its cost price and then allowed a discount of 10% while selling it. Find the selling price of the mixer grinder.

(A) 875 (B) 960
(C) 900 (D) 945
(E) 990

Q.050 Profit, Loss, Marked Price

Anita and Pallavi started a business. Their investment ratio was 5:8, and the ratio of their investment periods was 6:5. At the end of the year, 25% of the total profit was donated to an NGO, and the remaining profit was shared between them. If Pallavi's share is Rs. 8,400, find the total profit.

(A) Rs. 19,600 (B) Rs. 18,900
(C) Rs. 21,000 (D) Rs. 22,400
(E) Rs. 16,800

Q.051 Profit, Loss, Marked Price

A shopkeeper marked the price of a table 50% above its cost price. The ratio of the discount amount to the profit amount is 1 : 2. The selling price is how many times the profit earned?

(A) 3 (B) 5
(C) 6 (D) 4
(E) 7

Q.052 Profit, Loss, Marked Price

The cost price of article X is equal to the selling price of article Y. Article X is sold at a profit of 25%, and article Y is sold at a profit of 40%. If the sum of the cost price of article Y and the selling price of article X is Rs. 1760, find the cost price of article X.

(A) Rs. 840 (B) Rs. 960
(C) Rs. 896 (D) Rs. 1024
(E) Rs. 1120

Q.053 Profit, Loss, Marked Price

Riya and Madhav started a business with investments of Rs. 2400 and Rs. (x − 2400) respectively. After 8 months, Riya left the business. At the end of one year, Madhav's profit share was Rs. 3500 out of the total profit of Rs. 5500. Find x.

(A) 4800 (B) 5200
(C) 5600 (D) 6000
(E) 5000

Q.054 Profit, Loss, Marked Price

Shreya and Naman started a business with investments of Rs. 3000 and Rs. (x − 3000) respectively. After 6 months, Shreya left the business. At the end of one year, Naman's profit share was Rs. 3600 out of the total profit of Rs. 4800. Find x.

(A) 7000 (B) 7800
(C) 7200 (D) 7500
(E) 6800

Q.055 Profit, Loss, Marked Price

Ishan and Jatin started a business by investing in the ratio of 4 : 7 respectively. After 4 months, Lalit joined the business with twice the capital invested by Jatin. At the end of the year, the total profit was Rs. 122000. Find Ishan's profit share.

(A) Rs. 24000 (B) Rs. 30000
(C) Rs. 42000 (D) Rs. 56000
(E) Rs. 36000

Q.056 Profit, Loss, Marked Price

A shopkeeper marked an article 80% above its cost price and allowed a discount of 20%. He earned a profit of Rs. 220. If he had allowed a discount of 35%, find the profit he would have earned.

(A) Rs. 75 (B) Rs. 85
(C) Rs. 90 (D) Rs. 100
(E) Rs. 110

Q.057 Profit, Loss, Marked Price

Parth and Mitali started a business with initial investments of Rs. 5,000 and Rs. 7,200 respectively. After 6 months, Parth invested an additional amount of Rs. A. At the end of one year, Mitali's profit share was 80% of Parth's profit share. Find A.

(A) 9000 (B) 6000
(C) 10000 (D) 8000
(E) 7000

Q.058 Profit, Loss, Marked Price

The ratio of the cost prices of articles P and Q is 5:8 respectively. Article P is sold at a profit of 25%. The difference between the cost price of article Q and the selling price of article P is Rs. 140. If article Q is sold at a loss of 15%, find the selling price of article Q.

(A) 520
(B) 544
(C) 560
(D) 480
(E) 600

Q.059 Profit, Loss, Marked Price

Nilesh, Mukesh and Suresh invested Rs. 4000, Rs. 6000 and Rs. 5000 respectively in a business. After 6 months, Nilesh and Mukesh increased their investments by 25% of their initial investments, while Suresh decreased his investment by 20% of his initial investment. If Mukesh's profit share at the end of the first year is Rs. 9000, find the total profit.

(A) Rs. 18000
(B) Rs. 24000
(C) Rs. 19500
(D) Rs. 21000
(E) Rs. 27000

Q.060 Profit, Loss, Marked Price

There are two products X and Y. The marked price of X is 50% more than the marked price of Y. The cost price of X is equal to the marked price of Y. Product X is sold at 25% profit, and product Y is sold at 20% loss. If the discounts given on X and Y are Rs. 800 and Rs. 1280 respectively, find the cost price of Y.

(A) Rs. 2200
(B) Rs. 2500
(C) Rs. 2600
(D) Rs. 2000
(E) Rs. 2400

Q.061 Profit, Loss, Marked Price

The cost price of a tablet is Rs. 1250 and the cost price of a pair of headphones is Rs. 960. A shopkeeper sold the tablet at a profit of 24% and the headphones at a loss of 15%. Find the overall profit or loss made by the shopkeeper.

(A) Rs. 120 profit
(B) Rs. 144 loss
(C) Rs. 156 profit
(D) Rs. 180 profit
(E) Rs. 156 loss

Q.062 Profit, Loss, Marked Price

Two partners entered into a business by investing Rs. 5000 and Rs. 7000 respectively. After 6 months, the first partner increased his capital by 40%, while the second partner increased his capital by 20%. If the total profit at the end of one year is Rs. 6850, find the profit share of the second partner.

(A) Rs. 3200
(B) Rs. 3600
(C) Rs. 4200
(D) Rs. 3850
(E) Rs. 4500

Q.063 Profit, Loss, Marked Price

The marked price of a blazer is Rs. 1200 more than its cost price. After giving a discount of Rs. 750, the shopkeeper earns a profit of 10%. Find the marked price of the blazer.

(A) Rs. 5600
(B) Rs. 5400
(C) Rs. 5700
(D) Rs. 6000
(E) Rs. 5900

Q.064 Profit, Loss, Marked Price

A customer bought an article for Rs. 408 after getting two successive discounts on its marked price of Rs. 640. If the first discount was 25%, find the second discount percentage.

(A) 10%
(B) 12%
(C) 20%
(D) 25%
(E) 15%

Q.065 Profit, Loss, Marked Price

Two partners invested in a business in the ratio of 5:4. The first partner invested for 16 months and the second partner invested for 15 months. If 1/5 of the total profit goes to charity and the first partner's share from the remaining profit is Rs. 4800, find the second partner's profit share.

(A) 4200
(B) 3000
(C) 4800
(D) 3600
(E) 2400

Q.066 Profit, Loss, Marked Price

Sonal started a business by investing Rs. 90000. After 2 months, Swati joined her with Rs. 150000. After another 6 months, Sonal withdrew 1/3 of her investment, and Swati added Rs. 30000 more. At the end of the year, the total profit was Rs. 86000. Find Sonal's profit share.

(A) Rs. 28000
(B) Rs. 32000
(C) Rs. 36000
(D) Rs. 40000
(E) Rs. 30000

Q.067 Profit, Loss, Marked Price

A trader purchased a wristwatch for Rs. P and sold it for Rs. Q at a profit of 30%. If he had purchased it at 25% less and sold it for Rs. 96 less, he would have earned a profit of 20%. Find the value of Q − P.

(A) 60
(B) 90
(C) 84
(D) 72
(E) 96

Q.068 Profit, Loss, Marked Price

Nirav started a business with Rs. 60000. After 3 months, Pallav joined him with Rs. 72000. After 5 more months, Uday joined them with Rs. 90000. If the total profit at the end of one year is Rs. 144000, find Uday's profit share.

(A) Rs. 30000
(B) Rs. 27000
(C) Rs. 36000
(D) Rs. 24000
(E) Rs. 32000

Q.069 Profit, Loss, Marked Price

A trader marks his goods 50% above the cost price and allows a discount of 20%. He cheats by giving only 800 grams instead of 1 kg. Find his gain percent.

(A) 40% (B) 45%
(C) 50% (D) 55%
(E) 60%

Q.070 Profit, Loss, Marked Price

Partner U and Partner V started a business by investing Rs. 3x and Rs. 5x, respectively. After 4 months, Partner W joined the business with an investment of Rs. 6x. If the total profit at the end of the year is Rs. 66000, find the share of Partner V in the profit.

(A) Rs. 24000 (B) Rs. 27500
(C) Rs. 30000 (D) Rs. 32500
(E) Rs. 25000

Q.071 Profit, Loss, Marked Price

Keshav bought a second-hand camera and spent Rs. 2,000 on its repair. He sold it for Rs. 24,000 and earned a profit of 20%. If he had not repaired it and sold it at a profit of 15%, then find the new selling price of the camera.

(A) Rs. 20,400 (B) Rs. 19,800
(C) Rs. 21,600 (D) Rs. 20,250
(E) Rs. 20,700

Q.072 Profit, Loss, Marked Price

A retailer marks a bag 75% above its cost price and allows a discount of 20%. If he still earns a profit of Rs. 800, find the selling price of the bag.

(A) Rs. 2400 (B) Rs. 3200
(C) Rs. 3000 (D) Rs. 2800
(E) Rs. 2600

Q.073 Profit, Loss, Marked Price

Two partners M and N started a business by investing in the ratio of 5 : 8 respectively. M received Rs. 6000 as profit out of the total profit of Rs. 14000. If M invested for 12 months, find the number of months for which N invested.

(A) 8 months (B) 9 months
(C) 10 months (D) 12 months
(E) 15 months

Q.074 Profit, Loss, Marked Price

A dealer marked an item 75% above its cost price and allowed a discount of 30% on it. Had he allowed only 18% discount, he would have earned Rs. 210 more. Find the cost price of the item.

(A) Rs. 900 (B) Rs. 1000
(C) Rs. 1200 (D) Rs. 800
(E) Rs. 1100

Q.075 Profit, Loss, Marked Price

Sameer and Nilesh started a business by investing their capitals in the ratio of 4:7, respectively. After six months, Sameer increased his initial investment by 50%, while Nilesh withdrew 3/7 of his initial investment. If the difference between their profit shares at the end of the year is Rs. 1200, then find Nilesh's profit share.

(A) Rs. 10800 (B) Rs. 12000
(C) Rs. 14400 (D) Rs. 13200
(E) Rs. 15600

Q.076 Profit, Loss, Marked Price

Priyanka and Avani started a business with investments of Rs. P and Rs. (P + 4000), respectively. Priyanka invested for T years, while Avani invested for 2T years. If Priyanka's profit share is Rs. 5,000 out of the total profit of Rs. 17,000, find Avani's investment.

(A) Rs. 20000 (B) Rs. 22000
(C) Rs. 24000 (D) Rs. 26000
(E) Rs. 28000

Q.077 Profit, Loss, Marked Price

Tushar and Prateek started a business with investments of Rs. 30000 and Rs. 42000, respectively. After 4 months, Tushar withdrew 20% of his initial investment, Prateek withdrew his full investment, and Sagar joined the business. If at the end of the year the ratio of profit shares of Tushar, Prateek, and Sagar is 13:7:10, then find the investment made by Sagar.

(A) Rs. 36000 (B) Rs. 24000
(C) Rs. 30000 (D) Rs. 42000
(E) Rs. 48000

Q.078 Profit, Loss, Marked Price

Kabir sells an article to Eshaan at a loss of 25%. Eshaan then resells it to Sameer at Kabir's original cost price. The difference between the cost price of the article for Eshaan and Sameer is Rs. 200. If Sameer sells it to Mukesh at a profit of 30%, find the profit Kabir would make if he directly sold the article to Mukesh at the same selling price at which Sameer sold it.

(A) Rs. 200 (B) Rs. 220
(C) Rs. 260 (D) Rs. 280
(E) Rs. 240

Q.079 Profit, Loss, Marked Price

Sahil and Tapan started a business with investments of Rs. 9000 and Rs. 7000 respectively. After 3 months, Sahil withdrew one-third of his initial investment, while Tapan added the same amount that Sahil withdrew. At the same time, Uday joined the business with Rs. 12000. If the total profit after one year is Rs. 18000, find the difference between the profit shares of Tapan and Uday.

(A) Rs. 180 (B) Rs. 240
(C) Rs. 300 (D) Rs. 360
(E) Rs. 120

Q.080 Profit, Loss, Marked Price

Two partners started a business. The first partner invested Rs. Y, and the second partner invested Rs. 1500 more than the first partner for the same period. If the first partner received Rs. 4000 as profit share out of the total profit of Rs. 9000, find Y.

(A) 5000 (B) 7200
(C) 6000 (D) 6500
(E) 5500

Q.081 Profit, Loss, Marked Price

For two articles X and Y, the marked price of X is 50% higher than the marked price of Y. The cost price of X is equal to the marked price of Y. Article X is sold at 25% profit and article Y is sold at 10% loss. If the discounts given on X and Y are Rs. 800 and Rs. 500 respectively, find the cost price of Y.

(A) Rs. 2500 (B) Rs. 2700
(C) Rs. 3000 (D) Rs. 3200
(E) Rs. 3500

Q.082 Profit, Loss, Marked Price

Ishan, Keshav and Ayush invested Rs. 4000, Rs. 6000 and Rs. 5000 respectively in a business. After 4 months, Ishan and Keshav increased their investments by 25% of their initial investments each, while Ayush reduced his investment by 20% of his initial investment. If Keshav's profit share at the end of the first year is Rs. 12600, find the total profit.

(A) Rs. 24000 (B) Rs. 28800
(C) Rs. 31200 (D) Rs. 36000
(E) Rs. 32000

Q.083 Profit, Loss, Marked Price

A merchant purchased two bags for a total of Rs. 1750. He sold the first bag at a loss of 16% and the second bag at a profit of 16%. If the selling prices of both bags were equal, find the difference between their cost prices.

(A) Rs. 240 (B) Rs. 320
(C) Rs. 260 (D) Rs. 280
(E) Rs. 360

Q.084 Profit, Loss, Marked Price

A shopkeeper marked a jacket 50% above its cost price and sold it for Rs. 840 after allowing two successive discounts of 30% and y%. If he incurred a loss of 16%, find the second discount percentage.

(A) 12% (B) 15%
(C) 20% (D) 24%
(E) 25%

Q.085 Profit, Loss, Marked Price

Priyansh and Nilesh started a business by investing Rs. 6000 and Rs. 4500, respectively. After 5 months, Priyansh withdrew Rs. 2000 from his investment, while Nilesh added Rs. 1500 to his investment. Find Nilesh's share from the total annual profit of Rs. 24500.

(A) Rs. 11600 (B) Rs. 12400
(C) Rs. 14000 (D) Rs. 12900
(E) Rs. 13500

Q.086 Profit, Loss, Marked Price

On selling an article for Rs. 1700, a seller earns a profit which is twice the loss incurred when the same article is sold for Rs. 800. Find the selling price of the article to earn a profit of 40%.

(A) Rs. 1480 (B) Rs. 1500
(C) Rs. 1600 (D) Rs. 1560
(E) Rs. 1540

Q.087 Profit, Loss, Marked Price

On selling 18 diaries for Rs. 1680, a shopkeeper earns a profit equal to the cost price of 6 diaries. Find the cost price of each diary.

(A) Rs. 65 (B) Rs. 70
(C) Rs. 75 (D) Rs. 80
(E) Rs. 60

Q.088 Profit, Loss, Marked Price

Ayush and Siddharth started a business. Ayush invested Rs. 6000, and Siddharth invested 60% of Ayush's investment. Ayush invested for X months, while Siddharth invested for the whole year. At the end of the year, Siddharth received Rs. 12000 out of a total profit of Rs. 27000. Find X.

(A) 6 (B) 7
(C) 8 (D) 9
(E) 10

Q.089 Profit, Loss, Marked Price

The selling price of an article is Rs. 600 after allowing a discount of 25% on its marked price. The ratio of the cost price to the marked price is 5:8. If the cost price increases by 25% and the selling price remains the same, then find the profit or loss percentage.

(A) 5% loss (B) 4% loss
(C) 4% profit (D) 8% loss
(E) 6% profit

Q.090 Profit, Loss, Marked Price

A dealer sold an appliance to a customer at a profit equal to 20% of its selling price. The customer then sold it at L% loss for Rs. 1080. If the dealer's cost price of the appliance was Rs. 960, then find the customer's loss percentage.

(A) 12% (B) 8%
(C) 15% (D) 10%
(E) 5%

Q.091 Profit, Loss, Marked Price

Vikram, Ojas, and Pradeep invested Rs. 3000, Rs. 5000, and Rs. 4000 respectively in a business. The profit shares of Vikram and Ojas are equal, and together they receive 2/3 of the total profit of Rs. 9000. Find the sum of the terms of the simplest ratio of the investment periods of Vikram, Ojas and Pradeep, in that order.

(A) 42 (B) 45
(C) 50 (D) 48
(E) 47

Q.092 Profit, Loss, Marked Price

A shopkeeper sold three articles P, Q, and R. Article P was sold at 20% profit, article Q was sold at 25% loss, and article R was sold at 50% profit. If the ratio of their selling prices is 6:3:9, respectively, then find the overall profit or loss percentage of the shopkeeper.

(A) 10% loss (B) 15% profit
(C) 25% profit (D) 30% loss
(E) 20% profit

Q.093 Profit, Loss, Marked Price

A dealer purchased a mixer grinder for Rs. 18,750. He marked it 60% above its cost price and then sold it after giving a discount of 25%. Find the profit earned by the dealer.

(A) Rs. 4,250 (B) Rs. 3,250
(C) Rs. 4,500 (D) Rs. 3,750
(E) Rs. 3,000

Q.094 Profit, Loss, Marked Price

Aman and Adarsh started a business together. Aman's investment was Rs. 700 less than Adarsh's investment. After 8 months, Adarsh left the business and Hemant joined with an investment Rs. 700 more than Adarsh's investment. Aman stayed in the business for the whole year. If the annual profit shares of Aman, Adarsh and Hemant are in the ratio 9:7:4, respectively, find Aman's investment.

(A) Rs. 3,600 (B) Rs. 4,200
(C) Rs. 4,900 (D) Rs. 5,600
(E) Rs. 3,800

Q.095 Profit, Loss, Marked Price

The marked price of an article is Rs. 1350 more than its cost price. After giving a discount of Rs. 750, the shopkeeper earns a profit of 20%. Find the marked price of the article.

(A) Rs. 4350 (B) Rs. 4050
(C) Rs. 4500 (D) Rs. 4200
(E) Rs. 4650

Q.096 Profit, Loss, Marked Price

Yash started a business with an investment of Rs. (5P + 2000). After 6 months, Siddharth joined him with an investment of Rs. (8P − 4000). At the end of 16 months, Siddharth's profit share is Rs. 7200 out of the total profit of Rs. 18000. Find Yash's investment.

(A) Rs. 10500 (B) Rs. 15500
(C) Rs. 12500 (D) Rs. 13500
(E) Rs. 14500

Q.097 Profit, Loss, Marked Price

A shopkeeper marked an article 50% above its cost price and then allowed two successive discounts of d% and 20%, respectively. If he still earned a profit of 8%, find the first discount percentage.

(A) 15% (B) 20%
(C) 5% (D) 10%
(E) 25%

Q.098 Profit, Loss, Marked Price

Kabir and Sahil invested a total of Rs. 54,000 in a business. Kabir invested his amount for 9 months, while Sahil invested his amount for 12 months. At the end of the year, the ratio of their profit shares was 3:5, respectively. Find the amount invested by Kabir.

(A) Rs. 30000 (B) Rs. 18000
(C) Rs. 24000 (D) Rs. 27000
(E) Rs. 36000

Q.099 Profit, Loss, Marked Price

Tushar and Naman entered into a partnership with investments of Rs. 2400 and Rs. 1600 respectively. After 4 months, Parth joined them with Rs. 3600. At the end of one year, Parth's profit share is Rs. 720. Find the difference between the profit shares of Tushar and Naman.

(A) Rs. 300 (B) Rs. 480
(C) Rs. 360 (D) Rs. 600
(E) Rs. 240

Q.100 Profit, Loss, Marked Price

A vendor bought a used laptop for Rs. 34800 and spent Rs. 1200 on its repair. If he sold it for Rs. X and made a profit of 15%, then find X.

(A) Rs. 41400 (B) Rs. 40800
(C) Rs. 42300 (D) Rs. 39600
(E) Rs. 43200

Q.101 Profit, Loss, Marked Price

Three partners, Abhinav, Gaurav and Chirag, started a firm with investments of Rs. 24,000, Rs. X and Rs. 36,000 respectively. After 4 months, Abhinav withdrew Rs. 6,000 and Chirag withdrew Rs. 12,000. At the end of the year, Gaurav received Rs. 21,000 out of the total profit of Rs. 84,000. Find X.

(A) 12000 (B) 20000
(C) 18000 (D) 16000
(E) 24000

Q.102 Profit, Loss, Marked Price

The discount on a product is 30% of its selling price, and the profit earned is 25%. If its cost price is Rs. 360, find the marked price of the product.

(A) Rs. 585 (B) Rs. 540
(C) Rs. 620 (D) Rs. 560
(E) Rs. 600

Q.103 Profit, Loss, Marked Price

Akash and Ayush started a business by investing Rs. 800 and Rs. 1200 respectively. After 4 months, Ayush invested Rs. X more. At the end of the year, Akash's profit share was Rs. 2400 out of the total profit of Rs. 7200. Find X.

(A) 450 (B) 600
(C) 750 (D) 800
(E) 500

Q.104 Profit, Loss, Marked Price

A shopkeeper marked an article 50% above its cost price and then allowed a discount of 40% on the marked price. Due to this, he suffered a loss of Rs. 360. If he had sold the article at the marked price, then find the difference between the two selling prices.

(A) Rs. 2160 (B) Rs. 1800
(C) Rs. 2400 (D) Rs. 2000
(E) Rs. 2520

Q.105 Profit, Loss, Marked Price

Yash invested Rs. P in a business. After 3 months, Sohan joined him with Rs. 3P, and Yash increased his initial investment by 50%. If at the end of the year the difference between their profit shares was Rs. 2800, then find Yash's profit share.

(A) Rs. 5200 (B) Rs. 4400
(C) Rs. 6000 (D) Rs. 3600
(E) Rs. 4800

Q.106 Profit, Loss, Marked Price

A trader marks an article 80% above its cost price. The ratio of the discount amount to the profit amount is 4:1. If the cost price of the article is Rs. 1000, then find the profit earned by the trader.

(A) Rs. 180 (B) Rs. 120
(C) Rs. 240 (D) Rs. 200
(E) Rs. 160

Q.107 Profit, Loss, Marked Price

A shopkeeper sold article P and article Q for Rs. 1440 each. He suffered a loss of 10% on article P and earned a profit of 20% on article Q. Find the total cost price of both articles.

(A) Rs. 2800 (B) Rs. 2600
(C) Rs. 3000 (D) Rs. 2400
(E) Rs. 3200

Q.108 Profit, Loss, Marked Price

Four friends Ramesh, Ayush, Sohan, and Nilesh started a business by investing Rs. 24000, Rs. 30000, Rs. 45000, and Rs. 51000 respectively for the same period. If the total profit at the end of one year is Rs. 25000, find the difference between the profit shares of Ayush and Nilesh.

(A) Rs. 2500 (B) Rs. 3000
(C) Rs. 4500 (D) Rs. 3500
(E) Rs. 4000

Q.109 Profit, Loss, Marked Price

Sameer, Rajesh and Nitesh started a business with investments of Rs. (X + 800), Rs. X and Rs. 2400 respectively. After 8 months, Sameer and Nitesh left the business. At the end of the year, Rajesh received Rs. 900 out of the total profit of Rs. 2700. Find Sameer's initial investment.

(A) 2000 (B) 2600
(C) 2800 (D) 2200
(E) 2400

Q.110 Profit, Loss, Marked Price

Rahul and Siddharth invested in a business in the ratio of 5:6, respectively. Siddharth invested for 4 months more than Rahul. Rahul received Rs. 1500 out of a total profit of Rs. 4500. For how many months did Siddharth invest in the business?

(A) 8 months (B) 12 months
(C) 10 months (D) 9 months
(E) 6 months

Q.111 Profit, Loss, Marked Price

The sum of the cost prices of two articles A and B is Rs. 700. Article A is marked 20% above its cost price, while article B is marked 50% above its cost price. If the marked price of article B is Rs. 240 more than the marked price of article A, then find the difference between their cost prices.

(A) Rs. 140 (B) Rs. 80
(C) Rs. 160 (D) Rs. 120
(E) Rs. 100

Q.112 Profit, Loss, Marked Price

The cost price of a mixer is Rs. 200 more than the cost price of a grinder. The mixer is sold at 20% profit, and the grinder is sold at 50% profit. If the selling price of the mixer is Rs. 60 more than the selling price of the grinder, then find the cost price of the mixer.

(A) 760 (B) 800
(C) 900 (D) 720
(E) 840

Q.113 Profit, Loss, Marked Price

A, B and C entered into a partnership with investments of Rs. x, Rs. (x + 400) and Rs. (x + 1000) respectively. They invested their amounts for 6 months, 9 months and 8 months respectively. A received Rs. 18000 as profit share, while B received Rs. 30000 as profit share. Find the amount invested by C.

(A) Rs. 4200 (B) Rs. 5000
(C) Rs. 4600 (D) Rs. 4800
(E) Rs. 5400

Q.114 Profit, Loss, Marked Price

Three partners Ramesh, Ketan and Lalit started a business by investing Rs. 15000, Rs. 20000 and Rs. 24000 respectively. After 5 months, Lalit left the business and Ketan left 9 months after the business began. If the total profit at the end of the year is Rs. 72000, find Ketan's profit share.

(A) Rs. 27000
(B) Rs. 24000
(C) Rs. 30000
(D) Rs. 21000
(E) Rs. 18000

Q.115 Profit, Loss, Marked Price

A shopkeeper purchased two items P and Q, where the cost price of P was Rs. 300 more than that of Q. He sold item P at a loss of 25% and item Q at a profit of 50%. If the ratio of the selling price of item P to that of item Q was 5 : 6, find the selling price of item Q.

(A) Rs. 600
(B) Rs. 675
(C) Rs. 720
(D) Rs. 750
(E) Rs. 810

Q.116 Profit, Loss, Marked Price

Rahul and Aman started a business by investing Rs. x and Rs. (x − 1800), respectively. After 4 months, Akash joined them with an investment of Rs. 7200, and Aman increased his investment by 25%. At the end of one year, the total profit was Rs. 9150, out of which Rahul's share was Rs. 3600. Find the value of x.

(A) 6400
(B) 8000
(C) 6800
(D) 7200
(E) 7600

Q.117 Profit, Loss, Marked Price

Abhinav started a business with an investment of Rs. 2400. After Y months, Bharat joined him with an investment of Rs. 3600. If at the end of the year, the profit-sharing ratio of Abhinav to Bharat is 8 : 7, find the value of 4Y + 6.

(A) 18
(B) 26
(C) 30
(D) 22
(E) 34

Q.118 Profit, Loss, Marked Price

The marked price of a mixer is 200% above its cost price. The shopkeeper allows x% discount while selling it and earns x% profit. If the selling price of the mixer is Rs. 540, find the discount amount allowed by the shopkeeper.

(A) Rs. 420
(B) Rs. 480
(C) Rs. 600
(D) Rs. 540
(E) Rs. 450

Q.119 Profit, Loss, Marked Price

The ratio of the investment periods of Suresh and Suresh is 5 : 6. At the end of the year, the total profit is Rs. 64000, and Suresh's share is Rs. 16000. Find the ratio of Suresh's investment to Suresh's investment.

(A) 3 : 2
(B) 4 : 3
(C) 2 : 5
(D) 5 : 3
(E) 5 : 2

Q.120 Profit, Loss, Marked Price

Ravi purchased two articles A and B for Rs. 56000 and Rs. 64000 respectively. The ratio of marked price to cost price for articles A and B is 9 : 8 and 5 : 4 respectively. If Ravi gives a discount of 20% on article A and 15% on article B, then find his overall profit or loss.

(A) Rs. 2400 profit
(B) Rs. 1600 loss
(C) Rs. 2000 loss
(D) Rs. 3600 profit
(E) Rs. 1200 loss

Q.121 Profit, Loss, Marked Price

A shopkeeper marked the price of a jacket 60% above its cost price and allowed two successive discounts of x% and (x + 20)% respectively. If he incurred a loss of 34% on selling the jacket, find the value of x.

(A) 10
(B) 20
(C) 30
(D) 15
(E) 25

Q.122 Profit, Loss, Marked Price

Kritika and Gaurav started a business by investing Rs. 9,000 and Rs. 7,000 respectively. After 4 months, Kritika left the business, and after 5 more months, Vikram joined the business with some capital. At the end of one year, the ratio of the profit share of Kritika to Vikram was 3:2. Vikram's profit share is what percentage of the combined profit shares of Kritika and Gaurav?

(A) 15%
(B) 25%
(C) 18%
(D) 20%
(E) 30%

Q.123 Profit, Loss, Marked Price

Ojas and Kritika invested a total of Rs. 7200 in a business. Ojas invested for 18 months and Kritika invested for 20 months. If the total profit was Rs. 3400 and Ojas's profit share was Rs. 1800, find Kritika's investment.

(A) Rs. 2800
(B) Rs. 3600
(C) Rs. 3000
(D) Rs. 3200
(E) Rs. 3400

Q.124 Profit, Loss, Marked Price

A shopkeeper sold an article for Rs. 765 and incurred a loss of 15%. If he had sold the same article at a profit of 1/6 of its cost price, what would have been the selling price?

(A) Rs. 1050
(B) Rs. 1025
(C) Rs. 1125
(D) Rs. 975
(E) Rs. 1080

Q.125 Profit, Loss, Marked Price

A shopkeeper marks the price of a jacket 40% above its cost price. After allowing a discount of 25%, he sells the jacket for Rs. 2,940. Find the cost price of the jacket.

(A) Rs. 2,600
(B) Rs. 3,200
(C) Rs. 3,500
(D) Rs. 2,800
(E) Rs. 3,000

Q.126 Profit, Loss, Marked Price

Mayank, Hardik and Tushar entered into a partnership business. Mayank invested Rs. x, Hardik invested 20% more than Mayank, and Tushar invested 25% more than Hardik for the first 4 months. After that, Mayank left the business, Hardik withdrew 25% of his investment, and Tushar increased his investment by 20%. Hardik left the business after 4 more months, while Tushar continued till the end of the year. If, at the end of the year, the difference between Tushar's profit share and the combined profit share of Mayank and Hardik is Rs. 9,600, then find Tushar's profit share.

(A) Rs. 24,480
(B) Rs. 25,680
(C) Rs. 21,600
(D) Rs. 22,480
(E) Rs. 26,880

Q.127 Profit, Loss, Marked Price

Rajesh bought two articles at the same cost price and sold both articles to Sameer at Rs. 240 each, earning profit on each article. The profit percentage on the first article for Rajesh, calculated on cost price, is 50% more than the profit percentage on the second article for Rajesh, calculated on selling price. If Sameer sells the first article at Rs. _____ and earns a profit amount which is 25% less than the profit amount earned by Rajesh on the first article, then find the value in the blank.

(A) Rs. 280
(B) Rs. 300
(C) Rs. 310
(D) Rs. 320
(E) Rs. 290

Q.128 Profit, Loss, Marked Price

Ayush and Sameer started a business by investing their capitals in the ratio of 5 : 7, respectively. At the end of the year, Ayush got Rs. 3500 as his profit share out of the total profit of Rs. 9800. If the ratio of the investment period of Sameer to that of Ayush is x : y in its simplest form, then find the value of x + y.

(A) 14
(B) 18
(C) 12
(D) 16
(E) 20

Q.129 Profit, Loss, Marked Price

A shopkeeper marked an article 60% above its cost price and allowed a discount of 25% on it. Had he allowed only a 10% discount, he would have earned Rs. 168 more. Find the cost price of the article.

(A) Rs. 600
(B) Rs. 750
(C) Rs. 700
(D) Rs. 800
(E) Rs. 900

📖 Chapter 02 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.046 Correct Option: (D)

- Investment of Priya = Rs. 15000
- Investment of Pallavi = Rs. 10000
- So, investment Ratio (Priya : Pallavi) = 15000 : 10000 = 3 : 2
- Let the total profit be Rs. P.
- 60% of the profit is shared equally, so both get the same amount from this part. Hence,
- there is no difference in their shares from 60% of the profit.
- The remaining profit = 40% of P.
- This 40% of profit is divided in the ratio 3 : 2.
- Difference in ratio parts = 3 - 2 = 1 part
- Total ratio parts = 3 + 2 = 5 parts
- So, difference between their shares from the remaining profit = 40% of P × 1 / 5
- = 8% of P
- Given difference = Rs. 320
- So,
- 8% of P = 320
- P = 320 × 100 / 8
- P = 4000

Solution Q.047 Correct Option: (B)

- Let original cost price of the watch = x units
- Marked price is 60% above cost price.
- So, marked price = 160% of x.
- Priya got a discount of 25%.
- So, Priya's purchase price = 75% of marked price.
- Priya's purchase price = 75% of 160% of x.
- 75% of 160% = 120%.
- So, Priya's purchase price = 120% of x.
- Priya sold the watch to Shreya at 20% profit.
- So, Shreya's purchase price = 120% of Priya's purchase price.
- Shreya's purchase price = 120% of 120% of x.
- 120% of 120% = 144%.
- So, Shreya's purchase price = 144% of x.
- Given Shreya paid Rs. 4320.
- Therefore, 144% of x = 4320.
- x = 4320 × 100 / 144.
- x = 3000.

Solution Q.048 **Correct Option: (C)**

• In partnership, profit is divided in the ratio of capital × time.
• Raj invested Rs. 9000 for the first 6 months.
• After 6 months, he withdrew 20% of Rs. 9000.
• 20% of 9000 = 1800.
• So, Raj's new investment = 9000 - 1800 = Rs. 7200.
• Raj's capital-time = 9000 × 6 + 7200 × 6.
• Raj's capital-time = 54000 + 43200 = 97200.
• Naman invested Rs. 11000 for the first 6 months.
• After 6 months, he added Rs. 3000.
• So, Naman's new investment = 11000 + 3000 = Rs. 14000.
• Naman's capital-time = 11000 × 6 + 14000 × 6.
• Naman's capital-time = 66000 + 84000 = 150000.
• Profit Ratio (Raj : Naman) = 97200 : 150000.
• Divide both terms by 1200.
• Ratio = 81 : 125.
• Raj's profit share = Rs. 12150.
• So, 81 parts = 12150.
• 1 part = 12150 / 81 = 150.
• Total parts = 81 + 125 = 206.
• Total profit = 206 × 150 = 30900.

Solution Q.049 **Correct Option: (D)**

• Cost price of the mixer grinder = Rs. 750.
• Marked price is 40% above cost price.
• So, marked price = 750 + 40% of 750 = 750 + 300 = Rs. 1050.
• Discount = 10% of marked price.
• Selling price = 90% of 1050 = 1050 × 90 / 100 = Rs. 945.

Solution Q.050 **Correct Option: (A)**

• Rs. 19,600
• Solution:
• In partnership questions, profit is divided in the ratio of investment × time.
• Investment Ratio (Anita : Pallavi) = 5:8.
• Time Ratio (Anita : Pallavi) = 6:5.
• So, profit-sharing ratio = 5 × 6 : 8 × 5.
• Profit-sharing ratio = 30:40.
• Now simplify the ratio.
• 30:40 = 3:4.
• So, Anita and Pallavi share the remaining profit in the ratio 3:4.
• Pallavi's share = 4 parts.
• Given Pallavi's share = Rs. 8,400.
• So, 4 parts = Rs. 8,400.
• Therefore, 1 part = Rs. 8,400 / 4 = Rs. 2,100.
• Remaining profit shared between Anita and Pallavi = 3 + 4 = 7 parts.
• So, remaining profit = 7 × 2,100 = Rs. 14,700.
• Now, 25% of the total profit was donated to an NGO.
• So, remaining profit = 100% - 25% = 75% of total profit.
• Therefore, 75% of total profit = Rs. 14,700.
• Total profit = 14,700 × 100 / 75 = Rs. 19,600.

Solution Q.051 **Correct Option: (D)**

• Marked price = Cost price + Profit + Discount
• Since only percentage and ratio are given, take cost price = Rs. 300 for easy
• calculation.
• The table is marked 50% above its cost price.
• Marked price: 300 × 150/100
• = Rs. 450
• The ratio of discount allowed to profit earned = 1 : 2
• Let discount = 1x and profit = 2x.
• Now, using the shortcut:
• Marked price = Cost price + Profit + Discount
• 450 = 300 + 2x + 1x
• 450 = 300 + 3x
• 3x = 450 - 300
• 3x = 150
• Value of x: 150/3
• = 50
• So, discount = 1x = Rs. 50
• Profit = 2x = Rs. 100
• Selling price = Cost price + Profit
• Selling price = 300 + 100 = Rs. 400
• Now, selling price compared to profit: 400/100
• = 4

Solution Q.052 **Correct Option: (C)**

• Rs. 896
• Solution:
• Let cost price of article Y = x units
• Article Y is sold at 40% profit.
• Selling price of article Y = 140% of x.
• Selling price of article Y = 140x/100 = 7x/5.
• Cost price of article X is equal to selling price of article Y.
• So, cost price of article X = 7x/5.
• Article X is sold at 25% profit.
• Selling price of article X = 125% of 7x/5.
• Selling price of article X = 125/100 × 7x/5.
• Selling price of article X = 5/4 × 7x/5 = 7x/4.
• Given:
• Cost price of article Y + Selling price of article X = 1760.
• x + 7x/4 = 1760
• 11x/4 = 1760
• x = 1760 × 4/11
• x = 640.
• Cost price of article X = 7x/5.
• Cost price of article X = 7 × 640/5 = Rs. 896.

Solution Q.053 **Correct Option: (B)**

• Riya's investment = Rs. 2400
• Riya's time = 8 months
• Riya's capital-months = 2400 × 8 = 19200
• Total profit = Rs. 5500
• Madhav's profit share = Rs. 3500
• Riya's profit share = 5500 - 3500 = Rs. 2000
• Profit Ratio (Riya : Madhav) = 2000:3500 = 4:7
• Madhav's investment = Rs. (x - 2400)
• Madhav's time = 12 months
• Madhav's capital-months = 12 × (x - 2400)
• Profit ratio = capital × time ratio
• So, 19200 : 12(x - 2400) = 4 : 7
• 19200 × 7 = 4 × 12(x - 2400)
• 134400 = 48(x - 2400)
• x - 2400 = 134400/48
• x - 2400 = 2800
• x = 2800 + 2400
• x = 5200

Solution Q.054 **Correct Option: (D)**

• Shreya's investment = Rs. 3000
• Shreya's time = 6 months
• Shreya's capital-months = 3000 × 6 = 18000
• Total profit = Rs. 4800
• Naman's profit share = Rs. 3600
• Shreya's profit share = 4800 - 3600 = Rs. 1200
• Profit Ratio (Shreya : Naman) = 1200:3600 = 1:3
• Naman's investment = Rs. (x - 3000)
• Naman's time = 12 months
• Naman's capital-months = 12 × (x - 3000)
• Profit ratio = capital × time ratio
• So, 18000 : 12(x - 3000) = 1 : 3
• 18000 × 3 = 1 × 12(x - 3000)
• 54000 = 12(x - 3000)
• x - 3000 = 54000/12
• x - 3000 = 4500
• x = 4500 + 3000
• x = 7500

Solution Q.055 **Correct Option: (A)**

• Rs. 24000
• Solution:
• In partnership, profit is divided in the ratio of capital × time.
• Investment Ratio (Ishan : Jatin) = 4 : 7
• Let Ishan's capital = 4x
• Let Jatin's capital = 7x
• Ishan invested for 12 months.
• Ishan's capital-time value = 4x × 12 = 48x
• Jatin invested for 12 months.
• Jatin's capital-time value = 7x × 12 = 84x
• Lalit joined after 4 months, so he invested for 8 months.
• Lalit's capital = 2 × Jatin's capital = 2 × 7x = 14x
• Lalit's capital-time value = 14x × 8 = 112x
• Profit ratio of Ishan, Jatin and Lalit = 48x : 84x : 112x
• Cancel x and divide by 4.
• Profit ratio = 12 : 21 : 28
• Total ratio units = 12 + 21 + 28 = 61
• Total profit = Rs. 122000
• Value of 1 unit = 122000/61 = Rs. 2000
• Ishan's profit share = 12 × 2000 = Rs. 24000

Solution Q.056 **Correct Option: (B)**

• Rs. 85
• Solution:
• Let cost price of the article = x units
• Marked price is 80% above cost price.
• So, marked price = 180% of x = 180x/100.
• Marked price = 9x/5.
• A discount of 20% is allowed.
• So, selling price = 80% of marked price.
• Selling price = 80/100 × 9x/5.
• Selling price = 36x/25.
• Profit = Selling price - Cost price.
• Profit = 36x/25 - x.
• Profit = 36x/25 - 25x/25.
• Profit = 11x/25.
• Given profit = Rs. 220.
• So,
• 11x/25 = 220
• x = 220 × 25/11
• x = 500
• Cost price = Rs. 500.
• Now, if discount is 35%, selling price = 65% of marked price.
• New selling price = 65/100 × 9x/5.
• Put x = 500.
• New selling price = 65/100 × 9 × 500/5.
• New selling price = Rs. 585.
• New profit = 585 - 500 = Rs. 85.

Solution Q.057 **Correct Option: (D)**

• Profit share depends on capital × time.
• Parth's initial investment = Rs. 5,000
• Mitali's investment = Rs. 7,200
• Parth invested Rs. 5,000 for 12 months.
• Parth also invested an additional Rs. A after 6 months, so this additional amount
• remained invested for 6 months.
• Parth's capital-months = 5000 × 12 + A × 6
• Parth's capital-months = 60000 + 6A
• Mitali's capital-months = 7200 × 12 = 86400
• Mitali's profit share is 80% of Parth's profit share.
• So, Mitali's share : Parth's share = 80 : 100 = 4 : 5
• Therefore, Parth's share : Mitali's share = 5 : 4
• Now,
• (60000 + 6A) : 86400 = 5 : 4
• Cross multiply:
• 4 × (60000 + 6A) = 5 × 86400
• 240000 + 24A = 432000
• 24A = 432000 - 240000
• 24A = 192000
• A = 192000/24
• A = 8000

Solution Q.058 **Correct Option: (B)**

• Let the cost prices of articles P and Q be 5x and 8x respectively.
• Article P is sold at a profit of 25%.
• Selling price of P = 125% of 5x
• Selling price of P = 125 × 5x/100
• Selling price of P = 25x/4
• Cost price of Q = 8x
• Given difference between cost price of Q and selling price of P = Rs. 140
• 8x - 25x/4 = 140
• 32x/4 - 25x/4 = 140
• 7x/4 = 140
• 7x = 560
• x = 560/7
• x = 80
• Cost price of Q = 8x
• Cost price of Q = 8 × 80 = Rs. 640
• Article Q is sold at a loss of 15%.
• Selling price of Q = 85% of 640
• Selling price of Q = 85 × 640/100
• Selling price of Q = Rs. 544

Solution Q.059 **Correct Option: (D)**

• Rs. 21000
• Solution:
• Profit ratio is based on investment × time.
• Total duration = 12 months.
• For the first 6 months:
• Nilesh's investment = Rs. 4000.
• Mukesh's investment = Rs. 6000.
• Suresh's investment = Rs. 5000.
• After 6 months:
• Nilesh's new investment = 4000 + 25% of 4000 = 4000 + 1000 = Rs. 5000.
• Mukesh's new investment = 6000 + 25% of 6000 = 6000 + 1500 = Rs. 7500.
• Suresh's new investment = 5000 - 20% of 5000 = 5000 - 1000 = Rs. 4000.
• Nilesh's investment value = 4000 × 6 + 5000 × 6 = 24000 + 30000 = 54000.
• Mukesh's investment value = 6000 × 6 + 7500 × 6 = 36000 + 45000 = 81000.
• Suresh's investment value = 5000 × 6 + 4000 × 6 = 30000 + 24000 = 54000.
• Profit ratio = 54000 : 81000 : 54000.
• Simplified ratio = 2 : 3 : 2.
• Mukesh's share = 3 parts = Rs. 9000.
• 1 part = 9000/3 = Rs. 3000.
• Total ratio parts = 2 + 3 + 2 = 7.
• Total profit = 7 × 3000 = Rs. 21000.

Solution Q.060 **Correct Option: (E)**

• Rs. 2400
• Solution:
• Let marked price of product Y = x units
• Marked price of product X is 50% more than marked price of Y.
• So, marked price of X = 150% of x = 3x/2
• Cost price of X = Marked price of Y = x
• Product X is sold at 25% profit.
• So, selling price of X = 125% of x = 5x/4
• Discount on X = Marked price of X - Selling price of X
• Discount on X = 3x/2 - 5x/4
• Discount on X = 6x/4 - 5x/4 = x/4
• Given discount on X = Rs. 800
• So, x/4 = 800
• x = 800 × 4 = 3200
• Marked price of Y = Rs. 3200
• Discount on Y = Rs. 1280
• Selling price of Y = 3200 - 1280 = Rs. 1920
• Product Y is sold at 20% loss.
• So, selling price of Y = 80% of cost price of Y
• 80% of cost price of Y = 1920
• Cost price of Y = 1920 × 100/80
• Cost price of Y = Rs. 2400

Solution Q.061 **Correct Option: (C)**

• Rs. 156 profit
• Solution:
• Cost price of tablet = Rs. 1250.
• Profit on tablet = 24%.
• Profit amount on tablet = 1250 × 24/100 = Rs. 300.
• Cost price of headphones = Rs. 960.
• Loss on headphones = 15%.
• Loss amount on headphones = 960 × 15/100 = Rs. 144.
• Overall result = Profit on tablet - Loss on headphones.
• Overall profit = 300 - 144 = Rs. 156.

Solution Q.062 **Correct Option: (D)**

• Rs. 3850
• Solution:
• Profit is divided in the ratio of capital × time.
• First partner's initial capital = Rs. 5000
• Second partner's initial capital = Rs. 7000
• For the first 6 months:
• First partner's capital-months = 5000 × 6 = 30000
• Second partner's capital-months = 7000 × 6 = 42000
• After 6 months, the first partner increased his capital by 40%.
• New capital of first partner = 5000 + 40% of 5000
• New capital of first partner = 5000 + 2000 = Rs. 7000
• After 6 months, the second partner increased his capital by 20%.
• New capital of second partner = 7000 + 20% of 7000
• New capital of second partner = 7000 + 1400 = Rs. 8400
• For the next 6 months:
• First partner's capital-months = 7000 × 6 = 42000
• Second partner's capital-months = 8400 × 6 = 50400
• Total capital-months of first partner = 30000 + 42000 = 72000
• Total capital-months of second partner = 42000 + 50400 = 92400
• Profit ratio = 72000 : 92400
• Profit ratio = 60 : 77
• Total ratio parts = 60 + 77 = 137
• Total profit = Rs. 6850
• Second partner's share = 6850 × 77/137
• Second partner's share = Rs. 3850

Solution Q.063 **Correct Option: (C)**

• Rs. 5700
• Solution:
• Let cost price of the blazer = x units
• Marked price = x + 1200
• Discount = Rs. 750
• Selling price = Marked price - Discount
• Selling price = x + 1200 - 750
• Selling price = x + 450
• The shopkeeper earns 10% profit.
• So, selling price = 110% of cost price
• x + 450 = 110x/100
• x + 450 = 11x/10
• Multiply by 10:
• 10x + 4500 = 11x
• x = 4500
• Cost price = Rs. 4500
• Marked price = 4500 + 1200 = Rs. 5700

Solution Q.064 **Correct Option: (E)**

• %
• Solution:
• Marked price of the article = Rs. 640.
• First discount = 25%.
• Price after first discount = 640 × 75/100.
• Price after first discount = Rs. 480.
• Final selling price after second discount = Rs. 408.
• Second discount amount = 480 - 408 = Rs. 72.
• Second discount percentage = 72/480 × 100.
• Second discount percentage = 15%.

Solution Q.065 **Correct Option: (D)**

• Profit is divided in the ratio of investment × time.
• Investment Ratio (first partner : second partner) = 5:4.
• Time ratio = 16:15.
• Profit ratio = 5 × 16 : 4 × 15
• Profit ratio = 80:60
• Profit ratio = 4:3
• The charity amount does not change the ratio of the remaining profit shared between
• the partners.
• First partner's share = 4 parts = Rs. 4800.
• So, 1 part = 4800/4 = Rs. 1200.
• Second partner's share = 3 parts.
• Second partner's share = 3 × 1200 = Rs. 3600.

Solution Q.066 **Correct Option: (B)**

• Rs. 32000
• Solution:
• In partnership, profit is divided in the ratio of capital × time.
• Sonal invested Rs. 90000 for the first 8 months.
• Sonal's capital-time value for 8 months = 90000 × 8 = 720000
• After 8 months, Sonal withdrew 1/3 of her investment.
• Amount withdrawn = 90000 × 1/3 = Rs. 30000
• Remaining investment of Sonal = 90000 - 30000 = Rs. 60000
• Sonal's capital-time value for the last 4 months = 60000 × 4 = 240000
• Total capital-time value of Sonal = 720000 + 240000 = 960000
• Swati joined after 2 months with Rs. 150000.
• Swati invested Rs. 150000 for the next 6 months.
• Swati's capital-time value for 6 months = 150000 × 6 = 900000
• After that, Swati added Rs. 30000.
• New investment of Swati = 150000 + 30000 = Rs. 180000
• Swati's capital-time value for the last 4 months = 180000 × 4 = 720000
• Total capital-time value of Swati = 900000 + 720000 = 1620000
• Profit Ratio (Sonal : Swati) = 960000 : 1620000
• Divide both terms by 60000.
• Profit ratio = 16 : 27
• Total ratio units = 16 + 27 = 43
• Total profit = Rs. 86000
• Value of 1 unit = 86000/43 = Rs. 2000
• Sonal's profit share = 16 × 2000 = Rs. 32000

Solution Q.067 **Correct Option: (D)**

• Let the cost price of the wristwatch be Rs. P.
• Selling price = Rs. Q.
• Profit = 30%.
• So, Q = 130/100 × P
• Q = 13P/10
• In the second case, the cost price is 25% less.
• New cost price = 75/100 × P
• New cost price = 3P/4
• Profit in the second case = 20%.
• New selling price = 120/100 × 3P/4
• New selling price = 6/5 × 3P/4
• New selling price = 18P/20
• New selling price = 9P/10
• Given, new selling price is Rs. 96 less than Q.
• So, Q - 96 = 9P/10
• Put Q = 13P/10.
• 13P/10 - 96 = 9P/10
• 13P/10 - 9P/10 = 96
• 4P/10 = 96
• 2P/5 = 96
• P = 96 × 5/2
• P = 240
• Now, Q - P is the profit.
• Profit = 30% of P
• Q - P = 30/100 × 240
• Q - P = 72

Solution Q.068 **Correct Option: (A)**

• Rs. 30000
• Solution:
• Profit is divided in the ratio of capital × time.
• Nirav invested Rs. 60000 for 12 months.
• Nirav's capital-months = 60000 × 12 = 720000
• Pallav joined after 3 months, so he invested for 9 months.
• Pallav's capital-months = 72000 × 9 = 648000
• Uday joined after 3 + 5 = 8 months, so he invested for 4 months.
• Uday's capital-months = 90000 × 4 = 360000
• Profit ratio = 720000 : 648000 : 360000
• Profit ratio = 10 : 9 : 5
• Total ratio parts = 10 + 9 + 5 = 24
• Total profit = Rs. 144000
• Uday's share = 144000 × 5/24
• Uday's share = Rs. 30000

Solution Q.069 **Correct Option: (C)**

• %
• Solution:
• Let the cost price of 1 kg goods be Rs. 100.
• Marked price = 50% above cost price
• Marked price = Rs. 150
• Discount = 20%
• Selling price charged for 1 kg = 80% of 150
• Selling price = 80 × 150/100 = Rs. 120
• But the trader gives only 800 grams.
• Cost price of 800 grams = 800 × 100/1000
• Cost price of 800 grams = Rs. 80
• Profit = 120 - 80 = Rs. 40
• Gain percent = 40 × 100/80
• Gain percent = 50%

Solution Q.070 **Correct Option: (B)**

• Rs. 27500
• Solution:
• Profit is divided in the ratio of investment × time.
• Partner U invested Rs. 3x for 12 months.
• Partner U's investment value = 3x × 12 = 36x.
• Partner V invested Rs. 5x for 12 months.
• Partner V's investment value = 5x × 12 = 60x.
• Partner W joined after 4 months.
• So, Partner W invested for 8 months.
• Partner W's investment value = 6x × 8 = 48x.
• Profit ratio of U, V and W = 36x : 60x : 48x.
• Simplified ratio = 3 : 5 : 4.
• Total ratio parts = 3 + 5 + 4 = 12.
• Total profit = Rs. 66000.
• Partner V's share = 5/12 × 66000.
• Partner V's share = Rs. 27500.

Solution Q.071 **Correct Option: (E)**

• Rs. 20,700
• Solution:
• Selling price after repair = Rs. 24,000.
• Profit = 20%.
• So, selling price = 120% of total cost price.
• Total cost price = 24,000 × 100/120
• Total cost price = Rs. 20,000.
• Repair cost = Rs. 2,000.
• Original cost price of camera = 20,000 - 2,000
• Original cost price = Rs. 18,000.
• If the camera was not repaired, cost price = Rs. 18,000.
• Profit in the new case = 15%.
• New selling price = 18,000 × 115/100
• New selling price = Rs. 20,700.

Solution Q.072 **Correct Option: (D)**

• Rs. 2800
• Solution:
• Let cost price of the bag = x units
• Marked price is 75% above cost price.
• Marked price = 175% of x.
• Marked price = 175x/100 = 7x/4.
• Discount allowed = 20%.
• So, selling price = 80% of marked price.
• Selling price = 80/100 × 7x/4.
• Selling price = 7x/5.
• Profit = Selling price - Cost price.
• Profit = 7x/5 - x.
• Profit = 7x/5 - 5x/5.
• Profit = 2x/5.
• Given profit = Rs. 800.
• So,
• 2x/5 = 800
• 2x = 800 × 5
• 2x = 4000
• x = 4000/2
• x = 2000.
• Cost price = Rs. 2000.
• Selling price = 7x/5.
• Selling price = 7 × 2000/5 = Rs. 2800.

Solution Q.073 **Correct Option: (C)**

• months
• Solution:
• In partnership, profit is divided in the ratio of capital × time.
• Investment Ratio (M : N) = 5 : 8
• Let M's investment = 5x and N's investment = 8x.
• M's profit share = Rs. 6000
• Total profit = Rs. 14000
• N's profit share = 14000 - 6000 = Rs. 8000
• Profit Ratio (M : N) = 6000 : 8000 = 3 : 4
• M invested for 12 months.
• Let N's investment period be t months.
• So, capital-time ratio = 5x × 12 : 8x × t
• Capital-time ratio = 60x : 8xt
• This is equal to profit ratio 3 : 4.
• So, 60x/(8xt) = 3/4
• Cancel x.
• 60/(8t) = 3/4
• Cross multiply:
• 60 × 4 = 3 × 8t
• 240 = 24t
• t = 240/24 = 10

Solution Q.074 **Correct Option: (B)**

• Rs. 1000
• Solution:
• Let cost price of the item = x units
• Marked price is 75% above cost price.
• Marked price = 175% of x = 175x/100 = 7x/4
• Difference between two discounts = 30% - 18% = 12%
• If only 18% discount is given instead of 30%, the selling price increases by 12% of
• marked price.
• Given increase in earning = Rs. 210
• So, 12% of marked price = 210
• Marked price = 210 × 100/12
• Marked price = Rs. 1750
• Now, marked price = 7x/4
• 7x/4 = 1750
• x = 1750 × 4/7
• x = Rs. 1000

Solution Q.075 **Correct Option: (D)**

• Rs. 13200
• Solution:
• Initial investment Ratio (Sameer : Nilesh) = 4:7
• Let Sameer's initial investment = 4x
• Let Nilesh's initial investment = 7x
• For the first 6 months:
• Sameer's capital-time = 4x × 6 = 24x
• Nilesh's capital-time = 7x × 6 = 42x
• After 6 months, Sameer increased his initial investment by 50%.
• New investment of Sameer = 4x + 50 × 4x/100
• New investment of Sameer = 4x + 2x = 6x
• After 6 months, Nilesh withdrew 3/7 of his initial investment.
• Amount withdrawn by Nilesh = 3/7 × 7x = 3x
• New investment of Nilesh = 7x - 3x = 4x
• For the next 6 months:
• Sameer's capital-time = 6x × 6 = 36x
• Nilesh's capital-time = 4x × 6 = 24x
• Total capital-time of Sameer = 24x + 36x = 60x
• Total capital-time of Nilesh = 42x + 24x = 66x
• Profit ratio of Sameer to Nilesh = 60x : 66x = 10 : 11
• Difference between their profit shares = 11 parts - 10 parts = 1 part
• 1 part = Rs. 1200
• Nilesh's profit share = 11 parts = 11 × 1200
• Nilesh's profit share = Rs. 13200

Solution Q.076 **Correct Option: (C)**

• Rs. 24000
• Solution:
• Priyanka's investment = Rs. P.
• Avani's investment = Rs. (P + 4000).
• Priyanka's time = T years.
• Avani's time = 2T years.
• Total profit = Rs. 17,000.
• Priyanka's profit share = Rs. 5,000.
• Avani's profit share = 17,000 - 5,000 = Rs. 12,000.
• Profit Ratio (Priyanka : Avani) = 5000:12000
• Profit ratio = 5:12.
• Profit shares are proportional to the respective capital-time products.
• So, P × T : (P + 4000) × 2T = 5:12.
• Cancel T from both sides.
• P : 2(P + 4000) = 5:12.
• Cross multiply:
• 12P = 5 × 2(P + 4000)
• 12P = 10(P + 4000)
• 12P = 10P + 40000
• 12P - 10P = 40000
• 2P = 40000
• P = 40000/2
• P = 20000.
• Avani's investment = P + 4000
• Avani's investment = 20000 + 4000
• Avani's investment = Rs. 24000.

Solution Q.077 **Correct Option: (C)**

• Rs. 30000
• Solution:
• Tushar's investment for first 4 months = Rs. 30000
• After 4 months, Tushar withdrew 20% of Rs. 30000.
• Remaining investment of Tushar = 80% of 30000 = 30000 × 80/100 = Rs. 24000
• Tushar's capital × time = 30000 × 4 + 24000 × 8
• Tushar's capital × time = 120000 + 192000 = 312000
• Prateek invested Rs. 42000 for 4 months only.
• Prateek's capital × time = 42000 × 4 = 168000
• Ratio of Tushar's and Prateek's profit shares = 312000:168000 = 13:7
• Given profit ratio of Tushar, Prateek and Sagar = 13:7:10
• So, 13 parts = 312000
• 1 part = 312000/13 = 24000
• Sagar's 10 parts = 10 × 24000 = 240000
• Sagar invested for 8 months.
• So, Sagar's investment × 8 = 240000
• Sagar's investment = 240000/8 = Rs. 30000

Solution Q.078 **Correct Option: (E)**

• Rs. 240
• Solution:
• Let cost price of the article for Kabir = x units
• The cost price of the article is the same for Kabir and Sameer.
• So, Sameer's cost price = Rs. x.
• Kabir sells the article to Eshaan at 25% loss.
• Eshaan's cost price = 75% of x = 75x/100 = 3x/4.
• Difference between Sameer's cost price and Eshaan's cost price = Rs. 200.
• So,
• x - 3x/4 = 200
• x/4 = 200
• x = 200 × 4
• x = 800.
• So, Kabir's cost price = Rs. 800.
• Sameer sells the article to Mukesh at 30% profit.
• Sameer's selling price = 130% of 800.
• Sameer's selling price = 130/100 × 800 = Rs. 1040.
• If Kabir directly sold the article to Mukesh at Rs. 1040, then Kabir's profit = 1040 - 800.
• Kabir's profit = Rs. 240.

Solution Q.079 **Correct Option: (A)**

• Rs. 180
• Solution:
• Profit is divided in the ratio of investment × time.
• Sahil's initial investment = Rs. 9000.
• Tapan's initial investment = Rs. 7000.
• After 3 months, Sahil withdrew one-third of his initial investment.
• Amount withdrawn by Sahil = 9000 × 1/3 = Rs. 3000.
• So, Sahil's remaining investment = 9000 - 3000 = Rs. 6000.
• Tapan added the same amount, Rs. 3000.
• So, Tapan's new investment = 7000 + 3000 = Rs. 10000.
• Uday joined after 3 months with Rs. 12000.
• Now calculate investment × time:
• Sahil = 9000 × 3 + 6000 × 9
• Sahil = 27000 + 54000 = 81000.
• Tapan = 7000 × 3 + 10000 × 9
• Tapan = 21000 + 90000 = 111000.
• Uday = 12000 × 9 = 108000.
• Profit ratio = 81000 : 111000 : 108000.
• Divide by 3000:
• Profit ratio = 27 : 37 : 36.
• Total ratio parts = 27 + 37 + 36 = 100.
• Total profit = Rs. 18000.
• Difference between Tapan's and Uday's ratio parts = 37 - 36 = 1 part.
• Value of 1 part = 18000/100 = Rs. 180.

Solution Q.080 **Correct Option: (C)**

• Both partners invested for the same time, so profit ratio = investment ratio.
• Total profit = Rs. 9000
• First partner's profit = Rs. 4000
• Second partner's profit = 9000 - 4000 = Rs. 5000
• Profit ratio of first partner to second partner = 4000:5000 = 4:5
• Investment ratio of first partner to second partner = 4:5
• First partner's investment = Rs. Y
• Second partner's investment = Rs. (Y + 1500)
• So, Y:(Y + 1500) = 4:5
• 5Y = 4(Y + 1500)
• 5Y = 4Y + 6000
• Y = 6000

Solution Q.081 **Correct Option: (C)**

• Rs. 3000
• Solution:
• Let marked price of article Y = x units
• Cost price of article X = Marked price of article Y = Rs. x
• Marked price of article X = 50% more than marked price of Y
• Marked price of X = 150% of x = 150x/100 = 3x/2
• Article X is sold at 25% profit.
• Selling price of X = 125% of x = 125x/100 = 5x/4
• Discount on X = Marked price of X - Selling price of X
• Discount on X = 3x/2 - 5x/4
• Discount on X = 6x/4 - 5x/4 = x/4
• Given discount on X = Rs. 800
• So, x/4 = 800
• x = 800 × 4 = Rs. 3200
• Marked price of Y = Rs. 3200
• Discount on Y = Rs. 500
• Selling price of Y = 3200 - 500 = Rs. 2700
• Article Y is sold at 10% loss.
• So, selling price of Y = 90% of cost price of Y
• Cost price of Y = 2700 × 100/90
• Cost price of Y = Rs. 3000

Solution Q.082 **Correct Option: (B)**

• Rs. 28800
• Solution:
• Profit is divided in the ratio of investment × time.
• Ishan's initial investment = Rs. 4000
• Keshav's initial investment = Rs. 6000
• Ayush's initial investment = Rs. 5000
• After 4 months, Ishan increased his investment by 25%.
• Ishan's new investment = 4000 + 25% of 4000
• Ishan's new investment = 4000 + 1000 = Rs. 5000
• Ishan's investment-time value = 4000 × 4 + 5000 × 8
• Ishan's investment-time value = 16000 + 40000 = 56000
• After 4 months, Keshav increased his investment by 25%.
• Keshav's new investment = 6000 + 25% of 6000
• Keshav's new investment = 6000 + 1500 = Rs. 7500
• Keshav's investment-time value = 6000 × 4 + 7500 × 8
• Keshav's investment-time value = 24000 + 60000 = 84000
• After 4 months, Ayush reduced his investment by 20%.
• Ayush's new investment = 5000 - 20% of 5000
• Ayush's new investment = 5000 - 1000 = Rs. 4000
• Ayush's investment-time value = 5000 × 4 + 4000 × 8
• Ayush's investment-time value = 20000 + 32000 = 52000
• Profit ratio of Ishan, Keshav and Ayush = 56000 : 84000 : 52000
• Divide by 4000.
• Profit ratio = 14 : 21 : 13
• Total ratio parts = 14 + 21 + 13 = 48 parts
• Keshav's share = 21 parts = Rs. 12600
• 1 part = 12600/21 = Rs. 600
• Total profit = 48 × 600 = Rs. 28800

Solution Q.083 **Correct Option: (D)**

• Rs. 280
• Solution:
• Let the cost prices of the first and second bags be A and B.
• First bag is sold at 16% loss, so its selling price = 84% of A.
• Second bag is sold at 16% profit, so its selling price = 116% of B.
• Both selling prices are equal.
• So, 84% of A = 116% of B.
• A : B = 116 : 84 = 29 : 21.
• Total cost price = Rs. 1750.
• Total parts = 29 + 21 = 50.
• Value of 1 part = 1750 / 50 = 35.
• Difference between cost prices = (29 - 21) parts = 8 parts.
• Required difference = 8 × 35 = Rs. 280.

Solution Q.084 **Correct Option: (C)**

• %
• Solution:
• Selling price = Rs. 840
• Loss = 16%
• So, selling price = 84% of cost price
• Cost price = 840 × 100/84 = Rs. 1000
• Marked price = 150% of cost price
• Marked price = 1000 × 150/100 = Rs. 1500
• After 30% discount, price = 1500 × 70/100 = Rs. 1050
• Now, after y% discount, selling price becomes Rs. 840
• So, 1050 × (100 - y)/100 = 840
• 100 - y = 840 × 100/1050 = 80
• y = 100 - 80 = 20

Solution Q.085 **Correct Option: (D)**

• Rs. 12900
• Solution:
• Profit ratio is based on capital × time.
• Priyansh invested Rs. 6000 for the first 5 months and Rs. 4000 for the next 7 months.
• Priyansh's capital-time = 6000 × 5 + 4000 × 7 = 30000 + 28000 = 58000.
• Nilesh invested Rs. 4500 for the first 5 months and Rs. 6000 for the next 7 months.
• Nilesh's capital-time = 4500 × 5 + 6000 × 7 = 22500 + 42000 = 64500.
• Ratio of profit shares = 58000:64500 = 116:129.
• Total parts = 116 + 129 = 245.
• Nilesh's share = 129/245 × 24500 = Rs. 12900.

Solution Q.086 **Correct Option: (E)**

• Rs. 1540
• Solution:
• Let the cost price of the article = Rs. x.
• Profit when sold for Rs. 1700 = 1700 - x
• Loss when sold for Rs. 800 = x - 800
• Given, profit = 2 × loss
• 1700 - x = 2(x - 800)
• 1700 - x = 2x - 1600
• 3300 = 3x
• x = 1100
• Cost price = Rs. 1100
• Required selling price for 40% profit = 140% of cost price
• Required selling price = 140/100 × 1100
• Required selling price = Rs. 1540

Solution Q.087 **Correct Option: (B)**

• Rs. 70
• Solution:
• Let cost price of each diary = x units
• Cost price of 18 diaries = 18x
• Profit = cost price of 6 diaries = 6x
• Selling price of 18 diaries = cost price + profit
• 1680 = 18x + 6x
• 1680 = 24x
• x = 1680/24 = 70
• Cost price of each diary = Rs. 70

Solution Q.088 **Correct Option: (D)**

• Ayush's investment = Rs. 6000.
• Siddharth's investment = 60% of 6000 = Rs. 3600.
• Ayush invested for X months.
• Ayush's capital-time = 6000 × X.
• Siddharth invested for 12 months.
• Siddharth's capital-time = 3600 × 12 = 43200.
• Profit share ratio = capital × time ratio.
• Siddharth received Rs. 12000 out of Rs. 27000.
• So, Ayush's share = 27000 - 12000 = Rs. 15000.
• Profit ratio of Ayush to Siddharth = 15000:12000 = 5:4.
• So, 6000X:43200 = 5:4.
• 6000X × 4 = 43200 × 5.
• 24000X = 216000.
• X = 216000 / 24000 = 9.

Solution Q.089 **Correct Option: (B)**

• % loss
• Solution:
• Selling price = Rs. 600
• Discount = 25%
• So, selling price = 75% of marked price
• Marked price = 600 × 100/75 = Rs. 800
• Cost price : Marked price = 5:8
• Cost price = 800 × 5/8 = Rs. 500
• New cost price = 500 × 125/100 = Rs. 625
• Selling price remains Rs. 600
• Loss = 625 - 600 = Rs. 25
• Loss percentage = 25/625 × 100 = 4%

Solution Q.090 **Correct Option: (D)**

• %
• Solution:
• Dealer's cost price = Rs. 960
• Profit = 20% of selling price
• So, cost price = 80% of selling price
• Selling price for dealer = 960 × 100/80 = Rs. 1200
• The customer bought the article for Rs. 1200
• The customer sold it for Rs. 1080
• Loss = 1200 - 1080 = Rs. 120
• Loss percentage = 120/1200 × 100 = 10%
• So, L = 10

Solution Q.091 **Correct Option: (E)**

• Total profit = Rs. 9000
• Profit received by Vikram and Ojas together = 2/3 × 9000 = Rs. 6000
• Their profit shares are equal.
• So, Vikram's profit = Rs. 3000 and Ojas's profit = Rs. 3000
• Pradeep's profit = 9000 - 6000 = Rs. 3000
• Therefore, profit ratio of Vikram, Ojas and Pradeep = 3000:3000:3000 = 1:1:1
• Profit shares are proportional to the respective capital-time products.
• Investments are Rs. 3000, Rs. 5000 and Rs. 4000.
• So, time ratio = 1/3000 : 1/5000 : 1/4000
• Multiply by 60000:
• Time ratio = 20:12:15
• Sum of terms = 20 + 12 + 15 = 47

Solution Q.092 **Correct Option: (E)**

• % profit
• Solution:
• Let the selling prices of P, Q, and R be 6x, 3x, and 9x respectively.
• Article P is sold at 20% profit.
• So, CP of P = 6x × 100/120 = 5x.
• Article Q is sold at 25% loss.
• So, CP of Q = 3x × 100/75 = 4x.
• Article R is sold at 50% profit.
• So, CP of R = 9x × 100/150 = 6x.
• Total selling price = 6x + 3x + 9x = 18x.
• Total cost price = 5x + 4x + 6x = 15x.
• Profit = 18x - 15x = 3x.
• Profit percentage = 3x × 100 / (15x) = 20%.
• So, the shopkeeper earned an overall profit of 20%.

Solution Q.093 **Correct Option: (D)**

• Rs. 3,750
• Solution:
• Cost price of mixer grinder = Rs. 18,750.
• Marked price is 60% above cost price.
• Marked price = 18750 × 160/100 = Rs. 30,000.
• Discount = 25%.
• Selling price = 30000 × 75/100 = Rs. 22,500.
• Profit = Selling price - Cost price.
• Profit = 22500 - 18750 = Rs. 3,750.

Solution Q.094 **Correct Option: (B)**

• Rs. 4,200
• Solution:
• Let Aman's investment be Rs. x.
• Adarsh's investment = Rs. (x + 700).
• Hemant's investment = Rs. (x + 700 + 700) = Rs. (x + 1400).
• Aman stayed for 12 months, Adarsh stayed for 8 months and Hemant stayed for 4
• months.
• Profit shares are proportional to the respective capital-time products.
• Aman's share : Adarsh's share : Hemant's share = 12x : 8(x + 700) : 4(x + 1400).
• Given ratio = 9 : 7 : 4.
• Compare Aman and Adarsh:
• 12x : 8(x + 700) = 9 : 7.
• Using cross multiplication:
• 12x × 7 = 9 × 8(x + 700)
• 84x = 72x + 50400
• 84x - 72x = 50400
• 12x = 50400
• x = 4200
• So, Aman's investment = Rs. 4,200.
• Check all three capital-time products: 4200 × 12 : 4900 × 8 : 5600 × 4 = 50400 :
• 39200 : 22400 = 9 : 7 : 4.

Solution Q.095 **Correct Option: (A)**

• Rs. 4350
• Solution:
• Let cost price = x units
• Marked price = x + 1350
• Discount = Rs. 750
• Selling price = x + 1350 - 750 = x + 600
• Profit = 20%
• So, selling price = 120% of cost price
• x + 600 = 120x/100
• x + 600 = 1.2x
• 600 = 0.2x
• x = 3000
• Marked price = 3000 + 1350 = Rs. 4350

Solution Q.096 **Correct Option: (D)**

• Rs. 13500
• Solution:
• Yash invested for 16 months.
• Siddharth joined after 6 months, so Siddharth invested for 16 - 6 = 10 months.
• Siddharth's profit share = Rs. 7200
• Total profit = Rs. 18000
• Yash's profit share = 18000 - 7200 = Rs. 10800
• Profit Ratio (Yash : Siddharth) = 10800:7200 = 3:2
• Profit shares are proportional to the respective capital-time products.
• So, (5P + 2000) × 16 : (8P - 4000) × 10 = 3:2
• 2 × 16 × (5P + 2000) = 3 × 10 × (8P - 4000)
• 32(5P + 2000) = 30(8P - 4000)
• 160P + 64000 = 240P - 120000
• 240P - 160P = 64000 + 120000
• 80P = 184000
• P = 2300
• Yash's investment = 5P + 2000
• Yash's investment = 5 × 2300 + 2000 = 11500 + 2000 = Rs. 13500

Solution Q.097 **Correct Option: (D)**

• %
• Solution:
• Let the cost price be Rs. 100.
• Marked price = 100 + 50% of 100 = Rs. 150.
• After d% discount, price = 150 × (100 - d)/100.
• After 20% discount, selling price = 150 × (100 - d)/100 × 80/100.
• Profit is 8%, so selling price = Rs. 108.
• Now, 150 × 80/100 = 120.
• So, 120 × (100 - d)/100 = 108.
• (100 - d) = 108 × 100 / 120 = 90.
• d = 100 - 90 = 10.
• Therefore, the first discount is 10%.

Solution Q.098 **Correct Option: (C)**

• Rs. 24000
• Solution:
• Let Kabir's investment be Rs. x.
• Sahil's investment = Rs. (54000 - x).
• Profit ratio is based on investment × time.
• Kabir's profit share : Sahil's profit share = 9x : 12(54000 - x).
• Given ratio = 3 : 5.
• So, 9x : 12(54000 - x) = 3 : 5.
• Using cross multiplication:
• 9x × 5 = 3 × 12(54000 - x)
• 45x = 36(54000 - x)
• 45x = 1944000 - 36x
• 81x = 1944000
• x = 1944000 / 81 = 24000.
• Kabir's investment = Rs. 24000.

Solution Q.099 **Correct Option: (E)**

• Rs. 240
• Solution:
• Profit share is based on investment × time.
• Tushar's investment for 12 months = 2400 × 12 = 28800
• Naman's investment for 12 months = 1600 × 12 = 19200
• Parth joined after 4 months, so he invested for 8 months.
• Parth's investment for 8 months = 3600 × 8 = 28800
• Profit ratio = 28800 : 19200 : 28800
• Profit ratio = 3 : 2 : 3
• Parth's 3 parts = Rs. 720
• 1 part = 720 / 3 = Rs. 240
• Difference between Tushar's and Naman's shares = 3 parts - 2 parts = 1 part
• Required difference = Rs. 240

Solution Q.100 **Correct Option: (A)**

• Rs. 41400
• Solution:
• Total cost price = Purchase price + Repair cost
• Total cost price = 34800 + 1200 = Rs. 36000
• Profit = 15% of Rs. 36000
• Profit = 36000 × 15/100 = Rs. 5400
• Selling price = Cost price + Profit
• Selling price = 36000 + 5400 = Rs. 41400
• So, X = 41400

Solution Q.101 **Correct Option: (D)**

• Profit ratio is based on investment × time.
• Abhinav's investment for the first 4 months = 24000 × 4 = 96000
• After withdrawal, Abhinav's investment = 24000 - 6000 = 18000
• Abhinav's investment for the next 8 months = 18000 × 8 = 144000
• Total capital-months of Abhinav = 96000 + 144000 = 240000
• Gaurav's investment for 12 months = X × 12 = 12X
• Chirag's investment for the first 4 months = 36000 × 4 = 144000
• After withdrawal, Chirag's investment = 36000 - 12000 = 24000
• Chirag's investment for the next 8 months = 24000 × 8 = 192000
• Total capital-months of Chirag = 144000 + 192000 = 336000
• Gaurav's profit share = 21000 out of 84000
• So, Gaurav's share = 21000/84000 = 1/4
• Now, 12X/(240000 + 12X + 336000) = 1/4
• 12X/(576000 + 12X) = 1/4
• 48X = 576000 + 12X
• 36X = 576000
• X = 16000

Solution Q.102 **Correct Option: (A)**

• Rs. 585
• Solution:
• Cost price = Rs. 360
• Profit = 25%
• Selling price = 360 × 125/100 = Rs. 450
• Discount is 30% of selling price.
• Discount = 450 × 30/100 = Rs. 135
• Marked price = Selling price + Discount
• Marked price = 450 + 135 = Rs. 585

Solution Q.103 **Correct Option: (B)**

• Profit is distributed in the ratio of investment × time.
• Akash invested Rs. 800 for 12 months.
• Akash's capital-months = 800 × 12 = 9600.
• Ayush invested Rs. 1200 for the first 4 months.
• After 4 months, Ayush's investment became Rs. (1200 + X) for the remaining 8
• months.
• Ayush's capital-months = 1200 × 4 + (1200 + X) × 8.
• Ayush's capital-months = 4800 + 9600 + 8X = 14400 + 8X.
• Akash's profit share = Rs. 2400 out of Rs. 7200.
• Ayush's profit share = 7200 - 2400 = Rs. 4800.
• So, profit Ratio (Akash : Ayush) = 2400:4800 = 1:2.
• Therefore, capital-month ratio = 1:2.
• 9600 : (14400 + 8X) = 1:2.
• 14400 + 8X = 9600 × 2 = 19200.
• 8X = 19200 - 14400 = 4800.
• X = 4800 / 8 = 600.

Solution Q.104 **Correct Option: (A)**

• Rs. 2160
• Solution:
• Let cost price of the article = x units
• Marked price = 150% of x = 1.5x.
• Discount = 40%.
• Selling price after discount = 60% of marked price.
• Selling price after discount = 60% of 1.5x = 0.9x.
• This means the shopkeeper sold it at 90% of cost price.
• So, loss = 10% of cost price.
• Given loss = Rs. 360.
• 10% of x = 360.
• x = 360 × 100 / 10 = Rs. 3600.
• Marked price = 1.5 × 3600 = Rs. 5400.
• Selling price after discount = 0.9 × 3600 = Rs. 3240.
• Required difference = 5400 - 3240 = Rs. 2160.

Solution Q.105 **Correct Option: (B)**

• Rs. 4400
• Solution:
• Profit is divided in the ratio of investment × time.
• Yash invested P for the first 3 months.
• After 3 months, he increased his initial investment by 50%.
• So, Yash's new investment = P + 50% of P = 1.5P.
• Yash's total capital-time = P × 3 + 1.5P × 9 = 3P + 13.5P = 16.5P.
• Sohan joined after 3 months, so he invested for 9 months.
• Sohan's capital-time = 3P × 9 = 27P.
• Ratio of profit share of Yash to Sohan = 16.5P : 27P = 16.5 : 27 = 33 : 54 = 11 : 18.
• Difference in ratio parts = 18 - 11 = 7 parts.
• Given difference = Rs. 2800.
• So, 7 parts = 2800.
• 1 part = 2800 / 7 = 400.
• Yash's share = 11 × 400 = Rs. 4400.
• Therefore, Yash's profit share is Rs. 4400.

Solution Q.106 **Correct Option: (E)**

• Rs. 160
• Solution:
• Let cost price = Rs. 1000.
• Marked price is 80% above cost price.
• Marked price = 1000 + 80% of 1000 = 1000 + 800 = Rs. 1800.
• Let profit earned be Rs. x.
• Given, discount allowed : profit earned = 4:1.
• So, discount allowed = 4x.
• Selling price = cost price + profit = 1000 + x.
• Discount = marked price - selling price.
• So, 4x = 1800 - (1000 + x).
• 4x = 800 - x.
• 5x = 800.
• x = 160.
• Profit earned by the trader = Rs. 160.

Solution Q.107 **Correct Option: (A)**

• Rs. 2800
• Solution:
• Selling price of article P = Rs. 1440.
• Loss on article P = 10%.
• So, selling price = 90% of cost price.
• Cost price of P = 1440 × 100 / 90 = Rs. 1600.
• Selling price of article Q = Rs. 1440.
• Profit on article Q = 20%.
• So, selling price = 120% of cost price.
• Cost price of Q = 1440 × 100 / 120 = Rs. 1200.
• Total cost price = 1600 + 1200 = Rs. 2800.

Solution Q.108 **Correct Option: (D)**

• Rs. 3500
• Solution:
• Since all four friends invested for the same period, profit will be divided in the ratio of
• their investments.
• Investment ratio = 24000 : 30000 : 45000 : 51000.
• Total investment = 24000 + 30000 + 45000 + 51000 = 150000.
• Difference between investments of Nilesh and Ayush = 51000 - 30000 = 21000.
• So, difference between their profit shares = 25000 × 21000 / 150000.
• Difference = 25000 × 21 / 150 = 3500.

Solution Q.109 **Correct Option: (E)**

• Profit share is based on investment × time.
• Sameer's investment = Rs. (X + 800), and he invested for 8 months.
• Sameer's capital-months = (X + 800) × 8 = 8X + 6400
• Rajesh's investment = Rs. X, and he invested for 12 months.
• Rajesh's capital-months = 12X
• Nitesh's investment = Rs. 2400, and he invested for 8 months.
• Nitesh's capital-months = 2400 × 8 = 19200
• Total profit = Rs. 2700 and Rajesh's profit = Rs. 900.
• So, Rajesh's profit share = 900/2700 = 1/3
• Therefore, Rajesh's share of the total capital-months is also 1/3.
• 12X/(8X + 6400 + 12X + 19200) = 1/3
• 12X/(20X + 25600) = 1/3
• 36X = 20X + 25600
• 16X = 25600
• X = 1600
• Sameer's initial investment = X + 800 = 1600 + 800 = Rs. 2400

Solution Q.110 **Correct Option: (C)**

• months
• Solution:
• Investment Ratio (Rahul : Siddharth) = 5:6.
• Let Rahul invest for x months.
• Then Siddharth invests for (x + 4) months.
• Profit depends on capital × time.
• So, profit Ratio (Rahul : Siddharth) = 5x : 6(x + 4).
• Rahul's profit share = Rs. 1500.
• Total profit = Rs. 4500.
• So, Siddharth's profit share = 4500 - 1500 = Rs. 3000.
• Profit Ratio (Rahul : Siddharth) = 1500 : 3000 = 1 : 2.
• So, 5x : 6(x + 4) = 1 : 2.
• 10x = 6x + 24.
• 4x = 24.
• x = 6.
• Siddharth's time = x + 4 = 6 + 4 = 10 months.

Solution Q.111 **Correct Option: (E)**

• Rs. 100
• Solution:
• Let cost price of article A = x units
• Then the cost price of article B = Rs. (700 - x).
• Article A is marked 20% above cost price.
• Marked price of A = 120% of x = 6x/5.
• Article B is marked 50% above cost price.
• Marked price of B = 150% of (700 - x) = 3(700 - x)/2.
• Marked price of B is Rs. 240 more than marked price of A.
• So, 3(700 - x)/2 - 6x/5 = 240.
• Taking LCM 10:
• 15(700 - x) - 12x = 2400.
• 10500 - 15x - 12x = 2400.
• 10500 - 27x = 2400.
• 27x = 8100.
• x = 300.
• So, cost price of article A = Rs. 300.
• Cost price of article B = 700 - 300 = Rs. 400.
• Required difference = 400 - 300 = Rs. 100.

Solution Q.112 **Correct Option: (B)**

• Let the cost price of the grinder = Rs. x
• Cost price of the mixer = Rs. (x + 200)
• Selling price of mixer at 20% profit = 120% of (x + 200)
• Selling price of mixer = 120/100 × (x + 200) = 1.2x + 240
• Selling price of grinder at 50% profit = 150% of x
• Selling price of grinder = 1.5x
• Given difference in selling prices = Rs. 60
• 1.2x + 240 - 1.5x = 60
• 240 - 60 = 1.5x - 1.2x
• 180 = 0.3x
• x = 600
• Cost price of mixer = x + 200 = 600 + 200 = Rs. 800

Solution Q.113 **Correct Option: (C)**

• Rs. 4600
• Solution:
• In partnership, profit is proportional to investment × time.
• A's investment = x and time = 6 months.
• So A's money-months = 6x.
• B's investment = x + 400 and time = 9 months.
• So B's money-months = 9(x + 400).
• Given A's profit share = Rs. 18000 and B's profit share = Rs. 30000.
• So profit Ratio (A : B) = 18000 : 30000 = 3 : 5.
• Therefore,
• 6x : 9(x + 400) = 3 : 5
• Cross-multiply:
• 5 × 6x = 3 × 9(x + 400)
• 30x = 27x + 10800
• 3x = 10800
• x = 3600.
• C's investment = x + 1000 = 3600 + 1000 = Rs. 4600.

Solution Q.114 **Correct Option: (A)**

• Rs. 27000
• Solution:
• Profit share is divided in the ratio of investment × time.
• Ramesh invested Rs. 15000 for 12 months.
• Ramesh's capital-time = 15000 × 12 = 180000.
• Ketan invested Rs. 20000 for 9 months.
• Ketan's capital-time = 20000 × 9 = 180000.
• Lalit invested Rs. 24000 for 5 months.
• Lalit's capital-time = 24000 × 5 = 120000.
• Ratio of profit shares = 180000 : 180000 : 120000.
• After dividing by 60000, ratio = 3 : 3 : 2.
• Total parts = 3 + 3 + 2 = 8.
• Ketan's share = 3/8 of 72000.
• Ketan's share = 72000 × 3 / 8 = 27000.

Solution Q.115 **Correct Option: (B)**

• Rs. 675
• Solution:
• Let the cost price of item Q = Rs. x.
• Cost price of item P = Rs. (x + 300).
• Selling price of P = 75% of (x + 300) = 3/4 × (x + 300).
• Selling price of Q = 150% of x = 3/2 × x.
• Given Ratio (selling price of P : Q) = 5 : 6.
• So, 3/4 × (x + 300) : 3/2 × x = 5 : 6.
• 6 × 3/4 × (x + 300) = 5 × 3/2 × x.
• 18/4 × (x + 300) = 15x/2.
• 9/2 × (x + 300) = 15x/2.
• 9x + 2700 = 15x.
• 6x = 2700.
• x = 450.
• Selling price of item Q = 150% of 450 = 450 × 150 / 100 = Rs. 675.

Solution Q.116 **Correct Option: (D)**

• Profit shares are proportional to the respective capital-time products.
• Rahul's capital-time value = x × 12 = 12x.
• Aman invested (x - 1800) for the first 4 months.
• After 4 months, Aman's investment increased by 25%.
• New investment of Aman = 125% of (x - 1800) = 5/4 × (x - 1800).
• Aman's capital-time value = (x - 1800) × 4 + 5/4 × (x - 1800) × 8.
• Aman's capital-time value = 4(x - 1800) + 10(x - 1800) = 14(x - 1800).
• Akash's capital-time value = 7200 × 8 = 57600.
• Total capital-time value = 12x + 14(x - 1800) + 57600.
• Total capital-time value = 12x + 14x - 25200 + 57600 = 26x + 32400.
• Rahul's profit share = Rs. 3600 out of Rs. 9150.
• So, Rahul's ratio part = 3600 : 9150 = 24 : 61.
• 12x/(26x + 32400) = 24/61.
• 61 × 12x = 24 × (26x + 32400).
• 732x = 624x + 777600.
• 108x = 777600.
• x = 7200.

Solution Q.117 **Correct Option: (B)**

• Abhinav invested Rs. 2400 for 12 months.
• Bharat invested Rs. 3600 for (12 - Y) months.
• Profit shares are proportional to the respective capital-time products.
• Abhinav : Bharat = 2400 × 12 : 3600 × (12 - Y)
• = 28800 : 3600 × (12 - Y)
• = 8 : (12 - Y)
• Given ratio = 8 : 7
• So, 12 - Y = 7
• Y = 5
• Required value = 4Y + 6
• = 4 × 5 + 6
• = 26

Solution Q.118 **Correct Option: (D)**

• Rs. 540
• Solution:
• Let the cost price be Rs. 100.
• Marked price is 200% above the cost price.
• So, marked price = 100 + 200 = Rs. 300.
• Discount = x% and profit = x%.
• Selling price according to profit = Rs. (100 + x).
• Selling price according to discount = 300 × (100 - x)/100
• = 300 - 3x.
• Now,
• 100 + x = 300 - 3x
• 4x = 200
• x = 50
• So, selling price = 150% of cost price.
• Given selling price = Rs. 540.
• Cost price = 540 × 100/150 = Rs. 360.
• Marked price = 3 × 360 = Rs. 1080.
• Discount allowed = 1080 - 540 = Rs. 540.

Solution Q.119 **Correct Option: (E)**

• : 2
• Solution:
• Total profit = Rs. 64000.
• Suresh's profit share = Rs. 16000.
• Suresh's profit share = 64000 - 16000 = Rs. 48000.
• Profit Ratio (Suresh : Suresh) = 16000 : 48000 = 1 : 3.
• Time Ratio (Suresh : Suresh) = 5 : 6.
• Profit ratio = Investment ratio × Time ratio.
• Investment Ratio (Suresh : Suresh) = 1/5 : 3/6.
• Investment Ratio (Suresh : Suresh) = 1/5 : 1/2.
• Multiplying by 10, investment ratio = 2 : 5.
• Required ratio of Suresh's investment to Suresh's investment = 5 : 2.

Solution Q.120 **Correct Option: (B)**

• Rs. 1600 loss
• Solution:
• Cost price of article A = Rs. 56000.
• Marked price of article A = 56000 × 9/8 = Rs. 63000.
• Selling price of article A after 20% discount = 63000 × 80/100 = Rs. 50400.
• Loss on article A = 56000 - 50400 = Rs. 5600.
• Cost price of article B = Rs. 64000.
• Marked price of article B = 64000 × 5/4 = Rs. 80000.
• Selling price of article B after 15% discount = 80000 × 85/100 = Rs. 68000.
• Profit on article B = 68000 - 64000 = Rs. 4000.
• Overall result = Loss of Rs. 5600 - Profit of Rs. 4000 = Rs. 1600 loss.
• Therefore, Ravi incurs an overall loss of Rs. 1600.

Solution Q.121 **Correct Option: (E)**

• Let the cost price of the jacket be Rs. 100.
• Marked price is 60% above cost price.
• Marked price = Rs. 160
• The shopkeeper incurred a loss of 34%.
• Selling price = 100 - 34 = Rs. 66
• Two successive discounts are x% and (x + 20)%.
• After first discount, price = 160 × (100 - x)/100
• After second discount, price = 160 × (100 - x)/100 × (100 - x - 20)/100
• So:
• 160 × (100 - x) × (80 - x)/10000 = 66
• (100 - x)(80 - x) = 66 × 10000/160
• (100 - x)(80 - x) = 4125
• 8000 - 180x + x^2 = 4125
• x^2 - 180x + 3875 = 0
• x^2 - 25x - 155x + 3875 = 0
• x(x - 25) - 155(x - 25) = 0
• (x - 25)(x - 155) = 0
• x = 25 or x = 155
• Discount cannot be more than 100%, so x = 25.

Solution Q.122 **Correct Option: (D)**

• %
• Solution:
• Profit shares are proportional to capital × time.
• Kritika's investment = Rs. 9,000.
• Kritika's investment period = 4 months.
• Kritika's capital-months = 9,000 × 4 = 36,000.
• Gaurav's investment = Rs. 7,000.
• Gaurav's investment period = 12 months.
• Gaurav's capital-months = 7,000 × 12 = 84,000.
• Vikram joined after 4 + 5 = 9 months and invested for 3 months.
• The profit-sharing ratio of Kritika to Vikram is 3 : 2.
• Their capital-months are therefore also in the ratio 3 : 2.
• Vikram's capital-months = 36,000 × 2/3 = 24,000.
• Combined capital-months of Kritika and Gaurav = 36,000 + 84,000 = 1,20,000.
• Required percentage = 24,000/1,20,000 × 100 = 20%.

Solution Q.123 **Correct Option: (D)**

• Rs. 3200
• Solution:
• Total profit = Rs. 3400.
• Ojas's profit share = Rs. 1800.
• Kritika's profit share = 3400 - 1800 = Rs. 1600.
• So, profit Ratio (Ojas : Kritika) = 1800 : 1600 = 9 : 8.
• Time Ratio (Ojas : Kritika) = 18 : 20 = 9 : 10.
• Profit shares are proportional to the respective capital-time products.
• So, investment ratio = profit ratio/time ratio.
• Investment Ratio (Ojas : Kritika) = 9/9 : 8/10.
• Investment ratio = 1 : 4/5.
• Multiplying both terms by 5:
• Investment ratio = 5 : 4.
• Total investment = Rs. 7200.
• Total ratio parts = 5 + 4 = 9.
• Kritika's investment = 7200 × 4/9.
• Kritika's investment = Rs. 3200.

Solution Q.124 **Correct Option: (A)**

• Rs. 1050
• Solution:
• Selling price at 15% loss = Rs. 765.
• At 15% loss, selling price = 85% of cost price.
• So,
• 85% of cost price = 765
• Cost price = 765 × 100/85
• Cost price = Rs. 900.
• Now, profit = 1/6 of cost price.
• Profit = 900 × 1/6 = Rs. 150.
• New selling price = cost price + profit.
• New selling price = 900 + 150 = Rs. 1050.

Solution Q.125 **Correct Option: (D)**

• Rs. 2,800
• Solution:
• Let the cost price of the jacket = Rs. x.
• Marked price = 140% of x = 140x/100.
• Discount = 25%, so selling price = 75% of marked price.
• Selling price = 140x/100 × 75/100 = 105x/100.
• Given selling price = Rs. 2,940.
• 105x/100 = 2,940.
• x = 2,940 × 100/105 = 2,800.
• So, the cost price of the jacket is Rs. 2,800.

Solution Q.126 **Correct Option: (A)**

• Rs. 24,480
• Solution:
• Let Mayank's investment = Rs. x.
• Hardik invested 20% more than Mayank.
• Hardik's investment = 120x/100 = 6x/5.
• Tushar invested 25% more than Hardik.
• Tushar's investment = 6x/5 × 125/100 = 3x/2.
• Mayank invested for 4 months only.
• Mayank's capital-time value = x × 4 = 4x.
• Hardik invested 6x/5 for first 4 months.
• After that, Hardik withdrew 25%, so his remaining investment = 75% of 6x/5 = 9x/10.
• Hardik invested this amount for next 4 months.
• Hardik's capital-time value = 6x/5 × 4 + 9x/10 × 4 = 24x/5 + 18x/5 = 42x/5.
• Tushar invested 3x/2 for first 4 months.
• After that, Tushar increased his investment by 20%.
• New investment of Tushar = 3x/2 × 120/100 = 9x/5.
• Tushar invested this amount for next 8 months.
• Tushar's capital-time value = 3x/2 × 4 + 9x/5 × 8 = 6x + 72x/5 = 102x/5.
• Combined capital-time value of Mayank and Hardik = 4x + 42x/5 = 62x/5.
• Difference between Tushar's value and combined value of Mayank and Hardik = 102x/5
• - 62x/5 = 40x/5 = 8x.
• Given profit difference = Rs. 9,600.
• So, Tushar's profit share = 9,600 × (102x/5)/(8x).
• Tushar's profit share = 9,600 × 102/40 = Rs. 24,480.

Solution Q.127 **Correct Option: (B)**

• Rs. 300
• Solution:
• Let cost price of each article for Rajesh = x units
• Selling price of each article = Rs. 240
• Profit amount on each article = 240 - x
• Profit percentage on the first article calculated on CP = (240 - x)/x × 100
• Profit percentage on the second article calculated on SP = (240 - x)/240 × 100
• According to the question,
• Profit percentage on CP = 150% of profit percentage on SP
• (240 - x)/x × 100 = 150/100 × (240 - x)/240 × 100
• Since Rajesh earns profit, (240 - x) can be cancelled.
• 1/x = 150/(100 × 240)
• 1/x = 3/480
• x = 160
• So, cost price of each article for Rajesh = Rs. 160
• Profit earned by Rajesh on the first article = 240 - 160 = Rs. 80
• Sameer's profit amount is 25% less than Rajesh's profit amount.
• Sameer's profit amount = 80 × 75/100 = Rs. 60
• Sameer's cost price = Rs. 240
• Sameer's selling price = 240 + 60 = Rs. 300
• Therefore, the value in the blank is Rs. 300.

Solution Q.128 **Correct Option: (D)**

• Investment Ratio (Ayush : Sameer) = 5 : 7
• Total profit = Rs. 9800
• Ayush's profit share = Rs. 3500
• Sameer's profit share = 9800 - 3500 = Rs. 6300
• Profit Ratio (Ayush : Sameer) = 3500 : 6300 = 5 : 9
• Let the investment periods of Ayush and Sameer be a and b respectively.
• Profit shares are proportional to the respective capital-time products.
• So, 5 × a : 7 × b = 5 : 9
• 5a/(7b) = 5/9
• 45a = 35b
• b/a = 9/7
• So, time period ratio of Sameer to Ayush = 9 : 7
• Here, x = 9 and y = 7
• x + y = 9 + 7 = 16

Solution Q.129 **Correct Option: (C)**

• Rs. 700
• Solution:
• Let the cost price of the article = Rs. 100x
• Marked price = 160% of 100x = Rs. 160x
• Selling price after 25% discount = 75% of 160x = Rs. 120x
• Selling price after 10% discount = 90% of 160x = Rs. 144x
• Difference between both selling prices = 144x - 120x = 24x
• According to the question,
• 24x = 168
• x = 7
• Cost price = 100x = 100 × 7 = Rs. 700

CHAPTER 03: SIMPLE & COMPOUND INTEREST (SI & CI)

40 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Simple Interest (SI) = (P × R × T) / 100 | Amount = P + SI
- Compound Interest Amount: A = P(1 + R/100)^T
- Difference between CI and SI for 2 Years: Diff = P × (R / 100)^2
- Difference for 3 Years: Diff = P × (R/100)^2 × (3 + R/100)

Q.130 Simple

Prateek invested Rs. x at 10% per annum compound interest for 2 years, while Suresh invested Rs. 3x at 20% per annum compound interest for 2 years. If the sum of the amounts received by both of them was Rs. 16590, find the value of 5x. Compound interest is calculated with annual compounding.

(A) 12000 (B) 18000
(C) 16500 (D) 15000
(E) 21000

Q.131 Simple

A person invested Rs. P at simple interest at R% per annum for 2 years and received Rs. 960 as interest. If he invested the same sum at 6% per annum simple interest for 5 years and received Rs. 5200 as the amount, find R.

(A) 8 (B) 10
(C) 15 (D) 18
(E) 12

Q.132 Simple

A person invests Rs. 34,000 in two schemes, Scheme P and Scheme Q. Scheme P and Scheme Q offer compound interest at 25% p.a. and 30% p.a. respectively for 2 years. If the ratio of the amounts received from Scheme P and Scheme Q after 2 years is 1250:1521, find the amount invested in Scheme Q. Compound interest is calculated with annual compounding.

(A) Rs. 12,000 (B) Rs. 16,000
(C) Rs. 20,000 (D) Rs. 14,000
(E) Rs. 18,000

Q.133 Simple

Raghav invested Rs. P at the rate of 6% per annum for 3 years on simple interest. Mukesh invested Rs. (P + 2000) at the rate of 8% per annum for 4 years on simple interest. If the difference between their interests is Rs. 2040, find the value of P.

(A) Rs. 8000 (B) Rs. 12000
(C) Rs. 15000 (D) Rs. 9000
(E) Rs. 10000

Q.134 Simple

A person invested Rs. 16000 in Scheme A at (x + 3)% per annum and Rs. 20000 in Scheme B at (x - 1)% per annum. Both schemes offer simple interest. If the interest received from both schemes after 4 years is equal, find the value of x.

(A) 15 (B) 17
(C) 19 (D) 21
(E) 13

Q.135 Simple

A man borrowed Rs. 10000 at 10% compound interest per annum for 2 years. Immediately after borrowing, he added Rs. (2000 + x) from his own money and lent the entire amount at 15% simple interest per annum for 2 years. If he earned Rs. 3000 more interest than the interest he paid, find x. Compound interest is calculated with annual compounding.

(A) 3000 (B) 4000
(C) 4500 (D) 5000
(E) 6000

Q.136 Simple

The difference between compound interest and simple interest for 3 years on Rs. 40000 is Rs. 2720. The rates of interest for the first and second years are 10% and 15% respectively for both compound interest and simple interest. The same rate applies to both types of interest in the third year. Find this rate. Compound interest is calculated with annual compounding.

(A) 12% (B) 18%
(C) 20% (D) 15%
(E) 25%

Q.137 Simple

A man invests equal amounts at the same rate of simple interest for X years and (X + 6) years. If the ratio of interest received by him is 2 : 5 respectively, find the value of X + 6.

(A) 8 (B) 10
(C) 12 (D) 14
(E) 6

Q.138 Simple

A person invests Rs. 20,000 in Scheme A at r% per annum simple interest for 3 years. The interest earned from Scheme A is invested in Scheme B at 10% per annum compound interest for 2 years. If the amount received from Scheme B is Rs. 7,260, find the value of r. Compound interest is calculated with annual compounding.

(A) 8 (B) 12
(C) 15 (D) 9
(E) 10

Q.139 Simple

Two persons invested equal amounts. The first person invested the whole amount at 20% per annum compound interest for 2 years. The second person invested half of his amount at R% per annum simple interest for 2 years and the remaining half at 10% per annum simple interest for 2 years. If the total interest received by both persons is equal, find R. Compound interest is calculated with annual compounding.

(A) 30 (B) 28
(C) 34 (D) 36
(E) 32

Q.140 Simple

A person invests Rs. 6,000 in a scheme for 7 years. The scheme offers simple interest at 6% per annum for the first 2 years, 9% per annum for the next 3 years and 12% per annum for the period beyond 5 years. Find the total interest earned.

(A) Rs. 3420 (B) Rs. 3960
(C) Rs. 3600 (D) Rs. 3780
(E) Rs. 4080

Q.141 Simple

Arpit invested Rs. 500 for 10 months and Chirag invested Rs. 750 for 8 months in a business. The total profit earned by them was Rs. 2200. Chirag invested his profit at 10% per annum compounded annually for 2 years. Find the difference between the amount reinvested by Chirag from his profit and the compound interest received on that investment.

(A) Rs. 840 (B) Rs. 960
(C) Rs. 900 (D) Rs. 948
(E) Rs. 1050

Q.142 Simple

Ritika borrowed Rs. X from Meena at 20% per annum simple interest for 2 years. She also borrowed Rs. (X + 250) from Divya at 20% per annum compound interest for 2 years. If the compound interest paid to Divya is Rs. 410 more than the simple interest paid to Meena, find the value of X. Compound interest is calculated with annual compounding.

(A) Rs. 6000 (B) Rs. 6500
(C) Rs. 7500 (D) Rs. 8000
(E) Rs. 9000

Q.143 Simple

A sum of money is lent at a certain rate of simple interest. The interest received after 3 years is Rs. X, and the interest received after 7 years is Rs. (X + 2400). Find the interest for one year.

(A) Rs. 600 (B) Rs. 500
(C) Rs. 800 (D) Rs. 400
(E) Rs. 700

Q.144 Simple

The difference between compound interest and simple interest received in 3 years on a sum of Rs. 8000 is Rs. 720. If the rates of interest for the 1st and 2nd years are 15% and 25% respectively for both compound interest and simple interest, find the rate for the 3rd year, which is also the same for both types of interest. Compound interest is calculated with annual compounding.

(A) 10% (B) 15%
(C) 12% (D) 18%
(E) 20%

Q.145 Simple

A sum of Rs. x invested at r% per annum simple interest for 10 years becomes 4 times itself. Another sum of Rs. (x + 360), invested at (r + 20)% per annum compound interest compounded annually for 2 years, becomes Rs. 10125. Find the value of x.

(A) 3840 (B) 3960
(C) 4260 (D) 4500
(E) 4140

Q.146 Simple

Pallavi invested Rs. 16000 at R% per annum compound interest, compounded annually. If the compound interest earned in the second year is Rs. 5000 and in the third year is Rs. 6250, find the annual rate of interest.

(A) 20% (B) 25%
(C) 30% (D) 15%
(E) 22%

Q.147 Simple

Ramesh invested Rs. X in a scheme that offered compound interest at 25% per annum, compounded half-yearly, for 1 year. After 1 year, he received Rs. 850 as interest. If he invests Rs. (X − 700) in another bank at 8% simple interest per annum for 3 years, then find the total amount received from the bank.

(A) 2800 (B) 3100
(C) 3400 (D) 3600
(E) 3900

Q.148 Simple

A person invested a certain sum at simple interest. In 4 years, the interest received is 32% of the principal. If the same sum is invested for 7 years at the same rate, the interest becomes Rs. 6160. Find the principal.

(A) Rs. 9000 (B) Rs. 12500
(C) Rs. 10000 (D) Rs. 11000
(E) Rs. 14000

Q.149 Simple

Arpit invested Rs. 8,000 at 15% p.a. compound interest, compounded annually, for two years. Nilesh invested Rs. 9,000 at 12% p.a. simple interest for three years. Find the total interest earned by both of them together.

(A) Rs. 5,620 (B) Rs. 5,820
(C) Rs. 5,920 (D) Rs. 5,520
(E) Rs. 6,020

Q.150 Simple

Sandeep invested Rs. X at simple interest at the rate of 15% p.a. for 3 years and received Rs. 2700 as interest. If he invests the same amount at compound interest at the rate of 25% p.a. for 2 years, then find the amount received by him. Compound interest is calculated with annual compounding.

(A) Rs. 8400 (B) Rs. 9000
(C) Rs. 9375 (D) Rs. 9600
(E) Rs. 10000

Q.151 Simple

A sum of Rs. P is invested at compound interest at 15% per annum for 2 years, and the amount received is Rs. 10,580. If Rs. (3/2)P is invested at simple interest at 12% per annum for 4 years, find the simple interest received. Compound interest is calculated with annual compounding.

(A) 4800 (B) 5760
(C) 6400 (D) 5200
(E) 6000

Q.152 Simple

Rahul invested Rs. 8000 each in two different schemes. The first scheme offers simple interest at R% p.a. for 4 years, while the second scheme offers simple interest at (R − 4)% p.a. for 5 years. If the interest from the first scheme is Rs. 640 more than that from the second scheme, find the value of (R + 6).

(A) 14 (B) 16
(C) 20 (D) 22
(E) 18

Q.153 Simple

Rajesh invested Rs. X at simple interest for 4 years at R% p.a., and the interest earned became twice the principal. If Rs. (X − 600) is invested at simple interest for 3 years at (R − 35)% p.a. and the interest earned is Rs. 2520, then find the value of X.

(A) Rs. 5600 (B) Rs. 6200
(C) Rs. 6800 (D) Rs. 7200
(E) Rs. 7800

Q.154 Simple

Shreya invests the same principal at the same annual rate of simple interest for X years and (X + 6) years. The amounts of simple interest received are in the ratio 2:5, respectively. Find X.

(A) 3 (B) 5
(C) 6 (D) 4
(E) 8

Q.155 Simple

Ayush invests Rs. (X + 750) in Scheme A and Rs. X in Scheme B. Scheme A offers compound interest at 20% p.a. for 2 years, while Scheme B offers simple interest at 15% p.a. for 4 years. If the interest received from Scheme B is Rs. 1190 more than that from Scheme A, find the value of X. Compound interest is calculated with annual compounding.

(A) 8500 (B) 10250
(C) 9000 (D) 9500
(E) 11000

Q.156 Simple

A man invested Rs. X in Scheme A at 25% per annum compound interest for two years. The total amount received from Scheme A was then invested in Scheme B at 18% per annum simple interest for two years. If the interest received from Scheme B is Rs. 1800, then find the total amount invested in Scheme B. Compound interest is calculated with annual compounding.

(A) Rs. 5000 (B) Rs. 4200
(C) Rs. 5600 (D) Rs. 4800
(E) Rs. 6000

Q.157 Simple

A person invested Rs. 80000 in two schemes A and B. Scheme A offers simple interest at 20% per annum for 2 years, while Scheme B offers compound interest at 15% per annum for 2 years. If the interest received from Scheme B is Rs. 4125 more than the interest received from Scheme A, find the amount invested in Scheme A. Compound interest is calculated with annual compounding.

(A) 32000 (B) 28000
(C) 35000 (D) 30000
(E) 25000

Q.158 Simple

Priya has some money. She invests 25% of it in Scheme A at 8% per annum simple interest for 3 years, 35% in Scheme B at 10% per annum simple interest for 4 years, and the remaining money in Scheme C at 12% per annum simple interest for 2 years. If the total amount received from all three schemes is Rs. 12960, find the difference between the amounts invested in Scheme B and Scheme A.

(A) Rs. 600 (B) Rs. 800
(C) Rs. 1200 (D) Rs. 1500
(E) Rs. 1000

Q.159 Simple

Priya invests Rs. 7500 and Divya invests Rs. 12500 at the same rate of simple interest per annum. If after 5 years Divya receives Rs. 1500 more interest than Priya, find the rate of interest per annum.

(A) 5% p.a. (B) 6% p.a.
(C) 8% p.a. (D) 4% p.a.
(E) 7% p.a.

Q.160 Simple

Pranay invested Rs. 12,000 in Scheme A for 3 years at 12.5% per annum simple interest. The total amount received from Scheme A was then invested in Scheme B for 2 years at 10% per annum compound interest, compounded annually. Find the interest received from Scheme B.

(A) Rs. 3,300 (B) Rs. 3,565
(C) Rs. 3,630 (D) Rs. 3,345
(E) Rs. 3,465

Q.161 Simple

Gaurav invested Rs. x in scheme A, which offers simple interest at the rate of 16% per annum for 2 years. He also invested Rs. 12000 in scheme B, which offers compound interest at the rate of 10% per annum compounded annually for 2 years. If the amounts received from both schemes are equal, find the value of x.

(A) 10500
(B) 12500
(C) 11800
(D) 12000
(E) 11000

Q.162 Simple

A man invested Rs. 4000 in two schemes P and Q in the ratio 3 : 5. Scheme P offers simple interest at X% p.a. for 4 years, while Scheme Q offers simple interest at (X + 6)% p.a. for 2 years. If the interest received from Scheme Q is Rs. 10X more than that from Scheme P, then find the value of X.

(A) 10
(B) 18
(C) 12
(D) 15
(E) 20

Q.163 Simple

Parth invested Rs. X in a scheme that gives simple interest at 15% p.a. for 2 years, and Sameer invested Rs. Y in the same scheme for the same two-year period. If the interest received by Parth is Rs. 900 more than that received by Sameer, and X is 40% more than Y, find the total amount invested by Parth and Sameer together.

(A) Rs. 18000
(B) Rs. 15000
(C) Rs. 21000
(D) Rs. 20000
(E) Rs. 16500

Q.164 Simple

Harshit borrowed Rs. 30000 on compound interest at the rate of 10% per annum. At the end of 2 years, he again borrowed Rs. X more at the same rate of interest. He made no earlier repayments. If he settled both loans by paying a total amount of Rs. 42680 at the end of the third year, find the value of X. Compound interest is calculated with annual compounding.

(A) Rs. 3000
(B) Rs. 2000
(C) Rs. 2500
(D) Rs. 1800
(E) Rs. 3500

Q.165 Simple

Raghav invested Rs. P in scheme M, which offers 10% per annum compound interest for two years. He then invested the amount received from scheme M in scheme N, which offers 12.5% per annum simple interest for three years. If he received Rs. 1,815 as interest from scheme N, then find the value of P. Compound interest is calculated with annual compounding.

(A) 4000
(B) 3500
(C) 4500
(D) 3000
(E) 5000

Q.166 Simple

Rajesh invested Rs. 21600 in a bank at the rate of (50/3)% per annum compound interest, compounded annually. Find the interest earned by him after three years.

(A) Rs. 12000
(B) Rs. 13500
(C) Rs. 12700
(D) Rs. 14400
(E) Rs. 11500

Q.167 Simple

Ritesh invested Rs. X for three years at compound interest at the rate of 25% per annum. In the first and third years, interest was compounded annually, while in the second year, interest was compounded half-yearly. If the interest earned only in the third year is Rs. 4050, then find X.

(A) Rs. 9600
(B) Rs. 12000
(C) Rs. 10240
(D) Rs. 10800
(E) Rs. 15000

Q.168 Simple

A, B and C together invested Rs. 1,25,000 in a business. B invested 25% more than A, while C invested Rs. 15,000 less than B. After one year, B invested his profit share at 12% p.a. simple interest for two years and received total interest of Rs. 9,600. Find the total profit of the business for the first year. Assume all three partners kept their stated investments in the business for the entire first year.

(A) Rs. 80,000
(B) Rs. 1,20,000
(C) Rs. 90,000
(D) Rs. 1,10,000
(E) Rs. 1,00,000

Q.169 Simple

Mayank and Mukesh have equal amounts. Mayank invested his whole amount at 20% p.a. compound interest for two years. Mukesh invested 1/4 of his amount at R% p.a. simple interest for two years and the remaining amount at 10% p.a. simple interest for two years. If the total interest received by both is equal, then find the value of R. Compound interest is calculated with annual compounding.

(A) 64
(B) 52
(C) 58
(D) 48
(E) 60

📖 Chapter 03 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.130 — **Correct Option: (D)**

• Prateek invested Rs. x at 10% compound interest for 2 years.
• Amount received by Prateek = x × 110% × 110%.
• So, Prateek's amount = 121% of x = 1.21x.
• Suresh invested Rs. 3x at 20% compound interest for 2 years.
• Amount received by Suresh = 3x × 120% × 120%.
• 120% × 120% = 144%.
• So, Suresh's amount = 144% of 3x = 4.32x.
• Total amount received = 1.21x + 4.32x = 5.53x.
• Given total amount = Rs. 16590.
• So, 5.53x = 16590.
• x = 16590 / 5.53.
• x = 3000.
• Required value = 5x = 5 × 3000 = 15000.

Solution Q.131 — **Correct Option: (E)**

• First find the principal from the second condition.
• At 6% per annum for 5 years, simple interest = P × 6 × 5/100.
• Simple interest = 30P/100 = 3P/10.
• Amount = Principal + Interest.
• So, amount = P + 3P/10 = 13P/10.
• Given amount = Rs. 5200.
• 13P/10 = 5200
• P = 5200 × 10/13
• P = 4000.
• Now use the first condition.
• Simple interest = Rs. 960.
• Time = 2 years.
• Rate = R%.
• Simple interest = P × R × T/100.
• 960 = 4000 × R × 2/100
• 960 = 80R
• R = 960/80
• R = 12.
• Therefore, the value of R is 12.

Solution Q.132 — **Correct Option: (E)**

• Rs. 18,000
• Solution:
• Total investment = Rs. 34,000
• For Scheme P, rate = 25% p.a. for 2 years.
• Amount factor for Scheme P = 125/100 × 125/100 = 25/16
• For Scheme Q, rate = 30% p.a. for 2 years.
• Amount factor for Scheme Q = 130/100 × 130/100 = 169/100
• Amount Ratio (Scheme P : Scheme Q) = 1250:1521
• Investment ratio = 1250/(25/16) : 1521/(169/100)
• Investment ratio = 1250 × 16/25 : 1521 × 100/169
• Investment ratio = 800 : 900
• Investment ratio = 8 : 9
• Total ratio parts = 8 + 9 = 17
• Investment in Scheme Q = 34000 × 9/17
• Investment in Scheme Q = 2000 × 9
• Investment in Scheme Q = Rs. 18,000

Solution Q.133 — **Correct Option: (E)**

• Rs. 10000
• Solution:
• Simple interest = Principal × Rate × Time/100
• Raghav's principal = Rs. P
• Rate = 6% per annum
• Time = 3 years
• Raghav's interest = P × 6 × 3/100
• Raghav's interest = 18P/100
• Mukesh's principal = Rs. (P + 2000)
• Rate = 8% per annum
• Time = 4 years
• Mukesh's interest = (P + 2000) × 8 × 4/100
• Mukesh's interest = 32(P + 2000)/100
• Difference between their interests = Rs. 2040
• So, 32(P + 2000)/100 - 18P/100 = 2040
• 32P + 64000 - 18P = 204000
• 14P + 64000 = 204000
• 14P = 204000 - 64000
• 14P = 140000
• P = 140000/14 = 10000
• Therefore, the value of P is Rs. 10000.

Solution Q.134 — **Correct Option: (B)**

• Simple interest = Principal × Rate × Time/100.
• Interest from Scheme A = 16000 × (x + 3) × 4/100.
• Interest from Scheme B = 20000 × (x - 1) × 4/100.
• Both interests are equal.
• So,
• 16000 × (x + 3) × 4/100 = 20000 × (x - 1) × 4/100
• Cancel 4/100 from both sides.
• 16000(x + 3) = 20000(x - 1)
• Divide both sides by 4000.
• 4(x + 3) = 5(x - 1)
• 4x + 12 = 5x - 5
• 5x - 4x = 12 + 5
• x = 17

Solution Q.135 — **Correct Option: (D)**

• First find the compound interest paid by the man.
• Principal borrowed = Rs. 10000.
• Rate = 10% per annum.
• Time = 2 years.
• Amount after 2 years = 10000 × 110/100 × 110/100.
• Amount = Rs. 12100.
• Compound interest paid = 12100 - 10000 = Rs. 2100.
• Now, the man added Rs. (2000 + x) to the borrowed amount.
• Total amount lent = 10000 + 2000 + x = 12000 + x.
• He lent this amount at 15% simple interest for 2 years.
• Simple interest earned = (12000 + x) × 15 × 2/100.
• Simple interest earned = (12000 + x) × 30/100.
• He earned Rs. 3000 more interest than he paid.
• So, simple interest earned = 2100 + 3000 = Rs. 5100.
• Now,
• (12000 + x) × 30/100 = 5100
• 12000 + x = 5100 × 100/30
• 12000 + x = 17000
• x = 17000 - 12000
• x = 5000

Solution Q.136 **Correct Option: (C)**

• %
• Solution:
• Principal = Rs. 40000
• For compound interest, amount after the first year at 10%:
• 40000 × 110/100 = Rs. 44000
• Amount after the second year at 15%:
• 44000 × 115/100 = Rs. 50600
• Compound interest for first two years = 50600 - 40000 = Rs. 10600
• Simple interest for first two years:
• 40000 × (10 + 15)/100 = 40000 × 25/100 = Rs. 10000
• Difference after first two years = 10600 - 10000 = Rs. 600
• Total difference between compound interest and simple interest for 3 years = Rs. 2720
• Remaining difference due to third year = 2720 - 600 = Rs. 2120
• Let the rate of interest for the third year = x%.
• For the third year, compound interest is calculated on Rs. 50600, while simple interest
• is calculated on Rs. 40000.
• Difference in third-year interest = (50600 × x/100) - (40000 × x/100)
• Difference = 10600x/100
• So, 10600x/100 = 2120
• 106x = 2120
• x = 2120/106 = 20

Solution Q.137 **Correct Option: (B)**

• For simple interest, when principal and rate are the same, interest is directly
• proportional to time.
• Time periods are X years and (X + 6) years.
• Ratio of interest = 2 : 5.
• So,
• X/(X + 6) = 2/5
• 5X = 2(X + 6)
• 5X = 2X + 12
• 5X - 2X = 12
• 3X = 12
• X = 12/3
• X = 4
• Required value = X + 6.
• X + 6 = 4 + 6 = 10.

Solution Q.138 **Correct Option: (E)**

• Amount received from Scheme B = Rs. 7,260.
• Scheme B gives compound interest at 10% per annum for 2 years.
• Let the amount invested in Scheme B be Rs. I.
• Amount after 2 years = I × 110/100 × 110/100
• 7260 = I × 11/10 × 11/10
• 7260 = I × 121/100
• I = 7260 × 100/121
• I = 6000
• So, interest earned from Scheme A = Rs. 6000.
• In Scheme A:
• Principal = Rs. 20,000
• Time = 3 years
• Rate = r%
• Simple Interest = Principal × Rate × Time/100
• 6000 = 20000 × r × 3/100
• 6000 = 600r
• r = 6000/600
• r = 10

Solution Q.139 **Correct Option: (C)**

• Let each person invest Rs. 100.
• First person invests at 20% compound interest for 2 years.
• Amount after 2 years = 100 × 120/100 × 120/100
• Amount = Rs. 144
• Compound interest = 144 - 100 = Rs. 44
• So, first person gets 44% interest.
• Second person invests half amount at R% simple interest for 2 years.
• Interest on first half = 50 × R × 2/100 = R
• Second person invests remaining half amount at 10% simple interest for 2 years.
• Interest on second half = 50 × 10 × 2/100 = Rs. 10
• Total interest of second person = R + 10
• Both interests are equal.
• So, R + 10 = 44
• R = 44 - 10
• R = 34

Solution Q.140 **Correct Option: (D)**

• Rs. 3780
• Solution:
• Principal = Rs. 6,000.
• For the first 2 years, rate = 6% per annum.
• Interest for first 2 years = 6000 × 6 × 2/100
• Interest for first 2 years = Rs. 720.
• For the next 3 years, rate = 9% per annum.
• Interest for next 3 years = 6000 × 9 × 3/100
• Interest for next 3 years = Rs. 1620.
• Total period = 7 years.
• Period beyond 5 years = 7 - 5 = 2 years.
• For the period beyond 5 years, rate = 12% per annum.
• Interest for last 2 years = 6000 × 12 × 2/100
• Interest for last 2 years = Rs. 1440.
• Total interest = 720 + 1620 + 1440
• Total interest = Rs. 3780.

Solution Q.141 **Correct Option: (D)**

• Rs. 948
• Solution:
• Profit is divided in the ratio of investment × time.
• Arpit's investment value = 500 × 10 = 5000.
• Chirag's investment value = 750 × 8 = 6000.
• Profit Ratio (Arpit : Chirag) = 5000 : 6000.
• Simplified ratio = 5 : 6.
• Total ratio parts = 5 + 6 = 11.
• Total profit = Rs. 2200.
• Chirag's profit share = 6/11 × 2200.
• Chirag's profit share = Rs. 1200.
• Now Chirag invests Rs. 1200 at 10% per annum compound interest for 2 years.
• Compound interest for 2 years at 10% = Principal × 21/100.
• Compound interest = 1200 × 21/100.
• Compound interest = Rs. 252.
• Required difference = Amount invested - Compound interest.
• Required difference = 1200 - 252 = Rs. 948.

Solution Q.142 **Correct Option: (C)**

• Rs. 7500
• Solution:
• Ritika borrowed Rs. X from Meena at 20% simple interest for 2 years.
• Simple interest = X × 20 × 2/100
• Simple interest = 40X/100
• Simple interest = 2X/5.
• She borrowed Rs. (X + 250) from Divya at 20% compound interest for 2 years.
• At 20% compound interest for 2 years, amount factor = 120/100 × 120/100.
• Amount factor = 6/5 × 6/5 = 36/25.
• Compound interest factor = 36/25 - 1 = 11/25.
• Compound interest = 11/25 × (X + 250).
• Given, compound interest is Rs. 410 more than simple interest.
• So, 11(X + 250)/25 - 2X/5 = 410.
• Make denominator same:
• 11(X + 250)/25 - 10X/25 = 410.
• [11X + 2750 - 10X]/25 = 410.
• (X + 2750)/25 = 410.
• X + 2750 = 410 × 25
• X + 2750 = 10250
• X = 10250 - 2750
• X = 7500.

Solution Q.143 **Correct Option: (A)**

• Rs. 600
• Solution:
• In simple interest, interest for each year remains the same.
• Interest for 3 years = Rs. X
• Interest for 7 years = Rs. (X + 2400)
• Difference in time = 7 - 3 = 4 years
• Difference in interest = 2400
• Interest for 4 years = Rs. 2400
• Interest for 1 year = 2400/4 = Rs. 600

Solution Q.144 **Correct Option: (C)**

• %
• Solution:
• Principal = Rs. 8000
• Rate for 1st year = 15%
• Rate for 2nd year = 25%
• Let the rate for 3rd year be R%.
• Amount after 1st year on compound interest = 8000 × 115/100 = Rs. 9200
• Amount after 2nd year on compound interest = 9200 × 125/100 = Rs. 11500
• Amount after 3rd year on compound interest = 11500 × (100 + R)/100
• Compound interest = 11500 × (100 + R)/100 - 8000
• Compound interest = 11500 + 115R - 8000
• Compound interest = 3500 + 115R
• Simple interest for 3 years = 8000 × (15 + 25 + R)/100
• Simple interest = 8000 × (40 + R)/100
• Simple interest = 3200 + 80R
• Difference between CI and SI = Rs. 720
• So, (3500 + 115R) - (3200 + 80R) = 720
• 300 + 35R = 720
• 35R = 720 - 300
• 35R = 420
• R = 420/35 = 12

Solution Q.145 **Correct Option: (E)**

• Rs. x becomes 4 times itself in 10 years at r% simple interest.
• So, interest = 4x - x = 3x.
• Simple interest = x × r × 10 / 100.
• So, x × r × 10 / 100 = 3x.
• r/10 = 3.
• r = 30, so the simple-interest rate is 30% per annum.
• Now, the compound-interest rate = (r + 20)% = (30 + 20)% = 50% per annum.
• Amount after 2 years = Rs. 10125.
• So, x + 360 = 10125 × 100/150 × 100/150.
• x + 360 = 10125 × 4/9 = 4500.
• x = 4500 - 360 = 4140.

Solution Q.146 **Correct Option: (B)**

• %
• Solution:
• In compound interest, interest of the next year increases by the rate of interest.
• Second-year interest = Rs. 5000.
• Third-year interest = Rs. 6250.
• Increase in interest = 6250 - 5000 = Rs. 1250.
• Rate of increase = 1250 × 100 / 5000 = 25%.
• So, R = 25 and the annual rate is 25%.

Solution Q.147 **Correct Option: (B)**

• Rate is 25% per annum compounded half-yearly.
• So, half-yearly rate = 25% / 2 = 12.5%.
• Time = 1 year, so number of half-years = 2.
• For easy calculation, take principal = X.
• Amount after 1 year = X × 112.5/100 × 112.5/100.
• 112.5/100 = 9/8.
• So, amount = X × 9/8 × 9/8 = 81X/64.
• Compound interest = 81X/64 - X = 17X/64.
• Given interest = Rs. 850.
• So, 17X/64 = 850.
• X = 850 × 64 / 17 = 3200.
• Now, second investment = X - 700 = 3200 - 700 = Rs. 2500.
• Simple interest for 3 years at 8% = 2500 × 8 × 3 / 100 = Rs. 600.
• Total amount = 2500 + 600 = Rs. 3100.

Solution Q.148 **Correct Option: (D)**

• Rs. 11000
• Solution:
• In 4 years, interest is 32% of principal.
• So, rate of simple interest per annum = 32% / 4 = 8% p.a.
• For 7 years, interest = Rs. 6160
• Simple interest = Principal × Rate × Time/100
• 6160 = Principal × 8 × 7/100
• 6160 = Principal × 56/100
• Principal = 6160 × 100/56
• Principal = Rs. 11000

Solution Q.149 **Correct Option: (B)**

• Rs. 5,820
• Solution:
• Compound interest for Arpit:
• Principal = Rs. 8,000, rate = 15%, time = 2 years.
• Amount after 2 years = 8000 × 115/100 × 115/100
• Amount = 8000 × 23/20 × 23/20 = Rs. 10,580.
• Compound interest = 10580 - 8000 = Rs. 2,580.
• Simple interest for Nilesh:
• Principal = Rs. 9,000, rate = 12%, time = 3 years.
• Simple interest = 9000 × 12 × 3 / 100 = Rs. 3,240.
• Total interest earned = 2580 + 3240 = Rs. 5,820.

Solution Q.150 **Correct Option: (C)**

• Rs. 9375
• Solution:
• Simple interest = Principal × Rate × Time / 100
• 2700 = X × 15 × 3 / 100
• 2700 = X × 45/100
• X = 2700 × 100 / 45
• X = Rs. 6000
• Now, the same amount is invested at 25% compound interest for 2 years.
• Amount after 2 years = 6000 × 125/100 × 125/100
• Amount after 2 years = 6000 × 5/4 × 5/4
• Amount after 2 years = 9375

Solution Q.151 **Correct Option: (B)**

• For 2 years at 15% compound interest, amount factor = 115/100 × 115/100 =
• 13225/10000
• Amount received = Rs. 10580
• So, P × 13225/10000 = 10580
• P = 10580 × 10000 / 13225 = Rs. 8000
• Now, Rs. (3/2)P is invested at simple interest.
• New principal = 3/2 × 8000 = Rs. 12000
• Rate = 12% per annum
• Time = 4 years
• Simple interest = 12000 × 12 × 4 / 100 = Rs. 5760

Solution Q.152 **Correct Option: (E)**

• Principal in each scheme = Rs. 8000
• Interest from first scheme at R% for 4 years = 8000 × R × 4/100
• Interest from first scheme = 320R
• Interest from second scheme at (R - 4)% for 5 years = 8000 × (R - 4) × 5/100
• Interest from second scheme = 400(R - 4)
• Interest from second scheme = 400R - 1600
• Given, first interest is Rs. 640 more than second interest.
• So, 320R - (400R - 1600) = 640
• 320R - 400R + 1600 = 640
• 1600 - 80R = 640
• 80R = 960
• R = 12
• Required value = R + 6 = 12 + 6 = 18

Solution Q.153 **Correct Option: (B)**

• Rs. 6200
• Solution:
• Rajesh invested Rs. X at R% p.a. simple interest for 4 years.
• Interest earned became twice the principal.
• So, simple interest = 2X
• Using simple interest formula:
• SI = Principal × Rate × Time / 100
• 2X = X × R × 4 / 100
• 2 = 4R/100
• R = 50
• Now, the second annual rate = (R - 35)% = (50 - 35)% = 15%.
• Second principal = Rs. (X - 600)
• Time = 3 years
• Interest = Rs. 2520
• 2520 = (X - 600) × 15 × 3 / 100
• 2520 = (X - 600) × 45/100
• X - 600 = 2520 × 100 / 45
• X - 600 = 5600
• X = 5600 + 600 = 6200

Solution Q.154 **Correct Option: (D)**

• In simple interest, when principal and rate are the same, interest is directly
• proportional to time.
• So, X : (X + 6) = 2 : 5.
• 5X = 2(X + 6).
• 5X = 2X + 12.
• 3X = 12.
• X = 12 / 3 = 4.
• Therefore, the value of X is 4.

Solution Q.155 **Correct Option: (D)**

• For Scheme A, compound interest for 2 years at 20% p.a. means total interest
• percentage = 44%.
• Interest from Scheme A = 44% of (X + 750)
• Interest from Scheme A = 44/100 × (X + 750)
• Interest from Scheme A = 44X/100 + 330
• For Scheme B, simple interest for 4 years at 15% p.a. means total interest percentage =
• 15% × 4 = 60% of the principal.
• Interest from Scheme B = 60% of X
• Interest from Scheme B = 60X/100
• Given, Scheme B interest is Rs. 1190 more than Scheme A interest.
• 60X/100 - (44X/100 + 330) = 1190
• 16X/100 - 330 = 1190
• 16X/100 = 1520
• X = 1520 × 100 / 16
• X = 9500

Solution Q.156 **Correct Option: (A)**

• Rs. 5000
• Solution:
• The amount invested in Scheme B gives simple interest of Rs. 1800.
• Rate of simple interest = 18% per annum.
• Time = 2 years.
• Using the simple interest shortcut:
• Principal = Interest × 100 / (Rate × Time).
• Principal = 1800 × 100 / (18 × 2).
• Principal = 180000 / 36 = 5000.
• This principal is the total amount invested in Scheme B.
• Therefore, the total amount invested in Scheme B is Rs. 5000.

Solution Q.157 **Correct Option: (D)**

• Let amount invested in Scheme A = x units
• So, amount invested in Scheme B = Rs. (80000 - x).
• Scheme A offers simple interest at 20% per annum for 2 years.
• Interest from Scheme A = (20 × 2/100) × x = 0.40x.
• So, interest from Scheme A = 0.40x.
• Scheme B offers compound interest at 15% per annum for 2 years.
• At 15% compound interest for 2 years, total interest percentage = 32.25%.
• So, interest from Scheme B = 32.25% of (80000 - x).
• Interest from Scheme B = 0.3225 × (80000 - x).
• Given that Scheme B interest is Rs. 4125 more than Scheme A interest.
• So,
• 0.3225 × (80000 - x) - 0.40x = 4125
• 25800 - 0.3225x - 0.40x = 4125
• 25800 - 0.7225x = 4125
• 0.7225x = 21675
• x = 21675 / 0.7225
• x = 30000

Solution Q.158 **Correct Option: (E)**

• Rs. 1000
• Solution:
• Let total money with Priya = x units
• Investment in Scheme A = 25% of x
• Interest in Scheme A = 25% of x × 8 × 3 / 100
• = 6% of x
• Amount from Scheme A = 25% of x + 6% of x
• = 31% of x
• Investment in Scheme B = 35% of x
• Interest in Scheme B = 35% of x × 10 × 4 / 100
• = 14% of x
• Amount from Scheme B = 35% of x + 14% of x
• = 49% of x
• Investment in Scheme C = remaining 40% of x
• Interest in Scheme C = 40% of x × 12 × 2 / 100
• = 9.6% of x
• Amount from Scheme C = 40% of x + 9.6% of x
• = 49.6% of x
• Total amount received:
• 31% of x + 49% of x + 49.6% of x = 12960
• 129.6% of x = 12960
• x = 12960 × 100 / 129.6
• x = 10000
• Investment in Scheme A = 25% of 10000 = Rs. 2500
• Investment in Scheme B = 35% of 10000 = Rs. 3500
• Required difference = 3500 - 2500
• Required difference = Rs. 1000

Solution Q.159 **Correct Option: (B)**

• % p.a.
• Solution:
• Priya's principal = Rs. 7500
• Divya's principal = Rs. 12500
• Difference in principal = 12500 - 7500
• = Rs. 5000
• Both invested at the same rate for the same time.
• Time = 5 years
• Difference in interest = Rs. 1500
• Using simple interest:
• Difference in interest = Difference in principal × Rate × Time / 100
• So,
• 1500 = 5000 × Rate × 5 / 100
• 1500 = 250 × Rate
• Rate = 1500 / 250
• = 6%
• Hence, the rate of interest is 6% per annum.

Solution Q.160 **Correct Option: (E)**

• Rs. 3,465
• Solution:
• Principal in Scheme A = Rs. 12,000.
• Rate = 12.5% p.a. and time = 3 years.
• Simple interest = 12000 × 12.5 × 3 / 100 = Rs. 4,500.
• Amount received from Scheme A = 12,000 + 4,500 = Rs. 16,500.
• This amount is invested in Scheme B at 10% compound interest for 2 years.
• For 2 years at 10% compound interest, interest percentage = 10% + 10% + 1% = 21%
• of the principal. Compound interest = 16,500 × 21 / 100 = Rs. 3,465. Hence, the
• interest received from Scheme B is Rs. 3,465.

Solution Q.161 **Correct Option: (E)**

• Amount received from scheme B on compound interest:
• Principal = Rs. 12000.
• Rate = 10% per annum.
• Time = 2 years.
• Amount = 12000 × 110/100 × 110/100 = Rs. 14520.
• Amount received from scheme A on simple interest:
• Rate = 16% per annum.
• Time = 2 years.
• Total interest percentage = 16% × 2 = 32% of the principal.
• Amount = 132% of x.
• Since amounts from both schemes are equal:
• 132% of x = 14520.
• x = 14520 × 100 / 132 = 11000.

Solution Q.162 **Correct Option: (D)**

• Total investment = Rs. 4000.
• Investment in Scheme P = 4000 × 3/(3 + 5) = Rs. 1500.
• Investment in Scheme Q = 4000 × 5/(3 + 5) = Rs. 2500.
• Simple interest from Scheme P = 1500 × X × 4/100 = Rs. 60X.
• Simple interest from Scheme Q = 2500 × (X + 6) × 2/100 = Rs. (50X + 300).
• According to the question:
• Interest from Scheme Q - Interest from Scheme P = Rs. 10X.
• 50X + 300 - 60X = 10X.
• 300 - 10X = 10X.
• 300 = 20X.
• X = 15.
• Therefore, the value of X is 15.

Solution Q.163 **Correct Option: (A)**

• Rs. 18000
• Solution:
• Parth invested Rs. X and Sameer invested Rs. Y.
• X is 40% more than Y.
• So:
• X = 140Y/100
• Difference between their investments:
• X - Y = 140Y/100 - Y
• X - Y = 40Y/100
• Rate of simple interest = 15% p.a.
• Time = 2 years
• Difference between their interests = Rs. 900
• Interest difference = (X - Y) × Rate × Time/100
• 900 = (X - Y) × 15 × 2/100
• 900 = (X - Y) × 30/100
• X - Y = 900 × 100/30
• X - Y = 3000
• Now:
• 40Y/100 = 3000
• Y = 3000 × 100/40
• Y = 7500
• X = Y + 3000
• X = 7500 + 3000
• X = 10500
• Total investment = X + Y
• Total investment = 10500 + 7500
• Total investment = Rs. 18000

Solution Q.164 **Correct Option: (C)**

• Rs. 2500
• Solution:
• Harshit first borrowed Rs. 30000.
• Rate of compound interest = 10% p.a.
• Amount of first loan after 3 years:
• 30000 × 110/100 × 110/100 × 110/100
• = Rs. 39930
• At the end of 2 years, he borrowed Rs. X more.
• This second loan remained for 1 year.
• Amount of second loan after 1 year = X × 110/100
• Total amount paid at the end of third year = Rs. 42680
• So:
• 39930 + X × 110/100 = 42680
• X × 110/100 = 42680 - 39930
• X × 110/100 = 2750
• X = 2750 × 100/110
• X = 2500

Solution Q.165 **Correct Option: (A)**

• Amount invested in scheme M = Rs. P
• Rate of compound interest = 10% per annum
• Time = 2 years
• Amount after 2 years in scheme M = P × 110/100 × 110/100
• Amount after 2 years in scheme M = P × 11/10 × 11/10
• Amount after 2 years in scheme M = 121P/100
• This amount is invested in scheme N
• Rate of simple interest in scheme N = 12.5% per annum
• Time = 3 years
• Simple interest from scheme N = Principal × Rate × Time / 100
• 1815 = 121P/100 × 12.5 × 3 / 100
• 12.5 × 3 = 37.5
• 1815 = 121P/100 × 37.5/100
• 1815 = 121P × 375 / 100000
• 1815 = 45375P / 100000
• P = 1815 × 100000 / 45375
• P = 4000

Solution Q.166 **Correct Option: (C)**

• Rs. 12700
• Solution:
• Rate = (50/3)% = 1/6
• So, amount becomes 7/6 times every year.
• Principal = Rs. 21600
• Time = 3 years
• Amount after 3 years = 21600 × 7/6 × 7/6 × 7/6
• Amount = 21600 × 343/216
• Amount = Rs. 34300
• Compound interest = Amount - Principal
• Compound interest = 34300 - 21600
• Compound interest = Rs. 12700
• Therefore, the interest earned after three years is Rs. 12700.

Solution Q.167 **Correct Option: (C)**

• Rs. 10240
• Solution:
• Rate = 25% per annum.
• In the first year, amount = X × 125/100 = 5X/4.
• In the second year, interest is compounded half-yearly.
• Half-yearly rate = 25%/2 = 12.5%.
• So, half-yearly multiplier = 112.5/100 = 9/8.
• Amount at the end of second year = X × 5/4 × 9/8 × 9/8
• Amount at the end of second year = 405X/256.
• Interest earned only in the third year = 25% of 405X/256.
• 405X/256 × 25/100 = 4050
• 405X/1024 = 4050
• X = 4050 × 1024/405
• X = 10240
• So, the value of X is Rs. 10240.

Solution Q.168 **Correct Option: (E)**

• Rs. 1,00,000
• Solution:
• Let investment of A = x units
• B invested 25% more than A.
• Investment of B = 125% of x = 5x/4
• C invested Rs. 15,000 less than B.
• Investment of C = 5x/4 - 15,000
• Total investment = Rs. 1,25,000
• x + 5x/4 + 5x/4 - 15,000 = 1,25,000
• x + 10x/4 = 1,40,000
• x + 5x/2 = 1,40,000
• 7x/2 = 1,40,000
• x = 40,000
• Investment of A = Rs. 40,000
• Investment of B = 40,000 × 5/4 = Rs. 50,000
• Investment of C = 50,000 - 15,000 = Rs. 35,000
• Ratio of investments = 40,000 : 50,000 : 35,000 = 8 : 10 : 7
• Since all invested for one year, profit ratio = 8 : 10 : 7
• B invested his profit share at 12% p.a. simple interest for 2 years.
• Simple interest = Rs. 9,600
• B's profit share = 9,600 × 100/(12 × 2) = Rs. 40,000
• B's share = 10 parts = Rs. 40,000
• Total profit = 40,000 × 25/10 = Rs. 1,00,000
• Therefore, the total profit of the business is Rs. 1,00,000.

Solution Q.169 **Correct Option: (C)**

• Let the equal amount with Mayank and Mukesh = Rs. P
• Mayank invests the amount at 20% compound interest for 2 years.
• Compound interest percentage for 2 years = 20% + 20% + (20/100) × 20% = 44% of
• the principal.
• Interest received by Mayank = 44% of P
• Mukesh invests 1/4 of P at R% simple interest for 2 years.
• Interest on this part = P/4 × R × 2/100 = PR/200 = (R/2)% of P
• Mukesh invests the remaining 3/4 of P at 10% simple interest for 2 years.
• Interest on this part = 3P/4 × 10 × 2/100 = 15% of P
• Total interest received by Mukesh = (R/2)% of P + 15% of P
• According to the question,
• R/2 + 15 = 44
• R/2 = 29
• R = 58

CHAPTER 04: RATIO, PROPORTION & VARIATIONS

27 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Duplicate Ratio of a:b = a^2 : b^2 | Triplicate = a^3 : b^3
- Mean Proportional of a and b = √(ab)
- Third Proportional of a and b = b^2 / a | Fourth Proportional of a, b, c = (b × c) / a
- Cross-Multiplication Ratio Method: Useful for ratio addition/subtraction problems.

Q.170 Ratio, Proportion

In a garment store, the ratio of white shirts to black shirts was 4:7. After selling 15 white shirts and 30 black shirts, the ratio became 3:5. Find the initial number of white shirts in the store.

(A) 54 (B) 72
(C) 48 (D) 60
(E) 66

Q.171 Ratio, Proportion

The population of Sundarpur town in 2018 was 8000. The ratio of males to females was 3:5. In 2019, the number of males increased by 25% and the number of females increased by 16%. In 2020, the number of males increased by 28% and the number of females increased by 25%, each with respect to the previous year. Find the total population of Sundarpur at the end of 2020.

(A) 11850 (B) 12250
(C) 12400 (D) 11650
(E) 12050

Q.172 Ratio, Proportion

The ratio of monthly expenditure to monthly income of Riya is 5 : 11. If her income increases by 60% and her savings increase by 30%, find the percentage increase or decrease in her expenditure.

(A) 84% increase (B) 96% increase
(C) 90% decrease (D) 104% increase
(E) 88% increase

Q.173 Ratio, Proportion

Ritesh spends 12% of his monthly salary on rent and 25% of the remaining salary on food. He divides the remaining salary between commuting expenses and savings in the ratio 4 : 7. If the difference between his savings and rent is Rs. 15000, find his monthly salary.

(A) 45000 (B) 50000
(C) 60000 (D) 55000
(E) 40000

Q.174 Ratio, Proportion

There are two schools, P and Q. The sum of the number of boys in School P and the number of girls in School Q is 250. The ratio of boys in Schools P and Q is 5 : 7, and the ratio of girls in Schools P and Q is 4 : 5. If the total number of girls in both schools is 30 more than the total number of boys in both schools, find the number of boys in School Q.

(A) 140 (B) 120
(C) 150 (D) 180
(E) 160

Q.175 Ratio, Proportion

The length and breadth of a rectangular field are in the ratio 8 : 5. The cost of ploughing the field at the rate of Rs. 8 per sq. meter is Rs. 5120. Find the difference between the length and breadth of the field.

(A) 10 m (B) 15 m
(C) 12 m (D) 18 m
(E) 20 m

Q.176 Ratio, Proportion

In a town, the total population is X. Out of the total population, 34% are women, 46% are men and the rest are children. The ratio of boys to girls among the children is 3 : 2. If the number of men is 5100 more than the number of boys, find the value of X.

(A) 12000 (B) 15000
(C) 18000 (D) 20000
(E) 16000

Q.177 Ratio, Proportion

A man spends 30% of his monthly income on house rent and 25% of the remaining income on food. From the balance amount, the ratio of travelling expenses to savings is 5 : 2. If the difference between the amount spent on food and travelling is Rs. 3000, find his monthly income.

(A) Rs. 12000 (B) Rs. 15000
(C) Rs. 18000 (D) Rs. 20000
(E) Rs. 24000

Q.178 Ratio, Proportion

The ratio of the curved surface area to the total surface area of a cylinder is 5 : 6. If the volume of the cylinder is 5390 cubic cm, find the height of the cylinder. Take π = 22/7.

(A) 21 cm (B) 35 cm
(C) 28 cm (D) 42 cm
(E) 14 cm

Q.179 Ratio, Proportion

There are two companies P and Q. The number of male employees in both companies is the same. The number of female employees in company Q is 40% less than that in company P. If the number of female employees in company P increases by 20% and the number of male employees in company P decreases by 40%, then the total number of employees in both companies becomes equal. Find the initial ratio of total employees in company P to company Q.

(A) 5:3 (B) 21:25
(C) 6:5 (D) 25:21
(E) 7:6

Q.180 Ratio, Proportion

If the radius of a sphere becomes four times its original radius, find the sum of the terms in the simplest ratio of the original volume to the new volume.

(A) 17 (B) 81
(C) 64 (D) 27
(E) 65

Q.181 Ratio, Proportion

In a town, the total population is X. Out of the total population, 35% are women, 45% are men, and the remaining are children. Among the children, the ratio of boys to girls is 3:2. If the number of men is 4950 more than the number of boys, find X.

(A) 12000 (B) 18000
(C) 16000 (D) 15000
(E) 20000

Q.182 Ratio, Proportion

There are 180 sweets in a box, and 90 more sweets are added to it. If all the sweets are distributed between boys and girls in the ratio 5 : 4, find the difference between the number of sweets received by boys and girls.

(A) 25 (B) 35
(C) 30 (D) 45
(E) 40

Q.183 Ratio, Proportion

The curved surface area of a hemisphere is 7700 cm². The height of a cylinder is 20% less than the radius of the hemisphere. If the ratio of the radius to the height of the cylinder is 3:4, then find the volume of the cylinder in cm³. Take π = 22/7.

(A) 35280 (B) 41160
(C) 36960 (D) 38808
(E) 43120

Q.184 Ratio, Proportion

The sum of the lengths of two rectangular parks P and Q is 44 meters. Their breadths are 9 meters and 7 meters, respectively. If the ratio of the perimeter of park P to that of park Q is 11:9, then find the difference between their perimeters.

(A) 16 meters (B) 12 meters
(C) 20 meters (D) 18 meters
(E) 24 meters

Q.185 Ratio, Proportion

Ananya's monthly income is 20% more than Swati's monthly income. Swati and Ananya save Rs. 7000 and Rs. 9000 respectively. If the ratio of expenditure of Swati to that of Ananya is 6:7, then find their total expenditure.

(A) Rs. 42000 (B) Rs. 36000
(C) Rs. 45000 (D) Rs. 39000
(E) Rs. 48000

Q.186 Ratio, Proportion

The quantities of wheat in three containers P, Q, and R are in the ratio 5:3:4, respectively. The total quantity of wheat is 72 kg. If 18 kg of wheat is added to container R, find the sum of the terms of the simplest new ratio of the quantities of wheat in P, Q and R.

(A) 15 (B) 12
(C) 18 (D) 20
(E) 14

Q.187 Ratio, Proportion

In 2024, the salaries of Ayush and Sohan were in the ratio 4:5, respectively. The ratios of their salaries in 2024 to 2025 are 5:6 and 4:5, respectively. If the difference between their salaries in 2025 is Rs. 3480, find Sohan's salary in 2025.

(A) Rs. 12000 (B) Rs. 13500
(C) Rs. 15000 (D) Rs. 16500
(E) Rs. 18000

Q.188 Ratio, Proportion

The area of a square is 400 m². The length of a rectangle is 8 meters more than the side of the square. If the ratio of the length to the breadth of the rectangle is 7:5, then find the perimeter of the rectangle.

(A) 84 m (B) 90 m
(C) 96 m (D) 108 m
(E) 112 m

Q.189 Ratio, Proportion

The area of a right-angled triangular board is 270 cm². If its base and height are in the ratio 5:12, then find the height of the triangular board.

(A) 30 cm (B) 42 cm
(C) 24 cm (D) 36 cm
(E) 48 cm

Q.190 Ratio, Proportion

The monthly incomes of Hemant and Lakshya are in the ratio 10:13, and their monthly expenditures are in the ratio 7:9. If Hemant saves 30% of his income, then find the sum of the terms of the simplest ratio of their savings.

(A) 9 (B) 7
(C) 11 (D) 13
(E) 5

Q.191 Ratio, Proportion

The ratio of monthly expenditure to monthly income of Vikram is 4:9. If his income increases by 60% and his savings increase by 80%, then find the percentage increase or decrease in his monthly expenditure.

(A) 25% decrease (B) 35% increase
(C) 40% increase (D) 30% decrease
(E) 45% increase

Q.192 Ratio, Proportion

Series M consists of four consecutive even numbers, and Series N consists of four consecutive odd numbers. The ratio of the second smallest number of Series M to the largest number of Series N is 12:7. If the smallest number of Series M is 19 more than the smallest number of Series N, then find the sum of the second largest numbers of both series.

(A) 53 (B) 57
(C) 61 (D) 65
(E) 49

Q.193 Ratio, Proportion

Prateek, Mayank and Hemant spend 25%, 40% and 70% of their incomes respectively. The sum of their incomes is Rs. 31000 and the ratio of their savings is 9 : 12 : 9 respectively. Find the sum of their expenditures.

(A) 15000 (B) 16000
(C) 17500 (D) 18000
(E) 14500

Q.194 Ratio, Proportion

The ratio of deluxe-class fare to ordinary-class fare is 5 : 2. The number of deluxe-class tickets and ordinary-class tickets sold is in the ratio 3 : 4. If the total fare collected is Rs. 4,600, then find the fare collected from ordinary-class passengers.

(A) Rs. 1,200
(B) Rs. 1,800
(C) Rs. 2,000
(D) Rs. 1,600
(E) Rs. 2,400

Q.195 Ratio, Proportion

In a school, the ratio of boys to girls is 11 : 7. If 48 boys leave the school and 72 girls join the school, the ratio of boys to girls becomes 9 : 10. Find the difference between the initial number of boys and girls.

(A) 84
(B) 108
(C) 120
(D) 96
(E) 88

Q.196 Ratio, Proportion

A vessel contains 64 liters of pure milk. In each of three successive operations, x liters of the well-mixed liquid are removed and replaced with an equal volume of water. The liquid is mixed thoroughly after every replacement. If 27 liters of milk remain after the third operation, find x.

(A) 12 liters
(B) 20 liters
(C) 8 liters
(D) 16 liters
(E) 24 liters

📖 Chapter 04 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.170 — **Correct Option: (D)**

• The initial ratio of white shirts to black shirts is 4:7.
• Let the initial number of white shirts be 4x.
• Let the initial number of black shirts be 7x.
• After selling 15 white shirts, remaining white shirts = 4x - 15
• After selling 30 black shirts, remaining black shirts = 7x - 30
• The new ratio becomes 3:5.
• So, we can write:
• (4x - 15) : (7x - 30) = 3 : 5
• Now, by cross multiplication:
• 5 × (4x - 15) = 3 × (7x - 30)
• 20x - 75 = 21x - 90
• 21x - 20x = 90 - 75
• x = 15
• Initial number of white shirts = 4x
• Initial number of white shirts = 4 × 15 = 60

Solution Q.171 — **Correct Option: (E)**

• Total population of Sundarpur in 2018 = 8000.
• Ratio of males to females = 3:5.
• Total ratio parts = 3 + 5 = 8 parts.
• Males in 2018 = 3 parts out of 8 parts.
• So, males in 2018 = 8000 × 3 / 8 = 3000.
• Females in 2018 = 5 parts out of 8 parts.
• So, females in 2018 = 8000 × 5 / 8 = 5000.
• Now calculate males and females separately because their growth rates are different.
• Males increased by 25% in 2019.
• 25% of 3000 = 750.
• Males in 2019 = 3000 + 750 = 3750.
• Males increased by 28% in 2020.
• 28% of 3750 = 1050.
• Males in 2020 = 3750 + 1050 = 4800.
• Now calculate females.
• Females increased by 16% in 2019.
• 16% of 5000 = 800.
• Females in 2019 = 5000 + 800 = 5800.
• Females increased by 25% in 2020.
• 25% of 5800 = 1450.
• Females in 2020 = 5800 + 1450 = 7250.
• Total population at the end of 2020 = males in 2020 + females in 2020.
• Total population = 4800 + 7250 = 12050.

Solution Q.172 — **Correct Option: (B)**

• % increase
• Solution:
• Income = Expenditure + Savings
• Given ratio of expenditure to income = 5 : 11
• For easy calculation, take income = Rs. 1100 and expenditure = Rs. 500.
• So, savings = Income - Expenditure
• Savings = 1100 - 500 = Rs. 600
• Now, income is increased by 60%.
• Increase in income: 1100 × 60/100
• = Rs. 660
• New income = 1100 + 660 = Rs. 1760
• Savings are increased by 30%.
• Increase in savings: 600 × 30/100
• = Rs. 180
• New savings = 600 + 180 = Rs. 780
• Now, new expenditure = New income - New savings
• New expenditure = 1760 - 780 = Rs. 980
• Old expenditure = Rs. 500
• Increase in expenditure = 980 - 500 = Rs. 480
• Percentage increase in expenditure: 480 × 100/500
• = 96%

Solution Q.173 — **Correct Option: (B)**

• Treat the full monthly income as 100%; every percentage below is a share of that
• income.
• Amount spent on rent = 12%
• Remaining salary = 100% - 12% = 88%
• He spends 25% of the remaining salary on food.
• Food expense = (25/100) × 88% = 22% of the monthly salary.
• Remaining salary after rent and food = 100% - 12% - 22%
• Remaining salary = 66%
• This 66% is divided between commuting and savings in the ratio 4:7.
• Savings part = 7 parts out of 11 parts
• Savings = 66% × 7/11 = 42% of the monthly salary.
• Rent = 12%
• Difference between savings and rent = 42% - 12% = 30%
• Given difference = Rs. 15000
• So, 30% of salary = Rs. 15000
• Monthly salary = 15000 × 100/30
• Monthly salary = Rs. 50000

Solution Q.174 **Correct Option: (A)**

• Ratio (boys in School P : School Q) = 5 : 7.
• Let boys in School P = 5x and boys in School Q = 7x.
• Ratio (girls in School P : School Q) = 4 : 5.
• Let girls in School P = 4y and girls in School Q = 5y.
• Given:
• Boys in School P + Girls in School Q = 250.
• 5x + 5y = 250
• x + y = 50
• Total girls are 30 more than total boys.
• Total girls = 4y + 5y = 9y.
• Total boys = 5x + 7x = 12x.
• So,
• 9y - 12x = 30.
• From x + y = 50, y = 50 - x.
• Put y = 50 - x:
• 9(50 - x) - 12x = 30
• 450 - 9x - 12x = 30
• 450 - 21x = 30
• 21x = 420
• x = 420/21
• x = 20.
• Boys in School Q = 7x.
• Boys in School Q = 7 × 20 = 140.

Solution Q.175 **Correct Option: (C)**

• m
• Solution:
• Cost of ploughing = Rs. 5120.
• Rate of ploughing = Rs. 8 per sq. meter.
• Area of field = 5120/8 = 640 sq. meters.
• Length : Breadth = 8 : 5.
• Let length = 8x and breadth = 5x.
• Area = length × breadth.
• So,
• 8x × 5x = 640
• 40x² = 640
• x² = 640/40
• x² = 16
• x = 4
• Length = 8x = 8 × 4 = 32 m.
• Breadth = 5x = 5 × 4 = 20 m.
• Difference = 32 - 20 = 12 m.

Solution Q.176 **Correct Option: (B)**

• Total population = X.
• Women = 34% of X.
• Men = 46% of X.
• Children = 100% - 34% - 46% = 20% of X.
• Boys : Girls = 3 : 2.
• Total ratio parts = 3 + 2 = 5.
• Boys = 3/5 of children.
• Boys = 3/5 × 20% of X.
• Boys = 12% of X.
• Men = 46% of X.
• Given that men are 5100 more than boys.
• So,
• 46% of X - 12% of X = 5100
• 34% of X = 5100
• 34X/100 = 5100
• X = 5100 × 100/34
• X = 15000

Solution Q.177 **Correct Option: (B)**

• Rs. 15000
• Solution:
• Let monthly income = x units
• House rent = 30% of x.
• Remaining income after rent = 70% of x = 70x/100.
• Food expense = 25% of remaining income.
• Food expense = 25/100 × 70x/100 = 7x/40.
• Remaining income after food = 70x/100 - 7x/40.
• 70x/100 = 28x/40.
• Remaining income after food = 28x/40 - 7x/40 = 21x/40.
• This remaining amount is divided into travelling and savings in the ratio 5 : 2.
• Travelling expense = 5/7 × 21x/40.
• Travelling expense = 3x/8.
• Food expense = 7x/40.
• Difference between travelling and food = Rs. 3000.
• 3x/8 - 7x/40 = 3000
• 15x/40 - 7x/40 = 3000
• 8x/40 = 3000
• x/5 = 3000
• x = 3000 × 5 = 15000.

Solution Q.178 **Correct Option: (B)**

• cm
• Solution:
• Curved surface area of cylinder = 2πrh
• Total surface area of cylinder = 2πr(h + r)
• Given ratio of CSA to TSA = 5 : 6
• So, 2πrh/[2πr(h + r)] = 5/6
• h/(h + r) = 5/6
• Cross multiply:
• 6h = 5(h + r)
• 6h = 5h + 5r
• h = 5r
• Now, volume of cylinder = πr²h
• Put h = 5r.
• Volume = π × r × r × 5r
• 5390 = 5πr³
• Using π = 22/7:
• 5390 = 5 × 22/7 × r³
• 5390 = 110r³/7
• r³ = 5390 × 7/110
• r³ = 343
• r = 7 cm
• Height = 5r = 5 × 7 = 35 cm

Solution Q.179 **Correct Option: (D)**

• :21
• Solution:
• Let the number of male employees in each company be M.
• Let the number of female employees in company P be F.
• Female employees in company Q are 40% less than company P.
• So, female employees in company Q = 60% of F = 60F/100 = 3F/5
• Initial total employees in company P = M + F
• Initial total employees in company Q = M + 3F/5
• After change in company P:
• Male employees in company P decrease by 40%.
• New male employees in company P = 60M/100 = 3M/5
• Female employees in company P increase by 20%.
• New female employees in company P = 120F/100 = 6F/5
• Now total employees in company P become equal to total employees in company Q.
• So:
• 3M/5 + 6F/5 = M + 3F/5
• 6F/5 - 3F/5 = M - 3M/5
• 3F/5 = 2M/5
• 3F = 2M
• M/F = 3/2
• Let M = 3x and F = 2x.
• Initial total employees in company P = M + F = 3x + 2x = 5x
• Initial total employees in company Q = M + 3F/5
• Initial total employees in company Q = 3x + 3 × 2x/5
• Initial total employees in company Q = 3x + 6x/5
• Initial total employees in company Q = 15x/5 + 6x/5 = 21x/5
• Required ratio = 5x : 21x/5
• Required ratio = 25x : 21x
• Required ratio = 25:21

Solution Q.180 **Correct Option: (E)**

• Volume of a sphere is proportional to the cube of its radius.
• Let the original radius be r.
• Original volume is proportional to r^3.
• New radius = 4r.
• New volume is proportional to $(4r)^3$.
• New volume is proportional to $64r^3$.
• Ratio of original volume to new volume = $r^3 : 64r^3$.
• Ratio = 1 : 64.
• Sum of the terms = 1 + 64 = 65.

Solution Q.181 **Correct Option: (D)**

• Let the total population = X
• Women = 35% of X
• Men = 45% of X
• Children = 100% - 35% - 45% = 20% of X
• Boys:girls = 3:2
• Boys = 3/5 of children
• Boys = 3/5 × 20% of X = 12% of X
• According to the question:
• Men - boys = 4950
• 45% of X - 12% of X = 4950
• 33% of X = 4950
• X = 4950 × 100/33
• X = 15000

Solution Q.182 **Correct Option: (C)**

• First find the total number of sweets after adding more sweets.
• Initial number of sweets = 180
• More sweets added = 90
• Total sweets = 180 + 90 = 270
• The sweets are distributed between boys and girls in the ratio 5 : 4.
• Total ratio parts = 5 + 4 = 9.
• In banking exams, to find the difference directly, use:
• Difference = Total quantity × Difference of ratio parts/Total ratio parts
• Difference of ratio parts = 5 - 4 = 1.
• So,
• Required difference = 270 × 1/9
• Required difference = 30

Solution Q.183 **Correct Option: (D)**

• Curved surface area of hemisphere = $2\pi r^2$
• 7700 = 2 × 22/7 × r^2
• r^2 = 7700 × 7/44
• r^2 = 1225
• Radius of hemisphere = 35 cm
• Height of cylinder = 20% less than 35
• Height = 35 × 80/100 = 28 cm
• Ratio of radius to height of cylinder = 3:4
• 4 parts = 28
• 1 part = 28/4 = 7
• Radius of cylinder = 3 × 7 = 21 cm
• Volume of cylinder = $\pi r^2 h$
• Volume = 22/7 × 21 × 21 × 28
• Volume = 38808 cm^3

Solution Q.184 **Correct Option: (B)**

• meters
• Solution:
• Let the length of park P = x meters.
• Then the length of park Q = 44 - x meters.
• Breadth of park P = 9 meters
• Breadth of park Q = 7 meters
• Perimeter of rectangle = 2 × (Length + Breadth)
• Perimeter of park P = 2 × (x + 9)
• Perimeter of park Q = 2 × (44 - x + 7) = 2 × (51 - x)
• Given ratio of perimeters = 11:9
• So, 2(x + 9) : 2(51 - x) = 11:9
• (x + 9) : (51 - x) = 11:9
• 9(x + 9) = 11(51 - x)
• 9x + 81 = 561 - 11x
• 20x = 480
• x = 24
• Length of park P = 24 meters
• Length of park Q = 44 - 24 = 20 meters
• Perimeter of park P = 2 × (24 + 9) = 66 meters
• Perimeter of park Q = 2 × (20 + 7) = 54 meters
• Required difference = 66 - 54 = 12 meters

Solution Q.185 **Correct Option: (D)**

• Rs. 39000
• Solution:
• Let Swati's monthly income be 100x.
• Ananya's monthly income is 20% more than Swati's income.
• So, Ananya's monthly income = 120x
• Swati's expenditure = 100x - 7000
• Ananya's expenditure = 120x - 9000
• Ratio of Swati's expenditure to Ananya's expenditure = 6:7
• So,
• (100x - 7000) : (120x - 9000) = 6 : 7
• 7 × (100x - 7000) = 6 × (120x - 9000)
• 700x - 49000 = 720x - 54000
• 54000 - 49000 = 720x - 700x
• 5000 = 20x
• x = 250
• Swati's income = 100 × 250 = Rs. 25000
• Ananya's income = 120 × 250 = Rs. 30000
• Swati's expenditure = 25000 - 7000 = Rs. 18000
• Ananya's expenditure = 30000 - 9000 = Rs. 21000
• Total expenditure = 18000 + 21000 = Rs. 39000

Solution Q.186 **Correct Option: (A)**

• Ratio of wheat in P, Q, and R = 5:3:4
• Total ratio parts = 5 + 3 + 4 = 12
• Total quantity = 72 kg
• Value of 1 part = 72 / 12 = 6 kg
• Quantity in P = 5 × 6 = 30 kg
• Quantity in Q = 3 × 6 = 18 kg
• Quantity in R = 4 × 6 = 24 kg
• Now, 18 kg wheat is added to R.
• New quantity in R = 24 + 18 = 42 kg
• New quantities of P, Q, and R = 30 kg, 18 kg, and 42 kg
• New ratio = 30:18:42
• Dividing by 6, new ratio = 5:3:7
• Sum of the terms of new ratio = 5 + 3 + 7 = 15

Solution Q.187 **Correct Option: (C)**

• Rs. 15000
• Solution:
• In 2024, salary Ratio (Ayush : Sohan) = 4:5
• Let Ayush's salary in 2024 = 4x
• Let Sohan's salary in 2024 = 5x
• For Ayush, salary ratio of 2024 to 2025 = 5:6
• So, Ayush's salary in 2025 = 4x × 6/5 = 24x/5
• For Sohan, salary ratio of 2024 to 2025 = 4:5
• So, Sohan's salary in 2025 = 5x × 5/4 = 25x/4
• Given difference in 2025 salaries = Rs. 3480
• So, 25x/4 - 24x/5 = 3480
• LCM of 4 and 5 = 20
• 125x/20 - 96x/20 = 3480
• 29x/20 = 3480
• 29x = 69600
• x = 69600 / 29 = 2400
• Sohan's salary in 2025 = 25x/4
• Sohan's salary in 2025 = 25 × 2400 / 4 = Rs. 15000

Solution Q.188 **Correct Option: (C)**

• m
• Solution:
• Area of square = 400 m².
• Side of square = 20 m because 20 × 20 = 400.
• Length of rectangle = side of square + 8 = 20 + 8 = 28 m.
• Given, length : breadth = 7 : 5.
• Here 7 parts = 28 m.
• 1 part = 28 / 7 = 4 m.
• Breadth = 5 × 4 = 20 m.
• Perimeter of rectangle = 2 × (length + breadth).
• Perimeter = 2 × (28 + 20) = 2 × 48 = 96 m.
• Therefore, the perimeter of the rectangle is 96 m.

Solution Q.189 **Correct Option: (D)**

• cm
• Solution:
• Let the base and height of the triangle be 5x cm and 12x cm.
• Area of right-angled triangle = 1/2 × base × height
• 270 = 1/2 × 5x × 12x
• 270 = 30x²
• x² = 270 / 30 = 9
• x = 3
• Height = 12x = 12 × 3 = 36 cm
• So, the height of the triangular board is 36 cm.

Solution Q.190 **Correct Option: (B)**

• Let the monthly incomes of Hemant and Lakshya be 10x and 13x respectively.
• Hemant saves 30% of his income.
• Hemant's saving = 30% of 10x = 3x
• Hemant's expenditure = 10x - 3x = 7x
• Given, expenditure Ratio (Hemant : Lakshya) = 7:9
• So, Lakshya's expenditure = 9x
• Lakshya's saving = 13x - 9x = 4x
• Savings Ratio (Hemant : Lakshya) = 3x:4x = 3:4
• Sum of the terms = 3 + 4 = 7

Solution Q.191 **Correct Option: (B)**

• % increase
• Solution:
• Let Vikram's monthly income be 9x.
• His monthly expenditure = 4x.
• His monthly saving = 9x - 4x = 5x.
• New income after 60% increase = 9x × 160/100 = 14.4x.
• New saving after 80% increase = 5x × 180/100 = 9x.
• New expenditure = new income - new saving = 14.4x - 9x = 5.4x.
• Increase in expenditure = 5.4x - 4x = 1.4x.
• Percentage increase = 1.4x / (4x) × 100 = 35%.
• So, monthly expenditure increased by 35%.

Solution Q.192 **Correct Option: (B)**

• Let the four consecutive even numbers in Series M be a, a + 2, a + 4 and a + 6.
• Let the four consecutive odd numbers in Series N be b, b + 2, b + 4 and b + 6.
• The second smallest number of Series M = a + 2.
• The largest number of Series N = b + 6.
• Given, (a + 2) : (b + 6) = 12 : 7.
• Also, the smallest number of Series M is 19 more than the smallest number of Series
• N.
• So, a = b + 19.
• Put a = b + 19 in the ratio:
• (b + 19 + 2) : (b + 6) = 12 : 7.
• (b + 21) : (b + 6) = 12 : 7.
• 7(b + 21) = 12(b + 6).
• 7b + 147 = 12b + 72.
• 147 - 72 = 12b - 7b.
• 75 = 5b.
• b = 15.
• So, a = 15 + 19 = 34.
• Series M = 34, 36, 38, 40.
• Series N = 15, 17, 19, 21.
• Second largest number of Series M = 38.
• Second largest number of Series N = 19.
• Required sum = 38 + 19 = 57.

Solution Q.193 **Correct Option: (B)**

• Prateek spends 25%, so he saves 75% = 3/4 of his income.
• Mayank spends 40%, so he saves 60% = 3/5 of his income.
• Hemant spends 70%, so he saves 30% = 3/10 of his income.
• Let their savings be 9x, 12x and 9x respectively.
• Prateek's income = 9x / (3/4) = 9x × 4/3 = 12x.
• Mayank's income = 12x / (3/5) = 12x × 5/3 = 20x.
• Hemant's income = 9x / (3/10) = 9x × 10/3 = 30x.
• Total income = 12x + 20x + 30x = 62x.
• 62x = 31000.
• x = 500.
• Total savings = 9x + 12x + 9x = 30x = 30 × 500 = 15000.
• Sum of expenditures = Total income - Total savings.
• Sum of expenditures = 31000 - 15000 = 16000.

Solution Q.194 **Correct Option: (D)**

• Rs. 1,600
• Solution:
• Deluxe-class fare : Ordinary-class fare = 5 : 2
• Let deluxe-class fare = 5x
• Let ordinary-class fare = 2x
• Number of deluxe-class tickets : Number of ordinary-class tickets = 3 : 4
• Let deluxe-class tickets = 3y
• Let ordinary-class tickets = 4y
• Fare collected from deluxe class:
• 5x × 3y = 15xy
• Fare collected from ordinary class:
• 2x × 4y = 8xy
• Total fare ratio:
• Deluxe : Ordinary = 15 : 8
• Total parts = 15 + 8 = 23
• Total fare collected = Rs. 4,600
• Value of 1 part = 4600 / 23 = 200
• Fare collected from ordinary-class passengers:
• 8 × 200 = Rs. 1,600

Solution Q.195 **Correct Option: (D)**

• Let the initial number of boys and girls be 11x and 7x respectively.
• After 48 boys leave, number of boys = 11x - 48
• After 72 girls join, number of girls = 7x + 72
• New ratio of boys to girls = 9 : 10
• So, (11x - 48)/(7x + 72) = 9/10
• Cross multiply:
• 10 × (11x - 48) = 9 × (7x + 72)
• 110x - 480 = 63x + 648
• 110x - 63x = 648 + 480
• 47x = 1128
• x = 1128/47 = 24
• Initial number of boys = 11 × 24 = 264
• Initial number of girls = 7 × 24 = 168
• Required difference = 264 - 168 = 96

Solution Q.196 **Correct Option: (D)**

• liters
• Solution:
• Initial quantity of milk = 64 liters
• After each replacement, milk remaining = Milk present immediately before that
• replacement × (1 - x/64)
• The process is repeated three times
• Milk left after three replacements = 64 × $(1 - x/64)^3$
• According to the question,
• 64 × $(1 - x/64)^3$ = 27
• $(1 - x/64)^3$ = 27/64
• $(1 - x/64)^3$ = $(3/4)^3$
• 1 - x/64 = 3/4
• x/64 = 1 - 3/4
• x/64 = 1/4
• x = 64 × 1/4
• x = 16 liters

CHAPTER 05: PROBLEMS ON AGES & LINEAR EQUATIONS

54 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Age Difference remains Constant: The age gap between two persons never changes over time.
- If ratio of ages before t years was a:b and after t years is c:d, equate the age difference in ratio units.
- Average Age of Family: Total Age = Number of Members × Average Age.

Q.197 Problems on Ages

The average weight of five persons A, B, C, D and E is 68 kg. The weight of A is 76 kg. If a new person F, whose weight is 20 kg less than the weight of A, joins the group, find the new average weight of the group.

(A) 62 (B) 64
(C) 66 (D) 68
(E) 70

Q.198 Problems on Ages

The average of the first, second and third numbers is 34, and the average of the second, third and fourth numbers is 42. If the fourth number is 48, find the first number.

(A) 24 (B) 28
(C) 30 (D) 20
(E) 36

Q.199 Problems on Ages

Adarsh scored 25% more marks than Gaurav. Gaurav scored 20% fewer marks than Chirag, and Chirag scored 50% more marks than Mayank. Adarsh's marks are what percentage of Mayank's marks?

(A) 125% (B) 135%
(C) 160% (D) 144%
(E) 150%

Q.200 Problems on Ages

There are three consecutive odd numbers and four consecutive even numbers. The average of the three odd numbers is 35. The smallest even number is twice the largest odd number. Find the difference between the sum of the four even numbers and the sum of the three odd numbers.

(A) 197 (B) 203
(C) 209 (D) 215
(E) 221

Q.201 Problems on Ages

Five years ago, the ratio of the ages of Ishita and Naman was 3:4. Ten years from now, the ratio of their ages will be 6:7. Find Naman's present age.

(A) 22 years (B) 20 years
(C) 28 years (D) 30 years
(E) 25 years

Q.202 Problems on Ages

The average of 15 numbers is 82. The average of the first 6 numbers is 74, and the average of the last 5 numbers is 90. Find the average of the remaining 4 numbers.

(A) 78 (B) 90
(C) 81 (D) 84
(E) 87

Q.203 Problems on Ages

Ojas's age 6 years hence will be equal to Darshan's age 12 years ago. If the ratio of the ages of Ojas and Darshan after 3 years will be 5:8, find the sum of their present ages.

(A) 72 years (B) 60 years
(C) 66 years (D) 78 years
(E) 84 years

Q.204 Problems on Ages

A person spends 10% of his monthly salary on rent and 30% of the remaining salary on food. He divides the remaining salary between travelling expenses and savings in the ratio 3 : 4. If the difference between his savings and rent is Rs. 10400, find his monthly salary.

(A) Rs. 36000 (B) Rs. 45000
(C) Rs. 50000 (D) Rs. 48000
(E) Rs. 40000

Q.205 Problems on Ages

Mihir's age 4 years hence will be equal to Tanmay's age 8 years ago. If the ratio of the ages of Mihir and Tanmay after 6 years will be 5:8, find the sum of their present ages.

(A) 40 years (B) 48 years
(C) 52 years (D) 44 years
(E) 56 years

Q.206 Problems on Ages

The present age of an elder brother is 200% more than the present age of his younger brother. After 6 years, the elder brother's age will be 125% more than the younger brother's age. Find the present age of the elder brother.

(A) 24 years (B) 27 years
(C) 30 years (D) 36 years
(E) 40 years

Q.207 Problems on Ages

The average age of 35 students in a class is 18 years. If the age of the class teacher is included, the average age of the whole group becomes 19 years. Find the age of the teacher.

(A) 52 years (B) 60 years
(C) 48 years (D) 56 years
(E) 54 years

Q.208 Problems on Ages

The average of four consecutive even numbers is X. If the ratio of the smallest number to the largest number is 4 : 5, find the value of X.

(A) 24 (B) 25
(C) 30 (D) 28
(E) 27

Q.209 Problems on Ages

Five years ago, Mayank's age was 25% more than Prateek's age, and Prateek is 5 years younger than Mayank. If the average present age of Prateek, Prateek and Mayank is 24 years, find the difference between the ages of Prateek and Mayank six years hence.

(A) 10 (B) 11
(C) 12 (D) 13
(E) 15

Q.210 Problems on Ages

The ratio of the present ages of Aman and Sahil is 5 : 3. Five years ago, Aman's age was twice Sahil's age at that time. Four years hence, the sum of the ages of Aman and Sahil will be equal to Pawan's age at that time. Find the present age of Pawan.

(A) 44 years (B) 40 years
(C) 48 years (D) 42 years
(E) 36 years

Q.211 Problems on Ages

A person spends 20% of his monthly income on rent and 25% of the remaining income on groceries. He divides the remaining income between entertainment expenses and savings in the ratio 2 : 3. If the difference between his savings and rent is Rs. 8000, find his monthly income.

(A) Rs. 42000 (B) Rs. 45000
(C) Rs. 48000 (D) Rs. 52000
(E) Rs. 50000

Q.212 Problems on Ages

A person spends 25% of his monthly income on groceries, 16 2/3% of the remaining amount on rent, and 20% of the amount left after rent on transport. If he saves Rs. 30000, find the amount spent on groceries.

(A) Rs. 15000 (B) Rs. 18000
(C) Rs. 12000 (D) Rs. 20000
(E) Rs. 16000

Q.213 Problems on Ages

Four years from now, the ratio of the ages of Leela and Prateek will be 4:5. The present age ratio of Aman and Prateek is 5:6. If the sum of the ages of Leela and Aman after 6 years will be 70 years, find the sum of the present ages of Leela and Prateek.

(A) 56 years (B) 60 years
(C) 68 years (D) 64 years
(E) 72 years

Q.214 Problems on Ages

The average of five consecutive multiples of 10 is 70. The difference between the highest number and the second smallest number is equal to the largest even number in a set of four consecutive even numbers. Find the square of the smallest even number of that set.

(A) 484 (B) 576
(C) 400 (D) 625
(E) 676

Q.215 Problems on Ages

The average weight of a group is 45 kg. If a person weighing 63 kg leaves the group, then the average weight of the group decreases by 2 kg. Find the number of people in the group initially.

(A) 10 (B) 12
(C) 14 (D) 16
(E) 18

Q.216 Problems on Ages

The ratio of A to B is 5:3. If the average of both numbers is 48 less than A, find 25% of B.

(A) 30 (B) 48
(C) 42 (D) 36
(E) 60

Q.217 Problems on Ages

The average of 61 numbers is 54. If the average of the first 30 numbers is 50 and the average of the last 30 numbers is 56, find the 31st number in the given order.

(A) 108 (B) 114
(C) 120 (D) 96
(E) 102

Q.218 Problems on Ages

Arvind spends 35% of his monthly salary on rent and 25% on food. He divides the entire remaining amount between his father and mother in the ratio 7 : 5. If the difference between the amounts given to his father and mother is Rs. 10000, find Arvind's monthly salary.

(A) Rs. 150000 (B) Rs. 120000
(C) Rs. 180000 (D) Rs. 135000
(E) Rs. 160000

Q.219 Problems on Ages

The monthly income of a person is Rs. 50,000 and his monthly expenditure is Rs. 30,000. If his income increases by 24% and his savings increase by 30%, then find the percentage increase in his monthly expenditure.

(A) 16% (B) 18%
(C) 20% (D) 22%
(E) 25%

Q.220 Problems on Ages

Four years hence, the ratio of ages of Aniket and Divya will be equal to the ratio of ages of Sameer and Divya two years ago. If Aniket is 6 years older than Sameer and Divya's present age is 20 years, then find Sameer's present age.

(A) 34 years (B) 38 years
(C) 42 years (D) 36 years
(E) 40 years

Q.221 Problems on Ages

Three years ago, the average age of Shreya, Priya and Prateek was 28 years. If four years hence, the sum of the ages of Shreya and Prateek will be 68 years, then find the present age of Priya.

(A) 30 years (B) 31 years
(C) 33 years (D) 36 years
(E) 39 years

Q.222 Problems on Ages

Five years ago, the ratio of the ages of Sahil and Akash was 5:2. Seven years hence, Sahil's age will be 75% more than Akash's age at that time. If Kunal's present age is 100% more than Akash's present age, find Kunal's present age.

(A) 34 years (B) 30 years
(C) 38 years (D) 42 years
(E) 28 years

Q.223 Problems on Ages

A person spends 18% of his monthly salary on rent. From the remaining salary, he spends 25% on travel. He spends Rs. 6625 on food, and 35% of his monthly salary is saved. Find his annual income. He has no other expenses, and his monthly salary remains unchanged throughout the year.

(A) Rs. 300000 (B) Rs. 288000
(C) Rs. 312000 (D) Rs. 324000
(E) Rs. 336000

Q.224 Problems on Ages

Five years ago, Kunal's age was equal to the average of the present ages of Kunal, Akash, and Kabir. Kabir is 20% older than Akash and 4 years younger than Kunal. Find the present age of Kabir.

(A) 36 years (B) 45 years
(C) 48 years (D) 42 years
(E) 39 years

Q.225 Problems on Ages

Two years hence, Aman's age will be 60% of Aniket's present age. Aman's present age is 20% less than Sameer's present age. Sameer is 15 years younger than Aniket. Find Aman's age 5 years ago.

(A) 19 years (B) 21 years
(C) 25 years (D) 27 years
(E) 23 years

Q.226 Problems on Ages

Five years ago, Harita's age was 25% more than Kriti's age, and Kriti is 5 years younger than Harita. If the present average age of Kriti, Yamini and Harita is 30 years, find the difference between Yamini's and Harita's ages six years from now.

(A) 3 years (B) 4 years
(C) 5 years (D) 6 years
(E) 8 years

Q.227 Problems on Ages

The average weight of 29 students in a class is 30 kg. When the teacher's weight is also included, the average weight increases by 2 kg. The average weight of the teacher, his wife, and his daughter is 60 kg. If the daughter's weight is 50% less than her mother's weight, find the weight of the teacher's wife.

(A) 48 kg (B) 52 kg
(C) 56 kg (D) 58 kg
(E) 60 kg

Q.228 Problems on Ages

A person leaves Rs. 38400 to be divided among 3 sons, 2 daughters, and 4 nieces. Each daughter receives twice as much as each niece, and each son receives 4 times as much as each daughter. How much does each son receive?

(A) Rs. 8400 (B) Rs. 9600
(C) Rs. 10800 (D) Rs. 9000
(E) Rs. 10200

Q.229 Problems on Ages

The average of 15 numbers is 90. The average of the first seven numbers is 85, and the average of the last five numbers is 99. If the ninth number is 25% more than the eighth number and the tenth number is 20% less than the ninth number, then find the ninth number.

(A) 92 (B) 100
(C) 96 (D) 108
(E) 112

Q.230 Problems on Ages

Tushar's present age is 20% more than Mukesh's present age but 50% less than Akash's present age. If the ratio of Mukesh's age 17 years hence to Akash's age 6 years ago is 2:3, then find the difference between the present ages of Tushar and Akash.

(A) 48 years (B) 36 years
(C) 42 years (D) 54 years
(E) 45 years

Q.231 Problems on Ages

p years from now, the ratio of the ages of Aman and Siddharth will be 10:7. After (p + 5) years from now, the ratio of their ages will be 15:11. If the present age of Aman is 38 years, then find the present age of Siddharth.

(A) 22 years (B) 30 years
(C) 26 years (D) 24 years
(E) 28 years

Q.232 Problems on Ages

Among three numbers, the second number is twice the first number, and the third number is four times the second number. If the average of these three numbers is 55 and a fourth number is 8 more than this average, find the average of the first and the fourth numbers.

(A) 36 (B) 42
(C) 39 (D) 45
(E) 38

Q.233 Problems on Ages

The ratio of the present ages of Sameer and Hardik is 4:5. x years ago, the ratio of their ages was 3:4, and the sum of their ages after x years will be 99 years. Find Hardik's present age.

(A) 36 years (B) 40 years
(C) 48 years (D) 45 years
(E) 42 years

Q.234 Problems on Ages

The ratio of the ages of Anuj and Gaurav is 5:4, and the ratio of the ages of Gaurav and Chirag is 4:7. The average age of Gaurav and Chirag is 44 years. Find the sum of the terms of the simplest ratio of Chirag's age after 4 years to Anuj's age after 8 years.

(A) 11 (B) 8
(C) 9 (D) 10
(E) 13

Q.235 Problems on Ages

The ratio of the present ages of Avani and Mahi is 5:4, respectively. Megha is 9 years older than Mahi. After 14 years, the ratio of Mahi's age to Avani's age will become 6:7. Find the present age of Megha.

(A) 33 years (B) 35 years
(C) 41 years (D) 37 years
(E) 39 years

Q.236 Problems on Ages

List P contains eight consecutive integers. The average of the second smallest and the greatest integer of list P is 45. List Q contains five consecutive odd integers, and the smallest integer of list Q is equal to the second greatest integer of list P. Find the average of list Q.

(A) 49 (B) 53
(C) 51 (D) 55
(E) 47

Q.237 Problems on Ages

The average marks of 50 students in a class is 42. Out of them, 4 students scored zero. Among the remaining students, the first 30 students scored an average of 44 marks and the next 12 students scored an average of 35 marks. Find the total marks scored by the remaining 4 students.

(A) 340 (B) 400
(C) 320 (D) 360
(E) 380

Q.238 Problems on Ages

The total population of village K is 900, and the ratio of males to females in village K is 4:5. The number of males in village L is 25% more than the number of males in village K. If the total number of females in both villages together is 1000, then females in village L are what percentage of the total population of village L?

(A) 50% (B) 45%
(C) 40% (D) 55%
(E) 60%

Q.239 Problems on Ages

A total amount of Rs. 29,000 is divided among Shreya, Meena, and Rhea. If Shreya, Meena, and Rhea spend Rs. 800, Rs. 1000, and Rs. 1200 respectively from their initial shares, then the ratio of their remaining shares becomes 3:4:6. Find the average of the initial shares of Shreya and Rhea.

(A) Rs. 9800 (B) Rs. 10800
(C) Rs. 9600 (D) Rs. 10400
(E) Rs. 10000

Q.240 Problems on Ages

The present ages of Avani and Rhea are in the ratio 5:8. Yash's present age is 25% less than Rhea's present age. If the difference between the present ages of Avani and Rhea is 15 years, find the average of the present ages of Rhea and Yash.

(A) 30 years (B) 45 years
(C) 32 years (D) 40 years
(E) 35 years

Q.241 Problems on Ages

Three years ago, Ananya's age was 150% more than her daughter Tara's age at that time. Nine years from now, Ananya's age will be 75% more than Tara's age at that time. Find Tara's present age.

(A) 12 years (B) 18 years
(C) 15 years (D) 20 years
(E) 24 years

Q.242 Problems on Ages

The average marks of Swati, Divya and Ananya is 36. The average marks of Ananya, Radhika and Sona is 34. The average marks of Swati and Sona is 39, and the average marks of Divya and Sona is 35. If Radhika scored 8 fewer marks than Ananya, find Divya's marks.

(A) 28 (B) 30
(C) 36 (D) 32
(E) 34

Q.243 Problems on Ages

The average age of (m + 15) employees in an office is 36 years. Eight consultants with an average age of 57 years join them, and the average age of the whole group increases by 3 years. Find the value of m.

(A) 30 (B) 33
(C) 36 (D) 40
(E) 45

Q.244 Problems on Ages

P is 50% more than Q, and R is 40 less than P. If the average of P, Q and R is 200 and the ratio of Q to R is 4:x, then find the value of x.

(A) 4 (B) 5
(C) 7 (D) 6
(E) 3

Q.245 Problems on Ages

In a class, there are 8 boys and 8 girls. The average age of the whole class, including the teacher, is 15 years. The average age of the boys is 12 years, and the teacher's age is 6 years more than the average age of the girls. Find the average age of the girls.

(A) 14 years (B) 15 years
(C) 16 years (D) 17 years
(E) 18 years

Q.246 Problems on Ages

The average present age of Abhinav and Gaurav is 45 years. If Abhinav is older than Gaurav by 26 years, then find the present age of Abhinav.

(A) 54 years (B) 64 years
(C) 56 years (D) 58 years
(E) 60 years

Q.247 Problems on Ages

Priya is 5 years older than Kritika, and Shreya is 7 years younger than Priya. If the average age of Kritika and Shreya is 23 years, then find the sum of the ages of Priya, Kritika, and Shreya.

(A) 75 years (B) 72 years
(C) 78 years (D) 81 years
(E) 69 years

Q.248 Problems on Ages

The present ages of Akash and Rajesh are in the ratio 5:8. Yash is 10 years older than Akash and 5 years younger than Rajesh. Find Rajesh's age 6 years ago.

(A) 34 years (B) 30 years
(C) 28 years (D) 40 years
(E) 36 years

Q.249 Problems on Ages

The present average age of Nidhi, Rhea and Ananya is 29 years. Rhea is 6 years older than Ananya, and Nidhi is 9 years younger than Ananya. Find Nidhi's age after 4 years.

(A) 25 years (B) 28 years
(C) 31 years (D) 23 years
(E) 27 years

Q.250 Problems on Ages

The average present age of Kabir, Akash and Kunal is 35 years. Kabir is 5 years younger than Akash, and Kunal is 8 years older than Akash. Find Kabir's present age.

(A) 27 (B) 31
(C) 29 (D) 35
(E) 37

📖 Chapter 05 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.197 Correct Option: (C)

• Average weight of five persons = 68 kg
• So, total weight of five persons = 68 × 5 = 340 kg
• Weight of A = 76 kg
• Weight of new person F is 20 kg less than A.
• So, weight of F = 76 - 20 = 56 kg
• Now, total weight after F joins = 340 + 56 = 396 kg
• Total number of persons after F joins = 6
• New average weight = 396 / 6 = 66 kg

Solution Q.198 Correct Option: (A)

• Average of the first, second and third numbers = 34.
• So, sum of the first three numbers = 34 × 3 = 102.
• Average of the second, third and fourth numbers = 42.
• So, sum of the second, third and fourth numbers = 42 × 3 = 126.
• Fourth number = 48.
• Therefore, sum of the second and third numbers = 126 - 48 = 78.
• First number = sum of first three numbers - sum of second and third numbers.
• First number = 102 - 78 = 24.

Solution Q.199 Correct Option: (E)

• %
• Solution:
• Take Mayank's marks as 100.
• Chirag scored 50% more than Mayank, so Chirag's marks = 150.
• Gaurav scored 20% less than Chirag, so Gaurav's marks = 80% of 150 = 120.
• Adarsh scored 25% more than Gaurav, so Adarsh's marks = 125% of 120 = 150.
• Therefore, Adarsh's marks as compared to Mayank's marks = 150%.

Solution Q.200 Correct Option: (B)

• There are three consecutive odd numbers, and their average is 35.
• For three consecutive odd numbers, the middle number is equal to the average.
• So, the odd numbers are 33, 35 and 37.
• Largest odd number = 37.
• Smallest even number = 2 × 37 = 74.
• Therefore, the four consecutive even numbers are 74, 76, 78 and 80.
• Sum of four even numbers = 74 + 76 + 78 + 80 = 308.
• Sum of three odd numbers = 33 + 35 + 37 = 105.
• Required difference = 308 - 105 = 203.

Solution Q.201 Correct Option: (E)

• years
• Solution:
• Five years ago, the ratio of the ages of Ishita and Naman was 3:4.
• Let Ishita's age 5 years ago be 3x.
• Let Naman's age 5 years ago be 4x.
• So, Ishita's present age = 3x + 5
• Naman's present age = 4x + 5
• Ten years from now:
• Ishita's age = 3x + 5 + 10 = 3x + 15
• Naman's age = 4x + 5 + 10 = 4x + 15
• According to the question, their ratio 10 years from now will be 6:7.
• So, we can write:
• (3x + 15) : (4x + 15) = 6 : 7
• Now, by cross multiplication:
• 7 × (3x + 15) = 6 × (4x + 15)
• 21x + 105 = 24x + 90
• 24x - 21x = 105 - 90
• 3x = 15
• x = 15/3
• = 5
• Naman's present age = 4x + 5
• Naman's present age = 4 × 5 + 5
• Naman's present age = 20 + 5 = 25 years
• Therefore, Naman's present age is 25 years.

Solution Q.202 Correct Option: (D)

• Average of 15 numbers = 82
• So, total sum of 15 numbers = 82 × 15 = 1230.
• Average of the first 6 numbers = 74
• So, sum of the first 6 numbers = 74 × 6 = 444.
• Average of the last 5 numbers = 90
• So, sum of the last 5 numbers = 90 × 5 = 450.
• Now, number of remaining numbers = 15 - 6 - 5 = 4.
• Sum of the remaining 4 numbers = Total sum - Sum of first 6 numbers - Sum of last
• 5 numbers.
• Sum of the remaining 4 numbers = 1230 - 444 - 450.
• First add the known sums: 444 + 450 = 894.
• So, sum of the remaining 4 numbers = 1230 - 894 = 336.
• Average of the remaining 4 numbers = 336/4 = 84.

Solution Q.203 **Correct Option: (A)**

• years
• Solution:
• Let Ojas's present age be x years.
• Let Darshan's present age be y years.
• Ojas's age 6 years hence = x + 6
• Darshan's age 12 years ago = y - 12
• Given:
• x + 6 = y - 12
• y = x + 18
• After 3 years:
• Ojas's age = x + 3
• Darshan's age = y + 3
• Given ratio after 3 years = 5:8
• So:
• (x + 3) : (y + 3) = 5 : 8
• Put y = x + 18.
• (x + 3) : (x + 18 + 3) = 5 : 8
• (x + 3) : (x + 21) = 5 : 8
• Now cross multiply.
• 8(x + 3) = 5(x + 21)
• 8x + 24 = 5x + 105
• 8x - 5x = 105 - 24
• 3x = 81
• x = 81/3
• x = 27
• Ojas's present age = 27 years
• Darshan's present age = x + 18 = 27 + 18 = 45 years
• Sum of present ages = 27 + 45 = 72 years

Solution Q.204 **Correct Option: (E)**

• Rs. 40000
• Solution:
• Treat the full monthly income as 100%; every percentage below is a share of that
• income.
• Rent = 10%
• Remaining salary after rent = 100% - 10% = 90%
• Food expense = 30% of 90%
• Food expense = (30/100) × 90% = 27% of the monthly salary.
• Remaining salary after rent and food = 100% - 10% - 27%
• Remaining salary = 63%
• This 63% is divided between travelling and savings in the ratio 3:4.
• Savings part = 4 parts out of 7 parts
• Savings = 63% × 4/7 = 36% of the monthly salary.
• Rent = 10%
• Difference between savings and rent = 36% - 10% = 26%
• Given difference = Rs. 10400
• So, 26% of salary = Rs. 10400
• Monthly salary = 10400 × 100/26
• Monthly salary = Rs. 40000

Solution Q.205 **Correct Option: (A)**

• years
• Solution:
• Let Mihir's present age be x years.
• Let Tanmay's present age be y years.
• Mihir's age 4 years hence = x + 4
• Tanmay's age 8 years ago = y - 8
• Given:
• x + 4 = y - 8
• y = x + 12
• After 6 years:
• Mihir's age = x + 6
• Tanmay's age = y + 6
• Given ratio after 6 years = 5:8
• So:
• (x + 6) : (y + 6) = 5 : 8
• Put y = x + 12.
• (x + 6) : (x + 12 + 6) = 5 : 8
• (x + 6) : (x + 18) = 5 : 8
• Now cross multiply.
• 8(x + 6) = 5(x + 18)
• 8x + 48 = 5x + 90
• 8x - 5x = 90 - 48
• 3x = 42
• x = 42/3
• x = 14
• Mihir's present age = 14 years
• Tanmay's present age = x + 12 = 14 + 12 = 26 years
• Sum of present ages = 14 + 26 = 40 years

Solution Q.206 **Correct Option: (C)**

• years
• Solution:
• Let the present age of the younger brother = x years.
• The elder brother's present age is 200% more than the younger brother's age.
• 200% more means 3 times.
• So, elder brother's present age = 3x.
• After 6 years:
• Younger brother's age = x + 6.
• Elder brother's age = 3x + 6.
• After 6 years, elder brother's age will be 125% more than younger brother's age.
• 125% more means 225% of younger brother's age.
• So, elder brother's age after 6 years = 225/100 of younger brother's age.
• 225/100 = 9/4.
• Now,
• 3x + 6 = 9/4 × (x + 6)
• Multiply both sides by 4.
• 12x + 24 = 9x + 54
• 12x - 9x = 54 - 24
• 3x = 30
• x = 30/3
• x = 10
• Elder brother's present age = 3x = 3 × 10 = 30 years.

Solution Q.207 **Correct Option: (E)**

• years
• Solution:
• Number of students = 35
• Average age of students = 18 years
• Total age of 35 students = 35 × 18 = 630 years
• When the teacher is included, total number of persons = 35 + 1 = 36
• New average age = 19 years
• Total age of 36 persons = 36 × 19 = 684 years
• Teacher's age = total age of 36 persons - total age of 35 students
• Teacher's age = 684 - 630
• Teacher's age = 54 years

Solution Q.208 **Correct Option: (E)**

• Let the four consecutive even numbers = x, x + 2, x + 4 and x + 6.
• Smallest number = x.
• Largest number = x + 6.
• Given ratio:
• x : x + 6 = 4 : 5
• So,
• x/(x + 6) = 4/5
• 5x = 4(x + 6)
• 5x = 4x + 24
• x = 24
• The four numbers are 24, 26, 28 and 30.
• Average = (24 + 26 + 28 + 30)/4.
• Average = 108/4 = 27.
• Therefore, the value of X is 27.

Solution Q.209 **Correct Option: (D)**

• Let Prateek's present age be x years.
• Mayank is 5 years older than Prateek.
• So, Mayank's present age = x + 5
• Five years ago:
• Prateek's age = x - 5
• Mayank's age = x + 5 - 5 = x
• Five years ago, Mayank's age was 25% more than Prateek's age.
• So, Mayank's age = 125% of Prateek's age = 5/4 of Prateek's age
• x = 5/4 × (x - 5)
• 4x = 5x - 25
• x = 25
• Prateek's present age = 25 years
• Mayank's present age = 25 + 5 = 30 years
• Average present age of Prateek, Prateek and Mayank = 24 years
• Total present age = 24 × 3 = 72 years
• Prateek's present age = 72 - 25 - 30 = 17 years
• Difference between Mayank and Prateek = 30 - 17 = 13 years

Solution Q.210 **Correct Option: (A)**

• years
• Solution:
• The present age ratio of Aman and Sahil is 5 : 3.
• Let Aman's present age = 5x and Sahil's present age = 3x.
• Five years ago:
• Aman's age = 5x - 5.
• Sahil's age = 3x - 5.
• Aman's age was twice Sahil's age at that time.
• So,
• 5x - 5 = 2(3x - 5)
• 5x - 5 = 6x - 10
• 6x - 5x = 10 - 5
• x = 5
• Aman's present age = 5x = 5 × 5 = 25 years.
• Sahil's present age = 3x = 3 × 5 = 15 years.
• Four years hence:
• Aman's age = 25 + 4 = 29 years.
• Sahil's age = 15 + 4 = 19 years.
• Sum of their ages four years hence = 29 + 19 = 48 years.
• This is equal to Pawan's age four years hence.
• So, Pawan's present age = 48 - 4 = 44 years.

Solution Q.211 **Correct Option: (E)**

• Rs. 50000
• Solution:
• Treat the full monthly income as 100%; every percentage below is a share of that
• income.
• Amount spent on rent = 20%
• Remaining income after rent = 100% - 20% = 80%
• Amount spent on groceries = 25% of remaining income
• Groceries expense = (25/100) × 80%
• Groceries expense = 20%
• Remaining income after rent and groceries = 100% - 20% - 20%
• Remaining income = 60%
• This 60% is divided between entertainment and savings in the ratio 2:3.
• Total ratio parts = 2 + 3 = 5
• Savings = 3 parts out of 5 parts
• Savings = 60% × 3/5
• Savings = 36%
• Rent = 20%
• Difference between savings and rent = 36% - 20%
• Difference = 16%
• Given difference = Rs. 8000
• So, 16% of monthly income = Rs. 8000
• Monthly income = 8000 × 100/16
• Monthly income = Rs. 50000

Solution Q.212 **Correct Option: (A)**

• Rs. 15000
• Solution:
• Let the monthly income be Rs. 100.
• Amount spent on groceries = 25% of 100 = Rs. 25
• Remaining amount = 100 - 25 = Rs. 75
• Rent = 16 2/3% of remaining amount
• 16 2/3% = 1/6
• Rent = 75 × 1/6 = Rs. 12.5
• Remaining amount after rent = 75 - 12.5 = Rs. 62.5
• Transport expense = 20% of 62.5
• Transport expense = 62.5 × 20/100 = Rs. 12.5
• Savings = 62.5 - 12.5 = Rs. 50
• So, when income is Rs. 100, savings are Rs. 50.
• Given actual savings = Rs. 30000
• So, Rs. 50 corresponds to Rs. 30000.
• Monthly income = 30000 × 100/50 = Rs. 60000
• Amount spent on groceries = 25% of 60000
• Groceries expense = 60000 × 25/100 = Rs. 15000

Solution Q.213 **Correct Option: (D)**

• years
• Solution:
• Four years from now, the age ratio of Leela and Prateek will be 4:5.
• Let Leela's age after 4 years = 4x and Prateek's age after 4 years = 5x.
• Present age of Leela = 4x - 4.
• Present age of Prateek = 5x - 4.
• Present age Ratio (Aman : Prateek) = 5:6.
• So, present age of Aman = 5/6 × (5x - 4).
• After 6 years, the sum of the ages of Leela and Aman will be 70 years.
• So, present age of Leela + present age of Aman = 70 - 12 = 58.
• Now,
• 4x - 4 + 5/6 × (5x - 4) = 58
• 4x - 4 + (25x - 20)/6 = 58
• Multiply by 6:
• 24x - 24 + 25x - 20 = 348
• 49x - 44 = 348
• 49x = 392
• x = 392/49 = 8
• Present age of Leela = 4 × 8 - 4 = 28 years.
• Present age of Prateek = 5 × 8 - 4 = 36 years.
• Required sum = 28 + 36 = 64 years.

Solution Q.214 **Correct Option: (B)**

• Five consecutive multiples of 10 have an average of 70.
• So, the five numbers are:
• 50, 60, 70, 80, 90
• Highest number = 90
• Second smallest number = 60
• Difference = 90 - 60 = 30
• This difference is equal to the largest even number in a set of four consecutive even
• numbers.
• So, largest even number = 30
• The four consecutive even numbers are:
• 24, 26, 28, 30
• Smallest even number = 24
• Square of the smallest even number = 24 × 24
• Square = 576

Solution Q.215 **Correct Option: (A)**

• Let the initial number of people in the group be n.
• Initial average weight = 45 kg.
• Initial total weight = 45n kg.
• One person weighing 63 kg leaves the group.
• New number of people = n - 1.
• New average weight = 45 - 2 = 43 kg.
• New total weight = 43(n - 1).
• So,
• 45n - 63 = 43(n - 1)
• 45n - 63 = 43n - 43
• 45n - 43n = 63 - 43
• 2n = 20
• n = 20/2
• n = 10

Solution Q.216 **Correct Option: (D)**

• Ratio of A to B = 5:3.
• Let A = 5x and B = 3x.
• Average of A and B = (5x + 3x)/2
• Average = 8x/2 = 4x
• Given, average is 48 less than A.
• So, A - average = 48
• 5x - 4x = 48
• x = 48
• B = 3x = 3 × 48 = 144
• Required value = 25% of B
• 25% of 144 = 25/100 × 144
• 25% of 144 = 36

Solution Q.217 **Correct Option: (B)**

• Average of 61 numbers = 54.
• Total of 61 numbers = 61 × 54 = 3294.
• Average of first 30 numbers = 50.
• Sum of first 30 numbers = 30 × 50 = 1500.
• Average of last 30 numbers = 56.
• Sum of last 30 numbers = 30 × 56 = 1680.
• Sum of first 30 and last 30 numbers = 1500 + 1680 = 3180.
• Middle number = Total of 61 numbers - Sum of first 30 and last 30 numbers
• Middle number = 3294 - 3180
• Middle number = 114

Solution Q.218 **Correct Option: (A)**

• Rs. 150000
• Solution:
• Arvind spends 35% on rent and 25% on food.
• Total spending = 35% + 25% = 60%
• Remaining amount = 100% - 60% = 40% of salary
• The remaining amount is divided between father and mother in the ratio of 7 : 5.
• Difference in ratio parts = 7 - 5 = 2 parts
• Given difference = Rs. 10000
• So, 2 parts = Rs. 10000
• 1 part = 10000/2 = Rs. 5000
• Total ratio parts = 7 + 5 = 12 parts
• Remaining amount = 12 × 5000 = Rs. 60000
• This remaining amount is 40% of salary.
• So, 40% of salary = Rs. 60000
• Salary = 60000 × 100/40 = Rs. 150000

Solution Q.219 **Correct Option: (C)**

• %
• Solution:
• Initial monthly income = Rs. 50,000.
• Initial monthly expenditure = Rs. 30,000.
• Initial monthly savings = Income - Expenditure
• Initial savings = 50,000 - 30,000 = Rs. 20,000.
• Income increases by 24%.
• New income = 50,000 × 124/100
• New income = Rs. 62,000.
• Savings increase by 30%.
• New savings = 20,000 × 130/100
• New savings = Rs. 26,000.
• New expenditure = New income - New savings
• New expenditure = 62,000 - 26,000
• New expenditure = Rs. 36,000.
• Increase in expenditure = 36,000 - 30,000
• Increase in expenditure = Rs. 6,000.
• Percentage increase in expenditure = 6,000 × 100/30,000
• Percentage increase = 20%.

Solution Q.220 **Correct Option: (B)**

• years
• Solution:
• Let Sameer's present age be x years.
• Aniket is 6 years older than Sameer.
• So, Aniket's present age = x + 6 years.
• Divya's present age = 20 years.
• Four years hence:
• Aniket's age = x + 6 + 4 = x + 10 years.
• Divya's age = 20 + 4 = 24 years.
• Two years ago:
• Sameer's age = x - 2 years.
• Divya's age = 20 - 2 = 18 years.
• According to the question:
• (x + 10)/24 = (x - 2)/18
• Cross multiply:
• 18 × (x + 10) = 24 × (x - 2)
• 18x + 180 = 24x - 48
• 180 + 48 = 24x - 18x
• 228 = 6x
• x = 228/6
• x = 38.

Solution Q.221 **Correct Option: (C)**

• years
• Solution:
• Three years ago, average age of Shreya, Priya and Prateek = 28 years
• Sum of their ages three years ago = 28 × 3 = 84 years
• Present sum of their ages = 84 + 3 + 3 + 3
• Present sum of their ages = 93 years
• Four years hence, sum of ages of Shreya and Prateek = 68 years
• Present sum of ages of Shreya and Prateek = 68 - 4 - 4
• Present sum of ages of Shreya and Prateek = 60 years
• Present age of Priya = Present total age - Present age sum of Shreya and Prateek
• Present age of Priya = 93 - 60
• Present age of Priya = 33 years

Solution Q.222 **Correct Option: (A)**

• years
• Solution:
• Five years ago, the ratio of ages of Sahil and Akash was 5:2.
• Let Sahil's age five years ago = 5x years.
• Let Akash's age five years ago = 2x years.
• Present age of Sahil = 5x + 5.
• Present age of Akash = 2x + 5.
• Seven years hence:
• Sahil's age = 5x + 5 + 7 = 5x + 12.
• Akash's age = 2x + 5 + 7 = 2x + 12.
• Sahil's age will be 75% more than Akash's age.
• So, Sahil's age = 175% of Akash's age = 7/4 of Akash's age.
• 5x + 12 = 7/4 × (2x + 12)
• Cross multiply:
• 4 × (5x + 12) = 7 × (2x + 12)
• 20x + 48 = 14x + 84
• 20x - 14x = 84 - 48
• 6x = 36
• x = 36/6
• x = 6.
• Akash's present age = 2x + 5
• Akash's present age = 2 × 6 + 5
• Akash's present age = 17 years.
• Kunal's present age is 100% more than Akash's present age.
• So, Kunal's present age = 2 × 17
• Kunal's present age = 34 years.

Solution Q.223 **Correct Option: (A)**

• Rs. 300000
• Solution:
• Let the monthly salary be Rs. 100.
• Rent = 18% of 100 = Rs. 18
• Remaining salary after rent = 100 - 18 = Rs. 82
• Travel expense = 25% of remaining salary
• Travel expense = 82 × 25/100 = Rs. 20.50
• Savings = 35% of monthly salary = Rs. 35
• Food expense = 100 - 18 - 20.50 - 35
• Food expense = Rs. 26.50
• So, when monthly salary is Rs. 100, food expense is Rs. 26.50
• Actual food expense = Rs. 6625
• Monthly salary = 6625 × 100/26.50
• Monthly salary = Rs. 25000
• Annual income = 25000 × 12 = Rs. 300000

Solution Q.224 **Correct Option: (D)**

• years
• Solution:
• Let Akash's present age = x years.
• Kabir is 20% older than Akash.
• So, Kabir's present age = 120% of x = 120x/100 = 6x/5
• Kabir is 4 years younger than Kunal.
• So, Kunal's present age = 6x/5 + 4
• Five years ago, Kunal's age = 6x/5 + 4 - 5 = 6x/5 - 1
• Average of present ages of Kunal, Akash and Kabir = (6x/5 + 4 + x + 6x/5)/3
• According to the question:
• 6x/5 - 1 = (6x/5 + 4 + x + 6x/5)/3
• 6x/5 - 1 = (17x/5 + 4)/3
• 3(6x/5 - 1) = 17x/5 + 4
• 18x/5 - 3 = 17x/5 + 4
• 18x/5 - 17x/5 = 7
• x/5 = 7
• x = 35
• Kabir's present age = 6x/5 = 6 × 35/5 = 42 years

Solution Q.225 **Correct Option: (E)**

• years
• Solution:
• Let Aman's present age be x years.
• Aman's present age is 20% less than Sameer's present age.
• So, Aman's age = 80% of Sameer's age.
• x = 80/100 × Sameer's age
• x = 4/5 × Sameer's age.
• So, Sameer's present age = 5x/4.
• Sameer is 15 years younger than Aniket.
• So, Aniket's present age = 5x/4 + 15.
• Two years hence, Aman's age = x + 2.
• Given, x + 2 is 60% of Aniket's present age.
• So, x + 2 = 60/100 × (5x/4 + 15)
• x + 2 = 3/5 × (5x/4 + 15)
• x + 2 = 3x/4 + 9
• x - 3x/4 = 9 - 2
• x/4 = 7
• x = 7 × 4
• x = 28.
• Aman's present age = 28 years.
• Aman's age 5 years ago = 28 - 5 = 23 years.

Solution Q.226 **Correct Option: (C)**

• years
• Solution:
• Kriti is 5 years younger than Harita.
• So, the age difference between Harita and Kriti = 5 years
• Five years ago, Harita's age was 25% more than Kriti's age.
• 25% = 1/4
• So, five years ago, age Ratio (Kriti : Harita) = 4 : 5
• Difference in ratio parts = 5 - 4 = 1 part
• 1 part = 5 years
• Five years ago, Kriti's age = 4 × 5 = 20 years
• Five years ago, Harita's age = 5 × 5 = 25 years
• Present age of Kriti = 20 + 5 = 25 years
• Present age of Harita = 25 + 5 = 30 years
• Present average age of Kriti, Yamini and Harita = 30 years
• Total present age of all three = 30 × 3 = 90 years
• Present age of Yamini = 90 - 25 - 30 = 35 years
• Present difference between Yamini and Harita = 35 - 30 = 5 years
• After six years also, the difference will remain the same.

Solution Q.227 **Correct Option: (E)**

• kg
• Solution:
• Total weight of 29 students = 29 × 30 = 870 kg
• When teacher is included, total persons = 30
• New average = 30 + 2 = 32 kg
• Total weight of 30 persons = 30 × 32 = 960 kg
• Teacher's weight = 960 - 870 = 90 kg
• Average weight of teacher, wife, and daughter = 60 kg
• Total weight of teacher, wife, and daughter = 3 × 60 = 180 kg
• Wife + daughter = 180 - 90 = 90 kg
• Let wife's weight = 100 units
• Daughter's weight is 50% less than wife's weight, so daughter's weight = 50 units
• Total = 150 units = 90 kg
• Wife's weight = 90 × 100/150 = 60 kg

Solution Q.228 **Correct Option: (B)**

• Rs. 9600
• Solution:
• Let each niece receive Rs. x
• Each daughter receives twice the amount received by each niece, so each daughter =
• Rs. 2x
• Each son receives 4 times the amount received by each daughter, so each son = 4 × 2x
• = Rs. 8x
• Total amount = amount of 4 nieces + amount of 2 daughters + amount of 3 sons
• 38400 = 4x + 2 × 2x + 3 × 8x
• 38400 = 4x + 4x + 24x
• 38400 = 32x
• x = 38400/32 = 1200
• Each son receives = 8x = 8 × 1200 = Rs. 9600

Solution Q.229 **Correct Option: (B)**

• Total of 15 numbers = 15 × 90 = 1350.
• Sum of first seven numbers = 7 × 85 = 595.
• Sum of last five numbers = 5 × 99 = 495.
• So, sum of 8th, 9th and 10th numbers = 1350 - 595 - 495 = 260.
• Let the 8th number = 4x.
• Then 9th number = 25% more than 8th = 5x.
• 10th number = 20% less than 9th = 80% of 5x = 4x.
• So, 4x + 5x + 4x = 260.
• 13x = 260.
• x = 20.
• 9th number = 5x = 5 × 20 = 100.

Solution Q.230 **Correct Option: (C)**

• years
• Solution:
• Tushar is 20% more than Mukesh.
• So, Tushar:Mukesh = 120:100 = 6:5.
• Let Tushar's present age = 6x and Mukesh's present age = 5x.
• Tushar is 50% less than Akash, so Tushar is 50% of Akash.
• Therefore, Akash's present age = 12x.
• Given ratio:
• (Mukesh's age after 17 years):(Akash's age 6 years ago) = 2:3.
• So, (5x + 17):(12x - 6) = 2:3.
• 3 × (5x + 17) = 2 × (12x - 6).
• 15x + 51 = 24x - 12.
• 63 = 9x.
• x = 7.
• Tushar's age = 6 × 7 = 42 years.
• Akash's age = 12 × 7 = 84 years.
• Required difference = 84 - 42 = 42 years.

Solution Q.231 **Correct Option: (C)**

• years
• Solution:
• Let the present age of Siddharth = y years.
• Present age of Aman = 38 years
• p years from now:
• Age of Aman = 38 + p
• Age of Siddharth = y + p
• Given ratio = 10:7
• So, (38 + p)/(y + p) = 10/7
• 7(38 + p) = 10(y + p)
• 266 + 7p = 10y + 10p
• 10y = 266 - 3p ...(1)
• After (p + 5) years from now:
• Age of Aman = 38 + p + 5 = 43 + p
• Age of Siddharth = y + p + 5
• Given ratio = 15:11
• So, (43 + p)/(y + p + 5) = 15/11
• 11(43 + p) = 15(y + p + 5)
• 473 + 11p = 15y + 15p + 75
• 15y = 398 - 4p ...(2)
• From equation (1), 10y = 266 - 3p
• Multiplying by 3/2:
• 15y = 399 - 4.5p
• Now compare with equation (2):
• 399 - 4.5p = 398 - 4p
• 1 = 0.5p
• p = 2
• Put p = 2 in equation (1):
• 10y = 266 - 3 × 2
• 10y = 260
• y = 26

Solution Q.232 **Correct Option: (C)**

• Let the first number = x.
• Second number = 2x.
• Third number = 4 × 2x = 8x.
• Average of these three numbers = 55.
• Sum of three numbers = 55 × 3 = 165.
• So, x + 2x + 8x = 165.
• 11x = 165.
• x = 15.
• First number = 15.
• Fourth number = 55 + 8 = 63.
• Average of first and fourth numbers = (15 + 63) / 2 = 78 / 2 = 39.

Solution Q.233 **Correct Option: (D)**

• years
• Solution:
• Let the present ages of Sameer and Hardik be 4a and 5a years.
• x years ago, their ages were 4a - x and 5a - x.
• According to the question:
• (4a - x)/(5a - x) = 3/4
• 4(4a - x) = 3(5a - x)
• 16a - 4x = 15a - 3x
• a = x
• After x years, sum of their ages = 99
• (4a + x) + (5a + x) = 99
• 9a + 2x = 99
• Since x = a:
• 9a + 2a = 99
• 11a = 99
• a = 9
• Hardik's present age = 5a = 5 × 9 = 45 years

Solution Q.234 **Correct Option: (C)**

• Given Ratio (Anuj : Gaurav) = 5:4
• Given Ratio (Gaurav : Chirag) = 4:7
• So, Anuj : Gaurav : Chirag = 5:4:7
• Let their present ages be 5x, 4x and 7x.
• Average age of Gaurav and Chirag = 44 years
• So, total age of Gaurav and Chirag = 44 × 2 = 88 years
• 4x + 7x = 88
• 11x = 88
• x = 8
• Anuj's present age = 5 × 8 = 40 years
• Chirag's present age = 7 × 8 = 56 years
• Chirag's age after 4 years = 56 + 4 = 60 years
• Anuj's age after 8 years = 40 + 8 = 48 years
• Required ratio = 60:48 = 5:4
• Sum of terms = 5 + 4 = 9

Solution Q.235 **Correct Option: (D)**

• years
• Solution:
• Let the present ages of Avani and Mahi be 5x and 4x years.
• After 14 years, their ages will be 5x + 14 and 4x + 14 years.
• Given, after 14 years, Mahi : Avani = 6 : 7.
• So, (4x + 14) : (5x + 14) = 6 : 7.
• Using cross multiplication:
• 7 × (4x + 14) = 6 × (5x + 14)
• 28x + 98 = 30x + 84
• 30x - 28x = 98 - 84
• 2x = 14
• x = 7
• Mahi's present age = 4x = 4 × 7 = 28 years.
• Megha is 9 years older than Mahi.
• Megha's present age = 28 + 9 = 37 years.

Solution Q.236 **Correct Option: (C)**

• Let the eight consecutive integers in list P = x, x + 1, x + 2, x + 3, x + 4, x + 5, x + 6,
• and x + 7.
• Second smallest integer = x + 1.
• Greatest integer = x + 7.
• Their average = 45.
• So, (x + 1 + x + 7) / 2 = 45.
• (2x + 8) / 2 = 45.
• x + 4 = 45.
• x = 41.
• So, list P is 41, 42, 43, 44, 45, 46, 47, 48.
• Second greatest integer of list P = 47.
• The smallest integer of list Q = 47.
• List Q contains five consecutive odd integers, so list Q is 47, 49, 51, 53, 55.
• For five consecutive odd integers, average is the middle number.
• Average of list Q = 51.

Solution Q.237 **Correct Option: (D)**

• Total marks of 50 students = 50 × 42 = 2100.
• Marks of 4 students who scored zero = 0.
• Total marks of first 30 students = 30 × 44 = 1320.
• Total marks of next 12 students = 12 × 35 = 420.
• Total known marks = 1320 + 420 + 0 = 1740.
• Remaining marks = 2100 - 1740 = 360.
• So, the total marks scored by the remaining 4 students = 360.

Solution Q.238 **Correct Option: (A)**

• %
• Solution:
• Total population of village K = 900.
• Ratio of males to females in village K = 4:5.
• Total ratio parts = 4 + 5 = 9.
• Males in village K = 900 × 4 / 9 = 400.
• Females in village K = 900 × 5 / 9 = 500.
• Males in village L are 25% more than males in village K.
• Males in village L = 400 × 125 / 100 = 500.
• Total females in both villages = 1000.
• Females in village L = 1000 - 500 = 500.
• Total population of village L = males in L + females in L = 500 + 500 = 1000.
• Required percentage = 500 × 100 / 1000 = 50%.

Solution Q.239 **Correct Option: (E)**

• Rs. 10000
• Solution:
• Total initial amount = Rs. 29,000.
• Total amount spent = 800 + 1000 + 1200 = Rs. 3000.
• Total remaining amount = 29000 - 3000 = Rs. 26000.
• Remaining ratio of Shreya, Meena, and Rhea = 3:4:6.
• Sum of ratio parts = 3 + 4 + 6 = 13.
• Value of 1 part = 26000 / 13 = Rs. 2000.
• Shreya's remaining share = 3 × 2000 = Rs. 6000.
• Rhea's remaining share = 6 × 2000 = Rs. 12000.
• Shreya's initial share = 6000 + 800 = Rs. 6800.
• Rhea's initial share = 12000 + 1200 = Rs. 13200.
• Average of initial shares of Shreya and Rhea = (6800 + 13200) / 2.
• = 20000 / 2 = Rs. 10000.
• Therefore, the required average is Rs. 10000.

Solution Q.240 **Correct Option: (E)**

• years
• Solution:
• Let the present ages of Avani and Rhea be 5x and 8x respectively.
• Difference between their ages = 8x - 5x = 3x
• According to the question, 3x = 15
• x = 5
• Avani's present age = 5 × 5 = 25 years
• Rhea's present age = 8 × 5 = 40 years
• Yash's present age is 25% less than Rhea's age.
• So, Yash's age = 75% of 40 = 40 × 75/100 = 30 years
• Average of Rhea and Yash = (40 + 30) / 2 = 35 years

Solution Q.241 **Correct Option: (C)**

• years
• Solution:
• Let Tara's present age = x years and Ananya's present age = y years.
• Three years ago, Tara's age = x - 3 and Ananya's age = y - 3.
• Ananya's age was 150% more than Tara's age, so Ananya's age = 250% of Tara's age =
• 5/2 of Tara's age.
• So, y - 3 = 5/2 × (x - 3)
• Nine years from now, Tara's age = x + 9 and Ananya's age = y + 9.
• Ananya's age will be 75% more than Tara's age, so Ananya's age = 175% of Tara's age
• = 7/4 of Tara's age.
• So, y + 9 = 7/4 × (x + 9)
• Now subtract the first condition from the second condition.
• (y + 9) - (y - 3) = 12
• 7/4 × (x + 9) - 5/2 × (x - 3) = 12
• (7x + 63)/4 - (10x - 30)/4 = 12
• 93 - 3x = 48
• 3x = 45
• x = 15
• Tara's present age = 15 years

Solution Q.242 **Correct Option: (D)**

• Average marks of Swati, Divya and Ananya = 36.
• So, Swati + Divya + Ananya = 36 × 3 = 108.
• Average marks of Ananya, Radhika and Sona = 34.
• So, Ananya + Radhika + Sona = 34 × 3 = 102.
• Average marks of Swati and Sona = 39.
• So, Swati + Sona = 39 × 2 = 78.
• Average marks of Divya and Sona = 35.
• So, Divya + Sona = 35 × 2 = 70.
• Radhika scored 8 fewer marks than Ananya.
• So, Radhika = Ananya - 8.
• Let Divya's marks be K and Sona's marks be S.
• From Divya + Sona = 70, K + S = 70.
• From Swati + Sona = 78, Swati = 78 - S.
• From K + S = 70, S = 70 - K.
• So, Swati = 78 - (70 - K) = K + 8.
• Now, Swati + Divya + Ananya = 108.
• (K + 8) + K + Ananya = 108.
• 2K + Ananya = 100.
• So, Ananya = 100 - 2K.
• Also, Ananya + Radhika + Sona = 102.
• Ananya + (Ananya - 8) + Sona = 102.
• 2Meera + Sona = 110.
• Put Ananya = 100 - 2K and Sona = 70 - K.
• 2(100 - 2K) + (70 - K) = 110.
• 200 - 4K + 70 - K = 110.
• 270 - 5K = 110.
• 5K = 160.
• K = 32.
• Divya's marks = 32.

Solution Q.243 **Correct Option: (B)**

• Let the number of employees before joining be n.
• So, n = m + 15
• Average age of n employees = 36 years
• Total age of n employees = 36n
• Average age of 8 consultants = 57 years
• Total age of 8 consultants = 8 × 57 = 456 years
• New average age = 36 + 3 = 39 years
• Total number of people after joining = n + 8
• According to the question:
• 36n + 456 = 39 × (n + 8)
• 36n + 456 = 39n + 312
• 456 - 312 = 39n - 36n
• 144 = 3n
• n = 48
• Now, m + 15 = 48
• m = 48 - 15 = 33

Solution Q.244 **Correct Option: (B)**

• Let Q = y.
• P is 50% more than Q.
• So, P = 150% of y = 1.5y.
• R is 40 less than P.
• So, R = 1.5y - 40.
• Average of P, Q and R = 200.
• So, sum of P, Q and R = 200 × 3 = 600.
• Now, 1.5y + y + 1.5y - 40 = 600.
• 4y - 40 = 600.
• 4y = 640.
• y = 160.
• So, Q = 160.
• P = 1.5 × 160 = 240.
• R = 240 - 40 = 200.
• Ratio of Q to R = 160:200 = 4:5.
• So, x = 5.

Solution Q.245 **Correct Option: (D)**

• years
• Solution:
• Number of boys = 8
• Number of girls = 8
• Average age of boys = 12 years
• Total age of boys = 8 × 12 = 96 years
• Let the average age of girls = x years.
• Total age of girls = 8x
• Teacher's age = x + 6
• Total number of persons including teacher = 8 + 8 + 1 = 17
• Average age of all = 15 years
• Total age of all = 17 × 15 = 255 years
• So, 96 + 8x + x + 6 = 255
• 102 + 9x = 255
• 9x = 153
• x = 153 / 9 = 17
• Average age of girls = 17 years

Solution Q.246 **Correct Option: (D)**

• years
• Solution:
• Average age of Abhinav and Gaurav = 45 years.
• Sum of their ages = 45 × 2 = 90 years.
• Difference between their ages = 26 years.
• Since Abhinav is older, Abhinav's age = (sum + difference) / 2.
• Abhinav's age = (90 + 26) / 2 = 116 / 2 = 58 years.
• Therefore, the present age of Abhinav is 58 years.

Solution Q.247 **Correct Option: (A)**

• years
• Solution:
• Let Kritika's age = x years
• Priya is 5 years older than Kritika.
• So, Priya's age = x + 5 years
• Shreya is 7 years younger than Priya.
• So, Shreya's age = x + 5 - 7 = x - 2 years
• Average age of Kritika and Shreya = 23 years
• So, total age of Kritika and Shreya = 23 × 2 = 46 years
• x + x - 2 = 46
• 2x - 2 = 46
• 2x = 48
• x = 24
• Kritika's age = 24 years
• Priya's age = 24 + 5 = 29 years
• Shreya's age = 24 - 2 = 22 years
• Sum of their ages = 24 + 29 + 22 = 75 years

Solution Q.248 **Correct Option: (A)**

• years
• Solution:
• Let the present ages of Akash and Rajesh be 5x and 8x years respectively.
• Yash is 10 years older than Akash, so Yash's age = 5x + 10.
• Yash is 5 years younger than Rajesh, so Yash's age = 8x - 5.
• Now, 5x + 10 = 8x - 5.
• 15 = 3x.
• x = 5.
• Rajesh's present age = 8x = 8 × 5 = 40 years.
• Rajesh's age 6 years ago = 40 - 6 = 34 years.

Solution Q.249 **Correct Option: (A)**

• years
• Solution:
• Let Ananya's present age be x years.
• Rhea's present age = x + 6 years.
• Nidhi's present age = x - 9 years.
• Average age of Nidhi, Rhea and Ananya = 29 years.
• So, total present age of all three = 29 × 3 = 87 years.
• x + (x + 6) + (x - 9) = 87
• 3x - 3 = 87
• 3x = 90
• x = 30
• Nidhi's present age = x - 9 = 30 - 9 = 21 years.
• Nidhi's age after 4 years = 21 + 4 = 25 years.

Solution Q.250 **Correct Option: (C)**

• Let Akash's present age be x years.
• Kabir is 5 years younger than Akash.
• So, Kabir's age = x - 5.
• Kunal is 8 years older than Akash.
• So, Kunal's age = x + 8.
• Average age of Kabir, Akash and Kunal = 35 years.
• So, total age of all three = 35 × 3 = 105 years.
• Now,
• (x - 5) + x + (x + 8) = 105
• 3x + 3 = 105
• 3x = 102
• x = 34
• Therefore, Kabir's present age = 34 - 5 = 29 years.

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